

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\121-DQpeR6.pvdb

Date of Analysis : Sep 9, 2026 11:11am

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 7100.000 x 4600.000 mm ID	
Vessel Design Length, Tangent to Tangent	43739.99 mm
Distance of Bottom Tangent above Grade	9000.00 mm
Distance of Base above Grade	0.00 mm
Specified Datum Line Distance	9000.00 mm
Internal Design Temperature	130 °C
Internal Design Pressure	4.550 MPa
External Design Temperature	0 °C
Maximum Allowable Working Pressure	5 MPa
Shop Test Pressure	6.391 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-12.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Cone	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-765 I	...	...	K03046	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Btm skirt-1	...	...	...	1.6000	N/A	No
Btm skirt-2	...	...	...	1.6000	N/A	No
Btm skirt-3	...	...	...	1.6000	N/A	No
Top skirt	...	...	...	1.6000	N/A	No
BTM dish	4.687	0.00	5.000	6.0000	N/A	No
shell-1	4.650	0.00	5.000	6.0000	N/A	No
shell-2	4.619	0.00	5.000	6.0000	N/A	No
shell-3	4.588	0.00	5.000	6.0000	N/A	No
shell-4	4.563	0.00	5.000	6.0000	N/A	No
shell-5	4.563	0.00	5.000	6.0000	N/A	No
shell-6	4.563	0.00	5.000	6.0000	N/A	No
shell-7	4.563	0.00	5.000	6.0000	N/A	No
shell-8	4.563	0.00	5.000	6.0000	N/A	No
cone	4.563	0.00	5.000	6.0000	N/A	No
shell-9	4.563	0.00	5.000	3.0000	N/A	No
shell-10	4.563	0.00	5.000	3.0000	N/A	No
shell-11	4.563	0.00	5.000	3.0000	N/A	No
shell-12	4.563	0.00	5.000	3.0000	N/A	No



shell-13	4.563	0.00	5.000	3.0000	N/A	No
shell-14	4.563	0.00	5.000	3.0000	N/A	No
shell-15	4.563	0.00	5.000	3.0000	N/A	No
TOP dish	4.563	0.00	5.000	3.0000	N/A	No

Liquid Level: 11900.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.060

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-7000.0	2000.0	36.0	36.0	...	...	0.70	0.70
Skirt	-4000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	-1000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	34.0	34.0	...	...	0.70	0.55
Sphere	-0.0	0.0	70.0	66.0	60.0	...	1.00	1.00
Cylinder	3000.0	3000.0	114.0	114.0	113.3	...	1.00	1.00
Cylinder	6000.0	3000.0	114.0	114.0	112.6	...	1.00	1.00
Cylinder	9000.0	3000.0	114.0	114.0	111.9	...	1.00	1.00
Cylinder	12000.0	3000.0	112.0	112.0	111.3	...	1.00	1.00
Cylinder	15000.0	3000.0	112.0	112.0	111.3	...	1.00	1.00
Cylinder	18000.0	3000.0	112.0	112.0	111.3	...	1.00	1.00
Cylinder	20000.0	2000.0	112.0	112.0	111.3	...	1.00	1.00
Cylinder	21650.0	1650.0	112.0	112.0	111.3	...	1.00	1.00
Conical	24590.0	2940.0	144.0	122.0	120.4	...	1.00	1.00
Cylinder	27590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	30590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	33590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	36590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	39590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	42590.0	3000.0	72.0	72.0	71.2	...	1.00	1.00
Cylinder	43740.0	1150.0	72.0	72.0	71.2	...	1.00	1.00
Sphere	43740.0	0.0	46.0	40.0	37.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
C	WRC-107/537	0.907	n/a	Passed
A	WRC-297	0.978	Shell	Passed
P	WRC-297	0.991	Shell	Passed
D	WRC-297	0.991	Shell	Passed
B	WRC-107/537	0.928	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	240570	2706219	6.501E+09	84.58E+09
20	30	3500	230429	2705190	6.030E+09	79.16E+09
30	40	6500	217203	2701503	5.358E+09	71.05E+09
40	50	8500	205783	2694620	4.723E+09	62.95E+09
50	60	8999.98	201976	2691676	4.519E+09	60.26E+09
60	70	10500	201976	2596976	4.519E+09	60.26E+09
70	80	13500	185985	2466701	3.937E+09	52.66E+09
80	90	16500	171781	2310565	3.400E+09	45.49E+09
90	100	19500	157408	2117439	2.906E+09	38.85E+09
100	110	22500	145644	2007195	2.452E+09	32.66E+09
110	120	25500	131522	1841839	2035698944	26.88E+09
120	130	28000	117280	1601811	1662343936	21.71E+09
130	140	29825	109438	1466555	1435533056	18.64E+09
140	150	32120	93984.3	1350810	1267641472	16.32E+09
150	160	35090	81745.4	1060571	1009214080	12.77E+09
160	170	38090	73844.8	952618	775734208	9.752E+09



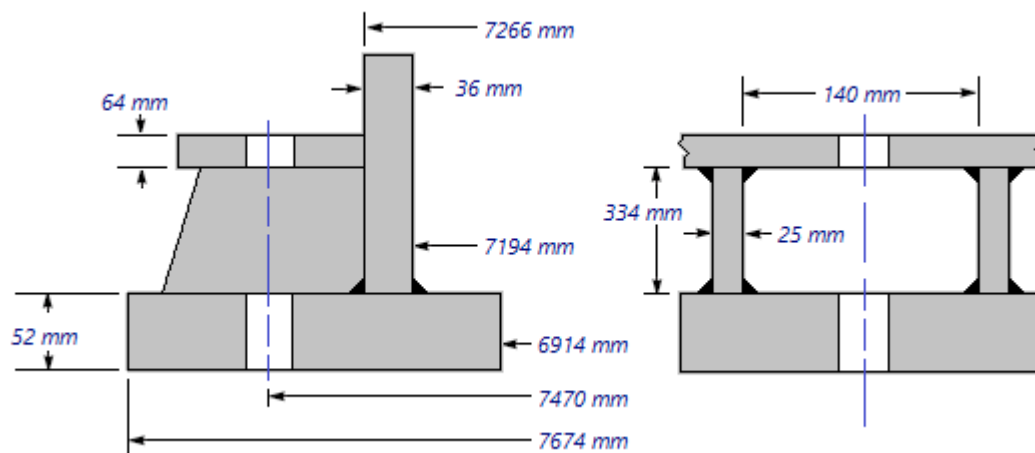
170	180	41090	63459.3	827797	569694528	7.081E+09
180	190	44090	53009.2	716114	394921088	4.764E+09
190	200	47090	42503.2	569663	251594448	2.834E+09
200	210	50090	34602.6	431772	135888896	1331734784
210	220	52165	24105.9	236877	47790548	328355936
220	230	52740	21077.3	119352	21800370	123445904

Abs Max of the all of the Stress Ratio's : 0.9413

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	52.0000	mm
Inside Diameter of Basing	6914.0005	mm
Outside Diameter of Basing	7673.9995	mm
Nominal Diameter of Bolts	76.0000	mm
Diameter of Bolt Circle	7469.9995	mm
Number of Bolts	52	
Thickness of Gusset Plates	25.0000	mm
Average Width of Gusset Plates	194.0000	mm
Height of Gussets	334.0000	mm
Distance between Gussets	140.0000	mm
Thickness of Top Plate or Ring	64.0000	mm
Radial Width of the Top Plate	204.0000	mm
Number of Gussets	104	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	301216.	N
Total Earthquake Shear on Support	2766866.	N
Wind Moment on Support	15800850432.	N-mm
Earthquake Moment on Support	93878001664.	N-mm
Maximum Initial Bolt Load	633196.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	878360.9	kgm
Shop Test - Fabricated + water (full)	2410680.5	kgm
Shipping - Fab. + rem. intls.+ shipping app.	1074875.8	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	1230512.5	kgm
Empty - Fab. + intls. + details + wghts.	1230512.6	kgm



Operating - Empty + operating liquid (no ca) 1817491.2 kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
C	5	...	...	OK	OK	Passed
A	5	...	...	OK	OK	Passed
MH-6	5	...	...	...	OK	Passed
MH-5	5	...	...	...	OK	Passed
MH-4	5	...	...	...	OK	Passed
E	5	...	...	OK	Failed	Passed
P	5	...	...	OK	OK	Passed
MH-3	5	...	...	...	OK	Passed
MH-2	5	...	...	...	OK	Passed
D	5	...	...	OK	OK	Passed
MH-1	5	...	...	...	OK	Passed
B	5	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5 Nozzle : MH-4
 Minimum MAWP Shells/Flanges : 5 Element : shell-1
 Minimum MAPnc Shells/Flanges : 5 Element : shell-15

Computed Vessel M.A.W.P. : 5 MPa

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- Error: Nozzle A, was too close to a gross structural discontinuity.
- Error: Nozzle MH-6, was too close to a gross structural discontinuity.
- Error: Nozzle E, was too close to a gross structural discontinuity.

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
50	C	0.000	0.000	463.200
80	A	8099.997	180.000	310.800
90	MH-6	9800.009	300.000	310.800
110	MH-5	16399.980	300.000	310.800
130	MH-4	20599.980	300.000	310.800
140	E	22623.781	180.000	539.400
150	P	24739.980	180.000	28.400
170	MH-3	32089.979	225.000	307.800
200	MH-2	40589.973	300.000	307.800
200	D	40089.973	180.000	231.600
210	MH-1	42889.969	300.000	307.800
220	B	0.000	0.000	307.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
P	2.000 in	LWN	LWN	2.787	10.000	...	...	323.24	600
D	18.000 in	Actual	WNF	19.575	20.000	...	...	492.80	600
A	24.000 in	Actual	WNF	31.402	94.000	...	...	537.39	600
MH-6	24.000 in	Actual	WNF	30.929	88.000	...	...	536.32	600
MH-5	24.000 in	Actual	WNF	30.929	88.000	...	...	536.32	600
MH-4	24.000 in	Actual	WNF	30.929	88.000	...	...	536.32	600
MH-3	24.000 in	Actual	WNF	29.512	70.000	...	...	507.19	600
MH-2	24.000 in	Actual	WNF	29.512	70.000	...	...	507.19	600
MH-1	24.000 in	Actual	WNF	29.512	70.000	...	...	509.71	600
B	24.000 in	Actual	WNF	27.150	40.000	...	...	473.04	600
C	36.000 in	Actual	WNF	43.717	98.000	...	...	612.59	600
E	42.000 in	Actual	WNF	44.362	30.000	...	...	1013.62	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
P	SA-266 2	10.000	10.000	...	...	...
D	SA-266 2	20.000	20.000	...	...	...
A	SA-266 2	37.000	10.000	...	...	...
MH-6	SA-266 2	37.000	10.000	...	...	...
MH-5	SA-266 2	37.000	10.000	...	...	...
MH-4	SA-266 2	37.000	10.000	...	...	...
MH-3	SA-266 2	37.000	10.000	...	...	...
MH-2	SA-266 2	37.000	10.000	...	...	...
MH-1	SA-266 2	37.000	10.000	...	...	...
B	SA-266 2	37.000	10.000	...	...	...
C	SA-266 2	37.000	10.000	...	...	...
E	SA-765 I	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
P	24739.977	180.0	250.00	0.00	shell-9
D	40089.969	180.0	400.00	0.00	shell-14
A	8099.996	180.0	400.00	0.00	shell-3
MH-6	9800.006	300.0	400.00	0.00	shell-4
MH-5	16399.977	300.0	400.00	0.00	shell-6
MH-4	20599.977	300.0	400.00	0.00	shell-8
MH-3	32089.977	225.0	400.00	0.00	shell-11
MH-2	40589.973	300.0	400.00	0.00	shell-14
MH-1	42889.973	300.0	400.00	0.00	shell-15
B	...	0.0	400.00	0.00	TOP dish
C	...	0.0	500.00	0.00	BTM dish
E	22623.779	180.0	800.00	0.00	cone

PV Elite 28 U2 Licensee: SL User
FileName : 121-D_Ope_R6
Nozzle Schedule:

Step: 54 11:11am

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 70.0mm THK X 7148.0mm ID X 0.0mm	SA-516 70
1	SPHERICAL HEAD: 46.0mm THK X 4632.0mm ID X 0.0mm	SA-516 70
13	PLATFORM: 1000mm X 1500mm WIDE	
1	PACKING: 3550.000mm	...
1	INSULATION: 3550mm X 80mm THK	insu
1	LADDER: 7000.0mm	...
1	PACKING: 1000.000mm	...
12	INSULATION: 3000mm X 80mm THK	insu
1	PACKING: 1350.000mm	...
1	PACKING: 2650.000mm	...
1	PACKING: 1250.000mm	...
1	PACKING: 1999.924mm	...
1	INSULATION: 2000mm X 80mm THK	insu
1	LADDER: 19600.0mm	...
1	PACKING: 750.003mm	...
1	INSULATION: 1650mm X 80mm THK	insu
1	INSULATION: 2940mm X 80mm THK	insu
1	PACKING: 2700.000mm	...
1	PACKING: 2999.921mm	...
1	PACKING: 150.007mm	...
1	PACKING: 749.991mm	...
1	PACKING: 2999.990mm	...
1	PACKING: 2100.010mm	...
3	TRAYS # 3 SPACING 11.811024 mm	
1	INSULATION: 1150mm X 80mm THK	insu
1	PLATFORM: 1000mm X 8000mm WIDE	
1	LADDER: 26000.0mm	...
1	INSULATION: 2300mm X 80mm THK	insu
6	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7674.0mm OD X 6914.0 ID X 52.0mm	SA-516 70
52	BASERING BOLTS: 76.0mm DIA	SA-36
52	BASERING BOLT NUTS: 76.0mm DIA	...
1	BASERING TOP RING: 7674.0mm OD X 7266.0 ID X 64.0mm	...



104	BASERING GUSSET PLATES: 194.0mm X 194.0mm X 334.0mm X 25.0mm	...
4	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3076.1mm X 2000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3013.4mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 21344.7mm X 2000.0mm	SA-516 70
10	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3015.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21105.3mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3077.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21356.5mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3077.9mm X 1000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3015.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21356.5mm X 1000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 114.0mm THK X 8860.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 114.0mm THK X 4584.9mm X 3000.0mm	SA-516 70
12	CYLINDRICAL SEGMENT 114.0mm THK X 3005.6mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 114.0mm THK X 19299.3mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 114.0mm THK X 1265.6mm X 3000.0mm	SA-516 70
18	CYLINDRICAL SEGMENT 112.0mm THK X 3005.6mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 112.0mm THK X 19299.3mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 112.0mm THK X 1265.6mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 112.0mm THK X 3005.6mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 112.0mm THK X 19299.3mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 112.0mm THK X 1265.6mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 112.0mm THK X 19299.3mm X 1650.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 112.0mm THK X 3005.6mm X 1650.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 112.0mm THK X 1265.6mm X 1650.0mm	SA-516 70
7	CON. CONICAL SEGMENT 144.0mm THK X 3005.6mm X 2940.0mm X 1947.3m	SA-516 70
1	CON. CONICAL SEGMENT 144.0mm THK X 21039.3mm X 2940.0mm X 13631.	SA-516 70
19	CYLINDRICAL SEGMENT 72.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 72.0mm THK X 2259.3mm X 3000.0mm	SA-516 70
5	CYLINDRICAL SEGMENT 72.0mm THK X 11403.1mm X 3000.0mm	SA-516 70
5	CYLINDRICAL SEGMENT 72.0mm THK X 6095.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 72.0mm THK X 11403.1mm X 1150.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 72.0mm THK X 3048.0mm X 1150.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 72.0mm THK X 6095.9mm X 1150.0mm	SA-516 70
1	NAMEPLATE	...

PV Elite 28 U2 Licensee: SL User
FileName : 121-D_Ope_R6
Bill of Material:

Step: 53 11:11am

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Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
BTM dish	[9]	!	-12	Yes			
shell-1	[8]	!	-12	Yes			
shell-2	[8]	!	-12	Yes			
shell-3	[8]	!	-12	Yes			
shell-4	[8]	!	-12	Yes			
shell-5	[8]	!	-12	Yes			
shell-6	[8]	!	-12	Yes			
shell-7	[8]	!	-12	Yes			
shell-8	[8]	!	-12	Yes			
cone	[8]	!	-12	Yes			
shell-9	[8]	!	-12	Yes			
shell-10	[8]	!	-12	Yes			
shell-11	[8]	!	-12	Yes			
shell-12	[8]	!	-12	Yes			
shell-13	[8]	!	-12	Yes			
shell-14	[8]	!	-12	Yes			
shell-15	[8]	!	-12	Yes			
TOP dish	[9]	!	-12	Yes			
C	[1]	!	-12	Yes			
Nozzle Flg A	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-6	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-5	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-4	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg E	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg P	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-3	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-2	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg D	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg MH-1	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg B	[5]	A	-18	-48	Yes		
	[1]	!	-12	Yes			
Nozzle Flg Nozl. Bolting[22]	[5]	A	-18	-48	No	1.000 64.000	1.00

Warmest MDMT: -12 -48

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable



- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	4.5503	MPa
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	0	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/22
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Btm skirt-1
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	7266 mm
Diameter of Skirt at Base	7266 mm
Skirt Thickness	36 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2



Material Name	SA-516 70	
Allowable Stress, Ambient	175	MPa
Allowable Stress, Operating	161.57	MPa
Allowable Stress, Hydrotest	247	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	242.6	MPa
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	
Element From Node	10	
Detail Type	Platform	
Detail ID	PLAT- SKIRT	
Dist. from "FROM" Node / Offset dist	1999	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	1499999	mm ²
Platform Weight	17700	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element#	2/22	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Btm skirt-2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	7266	mm
Diameter of Skirt at Base	7266	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT- SKIRT	
Dist. from "FROM" Node / Offset dist	3000	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	1499999	mm ²
Platform Weight	17692	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element#	3/22	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Btm skirt-3	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	7266	mm
Diameter of Skirt at Base	7266	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element From Node	30	
Detail Type	Weight	
Detail ID	skirt manwy/ven	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	7844.8	N
Offset from Element Centerline	1000	mm

Element#	4/22	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	Top skirt	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	7266	mm
Diameter of Skirt at Base	7266	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.55	
Weld is pre-Heated	Yes	

Element#	5/22	
Element From Node	50	
Element To Node	60	
Element Type	Spherical	
Description	BTM dish	
Distance "FROM" to "TO"	0	mm
Inside Diameter	7148	mm
Element Thickness	66	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	70	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Packing	
Detail ID	Pack 1: Btm dis	
Dist. from "FROM" Node / Offset dist	-3550	mm
Height of Packed Section	3550	mm
Density	0.000166	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	-3550	mm
Height/Length of Liquid	3550	mm
Liquid Density	0.001058	kg/cm³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	-3550	mm
Height/Length of Insulation	3550	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	C	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	164974	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT

Element#	6/22
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7100 mm
Element Thickness	114 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	114 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5503 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT-1	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	5844799	mm ²
Platform Weight	74048	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	7000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	60	
Detail Type	Packing	
Detail ID	pack#5	
Dist. from "FROM" Node / Offset dist	0.005233	mm
Height of Packed Section	1000	mm
Density	0.000166	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg/cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	60	
Detail Type	Weight	
Detail ID	VB	
Dist. from "FROM" Node / Offset dist	0.005233	mm
Miscellaneous Weight	2549.6	N
Offset from Element Centerline	0	mm

Element From Node	60	
Detail Type	Weight	
Detail ID	btm dish SF wt	
Dist. from "FROM" Node / Offset dist	0.005233	mm
Miscellaneous Weight	27163	N
Offset from Element Centerline	0	mm

Element From Node	60	
Detail Type	Weight	
Detail ID	CPP0VH	
Dist. from "FROM" Node / Offset dist	0.005233	mm
Miscellaneous Weight	51472	N
Offset from Element Centerline	5343	mm



Element To Node 80
 Element Type Cylinder
 Description shell-2
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 7100 mm
 Element Thickness 114 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 114 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.5503 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Platform
 Detail ID PLAT-2
 Dist. from "FROM" Node / Offset dist 1000 mm
 Platform Start Angle (degrees) 90.0
 Platform End Angle (degrees) 270.0
 Platform Wind Area 2999999 mm²
 Platform Weight 35754 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 135 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 0 mm
 Unit Weight of Ladder 0 N/mm
 Platform Length (top head platform) 0 mm

Element From Node 70
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 3000 mm
 Liquid Density 0.001058 kg/cm³

Element From Node 70
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element# 8/22
 Element From Node 80
 Element To Node 90
 Element Type Cylinder
 Description shell-3
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 7100 mm
 Element Thickness 114 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 114 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.5503 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa



Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Platform	
Detail ID	PLAT-3	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	2999999	mm ²
Platform Weight	35754	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2350	mm
Liquid Density	0.001058	kg/cm ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	80	
Detail Type	Nozzle	
Detail ID	A	
Dist. from "FROM" Node / Offset dist	2100	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11812	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT
Element From Node	80	
Detail Type	Weight	
Detail ID	FPA	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	32360	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	FPABolting ba	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	2451.5	N
Offset from Element Centerline	0	mm
Element From Node	80	



Detail Type	Weight	
Detail ID	APPOVH	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	208377	N
Offset from Element Centerline	5267	mm

Element#	9/22	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	shell-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7100	mm
Element Thickness	112	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	112	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Packing	
Detail ID	pack#4	
Dist. from "FROM" Node / Offset dist	1650	mm
Height of Packed Section	1350	mm
Density	0.000166	kg/cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	90	
Detail Type	Nozzle	
Detail ID	MH-6	
Dist. from "FROM" Node / Offset dist	800.01	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	17530	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT

Element From Node	90	
Detail Type	Weight	
Detail ID	B#4btm TSR	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm

Element From Node	90	
Detail Type	Weight	
Detail ID	B#4btm supp de	
Dist. from "FROM" Node / Offset dist	1650	mm



Miscellaneous Weight 43146 N
 Offset from Element Centerline 0 mm

Element# 10/22
 Element From Node 100
 Element To Node 110
 Element Type Cylinder
 Description shell-5
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 7100 mm
 Element Thickness 112 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 112 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.5503 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 100
 Detail Type Platform
 Detail ID PLAT-4
 Dist. from "FROM" Node / Offset dist 0.001161 mm
 Platform Start Angle (degrees) 90.0
 Platform End Angle (degrees) 270.0
 Platform Wind Area 2999999 mm²
 Platform Weight 35738 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 135 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 0 mm
 Unit Weight of Ladder 0 N/mm
 Platform Length (top head platform) 0 mm

Element From Node 100
 Detail Type Packing
 Detail ID pack#4
 Dist. from "FROM" Node / Offset dist 0.02325 mm
 Height of Packed Section 2650 mm
 Density 0.000166 kg/cm³
 Percent Volume Holdup 25.0
 Specific Gravity of Packing Liquid 1.0599999

Element From Node 100
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 100
 Detail Type Weight
 Detail ID B#4top HDG
 Dist. from "FROM" Node / Offset dist 2650 mm
 Miscellaneous Weight 9217.6 N
 Offset from Element Centerline 0 mm



Element From Node	100
Detail Type	Weight
Detail ID	B#4top dist. d
Dist. from "FROM" Node / Offset dist	2650 mm
Miscellaneous Weight	49128 N
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	B#4top dist. T
Dist. from "FROM" Node / Offset dist	2650 mm
Miscellaneous Weight	2941.8 N
Offset from Element Centerline	0 mm

Element#	11/22
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	shell-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7100 mm
Element Thickness	112 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	112 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5503 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Platform
Detail ID	PLAT-5
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	2999999 mm ²
Platform Weight	35738 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	110
Detail Type	Packing
Detail ID	pack#3
Dist. from "FROM" Node / Offset dist	1750 mm
Height of Packed Section	1250 mm
Density	0.000166 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm



Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	MH-5	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	17492	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	B#3btm TSR	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	B#3btm supp de	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	43146	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Nozzle wt	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	31869	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	10T margin	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	98060	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Cleat wt	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	140226	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Lifting wts	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	264762	N
Offset from Element Centerline	0	mm

Element#	12/22	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	shell-7	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	7100	mm
Element Thickness	112	mm
Internal Corrosion Allowance	6	mm



Nominal Thickness	112	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Packing	
Detail ID	pack#3	
Dist. from "FROM" Node / Offset dist	0.07441	mm
Height of Packed Section	1999.9	mm
Density	0.000166	kg/cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element#	13/22	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	shell-8	
Distance "FROM" to "TO"	1650	mm
Inside Diameter	7100	mm
Element Thickness	112	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	112	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	130	
Detail Type	Platform	
Detail ID	PLAT-6	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	10965440	mm²
Platform Weight	42852	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	19600	mm
Unit Weight of Ladder	0.363	N/mm



Platform Length (top head platform)	0	mm
Element From Node	130	
Detail Type	Packing	
Detail ID	pack#3	
Dist. from "FROM" Node / Offset dist	0.02325	mm
Height of Packed Section	750	mm
Density	0.000166	kg/cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	130	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1650	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	130	
Detail Type	Nozzle	
Detail ID	MH-4	
Dist. from "FROM" Node / Offset dist	599.98	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	17492	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top HDG	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	9217.6	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top distri.0	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	58836	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top dist. T	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm

Element#	14/22	
Element From Node	140	
Element To Node	150	
Element Type	Conical	
Description	cone	
Distance "FROM" to "TO"	2940	mm
Inside Diameter	7100	mm
Element Thickness	122	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	144	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	4600	mm
Design Length of Cone	2940	mm
Half Apex Angle of Cone	23.033741	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	140	
Detail Type	Platform	
Detail ID	PLAT-7	
Dist. from "FROM" Node / Offset dist	2350	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	2999999	mm ²
Platform Weight	35817	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	140	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2940	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	140	
Detail Type	Nozzle	
Detail ID	E	
Dist. from "FROM" Node / Offset dist	973.78	mm
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	31222	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	+IT

Element From Node	140	
Detail Type	Weight	
Detail ID	EPPOVH	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	557187	N
Offset from Element Centerline	5343	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	FPE	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	30889	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	FPEBolting ba	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	2451.5	N



Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	FPEinsert plt	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	16866	N
Offset from Element Centerline	0	mm

Element#	15/22	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description	shell-9	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	72	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	72	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Packing	
Detail ID	pack#2	
Dist. from "FROM" Node / Offset dist	299.98	mm
Height of Packed Section	2700	mm
Density	0.000153	kg/cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	P	
Dist. from "FROM" Node / Offset dist	149.98	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	293.88	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT

Element From Node	150	
Detail Type	Weight	
Detail ID	B#2btm TSR	
Dist. from "FROM" Node / Offset dist	299.98	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element From Node	150	
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Detail Type	Weight
Detail ID	B#2btm supp de
Dist. from "FROM" Node / Offset dist	299.98 mm
Miscellaneous Weight	17651 N
Offset from Element Centerline	0 mm

Element#	16/22
Element From Node	160
Element To Node	170
Element Type	Cylinder
Description	shell-10
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4600 mm
Element Thickness	72 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	72 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5503 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	160
Detail Type	Platform
Detail ID	PLAT-8
Dist. from "FROM" Node / Offset dist	850 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	2999999 mm ²
Platform Weight	25599 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	160
Detail Type	Packing
Detail ID	pack#2
Dist. from "FROM" Node / Offset dist	0.07674 mm
Height of Packed Section	2999.9 mm
Density	0.000153 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	160
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	17/22
Element From Node	170



Element To Node	180
Element Type	Cylinder
Description	shell-11
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4600 mm
Element Thickness	72 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	72 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5503 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	170
Detail Type	Platform
Detail ID	PLAT-9
Dist. from "FROM" Node / Offset dist	1850 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	2999999 mm ²
Platform Weight	25599 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	170
Detail Type	Packing
Detail ID	pack#2
Dist. from "FROM" Node / Offset dist	0.02325 mm
Height of Packed Section	150.01 mm
Density	0.000153 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	170
Detail Type	Packing
Detail ID	pack#1
Dist. from "FROM" Node / Offset dist	2250 mm
Height of Packed Section	749.99 mm
Density	0.000153 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	170
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	170
Detail Type	Nozzle
Detail ID	MH-3
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.



Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 225.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 15138 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element From Node 170
 Detail Type Weight
 Detail ID B#2top HDG
 Dist. from "FROM" Node / Offset dist 149.98 mm
 Miscellaneous Weight 2941.8 N
 Offset from Element Centerline 0 mm

Element From Node 170
 Detail Type Weight
 Detail ID B#2top distri.0
 Dist. from "FROM" Node / Offset dist 149.98 mm
 Miscellaneous Weight 18141 N
 Offset from Element Centerline 0 mm

Element From Node 170
 Detail Type Weight
 Detail ID B#2top dis. TA
 Dist. from "FROM" Node / Offset dist 149.98 mm
 Miscellaneous Weight 1961.2 N
 Offset from Element Centerline 0 mm

Element From Node 170
 Detail Type Weight
 Detail ID B#1btm TSR
 Dist. from "FROM" Node / Offset dist 2250 mm
 Miscellaneous Weight 1961.2 N
 Offset from Element Centerline 0 mm

Element From Node 170
 Detail Type Weight
 Detail ID B#1btm supp de
 Dist. from "FROM" Node / Offset dist 2250 mm
 Miscellaneous Weight 17651 N
 Offset from Element Centerline 0 mm

Element# 18/22
 Element From Node 180
 Element To Node 190
 Element Type Cylinder
 Description shell-12
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 4600 mm
 Element Thickness 72 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 72 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.5503 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.23
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 180
 Detail Type Platform
 Detail ID PLAT-10
 Dist. from "FROM" Node / Offset dist 2850 mm



Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	2999999	mm ²
Platform Weight	25599	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	180	
Detail Type	Packing	
Detail ID	pack#1	
Dist. from "FROM" Node / Offset dist	0.0186	mm
Height of Packed Section	3000	mm
Density	0.000153	kg/cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	19/22	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description	shell-13	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	72	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	72	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	190	
Detail Type	Packing	
Detail ID	pack#1	
Dist. from "FROM" Node / Offset dist	0.0186	mm
Height of Packed Section	2100	mm
Density	0.000153	kg/cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	190	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm



Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top HDG	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top distri.0	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	22162	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top dis. TA	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	20/22
Element From Node	200
Element To Node	210
Element Type	Cylinder
Description	shell-14
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4600 mm
Element Thickness	72 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	72 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5503 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	200
Detail Type	Platform
Detail ID	PLAT-11
Dist. from "FROM" Node / Offset dist	2850 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	2999999 mm ²
Platform Weight	25599 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	200
Detail Type	Insulation
Detail ID	Ins: 50



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	200	
Detail Type	Nozzle	
Detail ID	MH-2	
Dist. from "FROM" Node / Offset dist	999.98	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	15138	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	200	
Detail Type	Nozzle	
Detail ID	D	
Dist. from "FROM" Node / Offset dist	499.97	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4737.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	200	
Detail Type	Tray	
Detail ID	Tray#1-2-3	
Dist. from "FROM" Node / Offset dist	1900	mm
Tray Spacing	300	mm
Number of Trays	3	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	2647.6	N

Element From Node	200	
Detail Type	Weight	
Detail ID	FPD	
Dist. from "FROM" Node / Offset dist	699.98	mm
Miscellaneous Weight	9806	N
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	FPDBolting ba	
Dist. from "FROM" Node / Offset dist	699.98	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	DPPQVH	
Dist. from "FROM" Node / Offset dist	499.98	mm
Miscellaneous Weight	168143	N
Offset from Element Centerline	3779	mm

Element#	21/22
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	shell-15



Distance "FROM" to "TO"	1150	mm
Inside Diameter	4600	mm
Element Thickness	72	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	72	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.5503	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	210	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1150	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	210	
Detail Type	Nozzle	
Detail ID	MH-1	
Dist. from "FROM" Node / Offset dist	299.98	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	15166	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT
Element From Node	210	
Detail Type	Weight	
Detail ID	Demister dead	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	4020.5	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	Demister TA	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	13728	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	Tray Truss	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	11865	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	davit	
Dist. from "FROM" Node / Offset dist	499.98	mm
Miscellaneous Weight	8825.4	N
Offset from Element Centerline	2300	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	BPPQVH	
Dist. from "FROM" Node / Offset dist	574.98	mm
Miscellaneous Weight	172095	N



Offset from Element Centerline 3841 mm

Element# 22/22
 Element From Node 220
 Element To Node 230
 Element Type Spherical
 Description TOP dish
 Distance "FROM" to "TO" 0 mm
 Inside Diameter 4632 mm
 Element Thickness 40 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 46 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.5503 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.23
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 220
 Detail Type Platform
 Detail ID TOP PLT
 Dist. from "FROM" Node / Offset dist 2500 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 18566400 mm²
 Platform Weight 116181 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 8000 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 4000 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 270.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 26000 mm
 Unit Weight of Ladder 0.363 N/mm
 Platform Length (top head platform) 8000 mm

Element From Node 220
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 2300 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 220
 Detail Type Nozzle
 Detail ID B
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 7295.7 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element From Node 220
 Detail Type Weight
 Detail ID top dish SF wt
 Dist. from "FROM" Node / Offset dist 0 mm



Miscellaneous Weight	10296	N
Offset from Element Centerline	0	mm
Element From Node	220	
Detail Type	Weight	
Detail ID	PSV-1	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	4000	mm
Element From Node	220	
Detail Type	Weight	
Detail ID	PSV-2	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	4000	mm
Element From Node	220	
Detail Type	For./Mom.	
Detail ID	BFxFz	
Dist. from "FROM" Node / Offset dist	2300	mm
Force in X Direction	42883	N
Force in Y Direction	-59369	N
Force in Z Direction	42883	N
Moment about X Axis	79204152	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	79204152	N-mm
Force/Moment Combination Method	SRSS	

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Btm skirt-1	...	...	2000	...	2000
Btm skirt-2	...	...	5000	...	3000
Btm skirt-3	...	...	8000	...	3000
Top skirt	...	...	9000	...	1000
BTM dish	...	...	9000	...	...
shell-1	...	...	12000	...	3000
shell-2	...	...	15000	...	3000
shell-3	...	...	18000	...	3000
shell-4	...	...	21000	...	3000
shell-5	...	...	24000	...	3000
shell-6	...	...	27000	...	3000
shell-7	...	...	29000	...	2000
shell-8	...	...	30650	...	1650
cone	...	...	33590	...	2940
shell-9	...	...	36590	...	3000
shell-10	...	...	39590	...	3000
shell-11	...	...	42590	...	3000
shell-12	...	...	45590	...	3000
shell-13	...	...	48590	...	3000
shell-14	...	...	51590	...	3000
shell-15	...	...	52740	...	1150
TOP dish	...	...	52740	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Btm skirt-1	...		36	1.6	7194	...	1
Btm skirt-2	...		34	1.6	7198	...	2
Btm skirt-3	...		34	1.6	7198	...	3
Top skirt	...		34	1.6	7198	...	4
BTM dish	4.6869		70	6	7148	156.4	5
shell-1	4.65		114	6	7100	156.4	6
shell-2	4.6189		114	6	7100	156.4	7
shell-3	4.5877		114	6	7100	156.4	8
shell-4	4.5633		112	6	7100	156.4	9
shell-5	4.5633		112	6	7100	156.4	10
shell-6	4.5633		112	6	7100	156.4	11
shell-7	4.5633		112	6	7100	156.4	12
shell-8	4.5633		112	6	7100	156.4	13
cone	4.5633		144	6	7100	156.4	14
shell-9	4.5633		72	3	4600	156.4	15
shell-10	4.5633		72	3	4600	156.4	16
shell-11	4.5633		72	3	4600	156.4	17
shell-12	4.5633		72	3	4600	156.4	18
shell-13	4.5633		72	3	4600	156.4	19
shell-14	4.5633		72	3	4600	156.4	20
shell-15	4.5633		72	3	4600	156.4	21
TOP dish	4.5633		46	3	4632	156.4	22

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Btm skirt-1	...		No Calc	No Calc	36	No Calc	1
Btm skirt-2	...		No Calc	No Calc	34	No Calc	2
Btm skirt-3	...		No Calc	No Calc	34	No Calc	3
Top skirt	...		No Calc	No Calc	34	No Calc	4
BTM dish	4.55026	5.06235	6.4044	66	60.0456		5
shell-1	4.55026	4.57959	5.53138	114	113.313		6
shell-2	4.55026	4.61076	5.53138	114	112.583		7
shell-3	4.55026	4.64193	5.53138	114	111.853		8
shell-4	4.55026	4.58095	5.43583	112	111.281		9
shell-5	4.55026	4.58095	5.43583	112	111.281		10
shell-6	4.55026	4.58095	5.43583	112	111.281		11
shell-7	4.55026	4.58095	5.43583	112	111.281		12
shell-8	4.55026	4.58095	5.43583	112	111.281		13
cone	4.55026	4.61307	5.44889	122	120.402		14
shell-9	4.55026	4.60407	5.39426	72	71.1842		15
shell-10	4.55026	4.60407	5.39426	72	71.1842		16
shell-11	4.55026	4.60407	5.39426	72	71.1842		17
shell-12	4.55026	4.60407	5.39426	72	71.1842		18
shell-13	4.55026	4.60407	5.39426	72	71.1842		19
shell-14	4.55026	4.60407	5.39426	72	71.1842		20
shell-15	4.55026	4.60407	5.39426	72	71.1842		21
TOP dish	4.55026	4.93839	5.9933	40	37.0786		22

Minimum

5

5

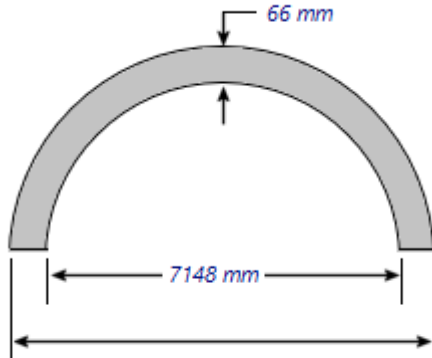
MAWP: 5 MPa, limited by: shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: BTM dish

Spherical Head Design Information:					
Design Pressure	P	4.687 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7148.000 mm	Actual Stress		138.7 MPa
Spec. Min. Thk t,e, or sa		66.0000 mm	Spec. Nominal Thk.		70.000 mm
Surface Area		83432760.0 mm²	SG of Contents		1.060
Element Volume		95631.3 ltr	Weight of Contents		993348.7 N
Empty Weight		435375.9 N	Operating Weight		1428724.5 N
			Comp. All. Ope. Stress		104.4 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 89.350 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 44.675 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + ci + co$
 $= 7160.0/2(\exp(0.5 \cdot 4.687/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 54.0456 + 6.0 + 0.0 = 60.0456 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 60.0 + 7160.0) / 7160.0) - 0.137$
 $= 5.199 - 0.137 = 5.0624 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 66.0 + 7148.0) / 7148.0)$
 $= 6.4044 \text{ MPa}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4.687 \cdot 7160.0^2 / (1.0 (7280.0^2 - 7160.0^2)) \\ &= 138.665 \text{ MPa} \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4.687 / (1.0 \cdot \ln((2 \cdot 60.0 + 7160.0) / 7160.0)) \\ &= 140.995 \text{ MPa} \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.455 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.597 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 83432760 / 3.14159) \\ &= 10306.788 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

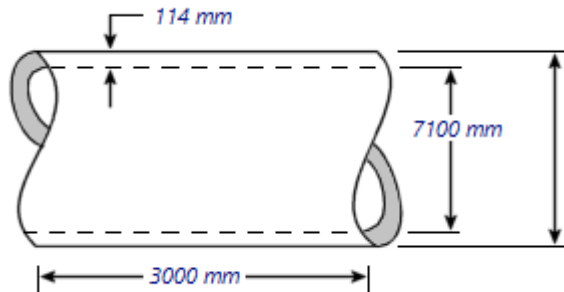
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 89.350 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 44.675 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4.650 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e , or s _a		114.0000 mm	Spec. Nominal Thk.		114.000 mm
Surface Area		69064776.0 mm ²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		1234060.8 N
Empty Weight		589049.0 N	Operating Weight		1823109.8 N
			Comp. All. Ope. Stress		115.3 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.65/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 107.3132 + 6.0 + 0.0 = 113.3132 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0) - 0.1$$

$$= 4.679 - 0.1 = 4.5796 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 114.0 + 7100.0) / 7100.0)$$

$$= 5.5314 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.65 \cdot 7112.0 / (1.0 (7328.0 - 7112.0))$$

$$= 153.1065 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.65 / (1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0))$
= 155.4199 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.580 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

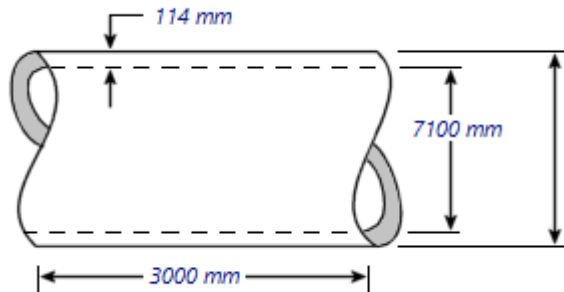
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	4.619 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		152.1 MPa
Spec. Min. Thk t _e or s _a		114.0000 mm	Spec. Nominal Thk.		114.000 mm
Surface Area		69064776.0 mm ²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		1234060.8 N
Empty Weight		589049.0 N	Operating Weight		1823109.8 N
			Comp. All. Ope. Stress		115.3 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.619/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.5828 + 6.0 + 0.0 = 112.5828 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0) - 0.069$$

$$= 4.679 - 0.069 = 4.6108 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 114.0 + 7100.0) / 7100.0)$$

$$= 5.5314 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.619 \cdot 7112.0 / (1.0 (7328.0 - 7112.0))$$

$$= 152.0801 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.619 / (1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0))$
= 154.3781 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.580 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

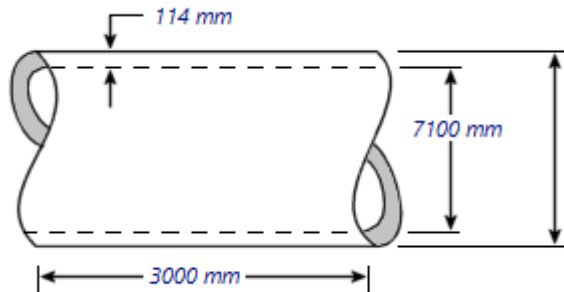
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	4.588 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		151.1 MPa
Spec. Min. Thk t _e , or s _a		114.0000 mm	Spec. Nominal Thk.		114.000 mm
Surface Area		69064776.0 mm ²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		966681.0 N
Empty Weight		589049.0 N	Operating Weight		1555730.0 N
			Comp. All. Ope. Stress		115.3 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.588/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.8533 + 6.0 + 0.0 = 111.8533 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0) - 0.037$$

$$= 4.679 - 0.037 = 4.6419 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 114.0 + 7100.0) / 7100.0)$$

$$= 5.5314 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.588 \cdot 7112.0 / (1.0 (7328.0 - 7112.0))$$

$$= 151.0537 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.588 / (1.0 \cdot \ln((2 \cdot 108.0 + 7112.0) / 7112.0))$
= 153.3362 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.580 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

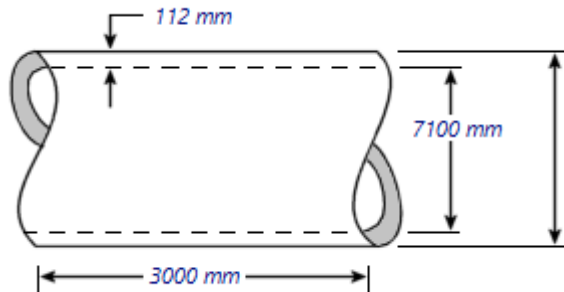
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e or s _a		112.0000 mm	Spec. Nominal Thk.		112.000 mm
Surface Area		69027080.0 mm ²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		578555.4 N	Operating Weight		578555.4 N
			Comp. All. Ope. Stress		115.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.2814 + 6.0 + 0.0 = 111.2814 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0) - 0.013$$

$$= 4.594 - 0.013 = 4.581 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 112.0 + 7100.0) / 7100.0)$$

$$= 5.4358 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 7112.0 / (1.0 (7324.0 - 7112.0))$$

$$= 153.0844 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0))$
= 155.3552 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.553 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

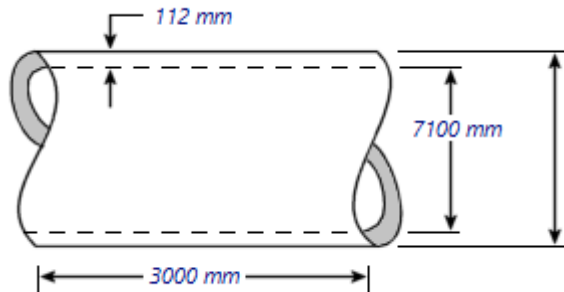
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e , or s _a		112.0000 mm	Spec. Nominal Thk.		112.000 mm
Surface Area		69027080.0 mm ²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		578555.4 N	Operating Weight		578555.4 N
			Comp. All. Ope. Stress		115.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
100 %		100 %	

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.2814 + 6.0 + 0.0 = 111.2814 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0) - 0.013$$

$$= 4.594 - 0.013 = 4.581 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 112.0 + 7100.0) / 7100.0)$$

$$= 5.4358 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 7112.0 / (1.0 (7324.0 - 7112.0))$$

$$= 153.0844 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0))$
= 155.3552 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.553 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

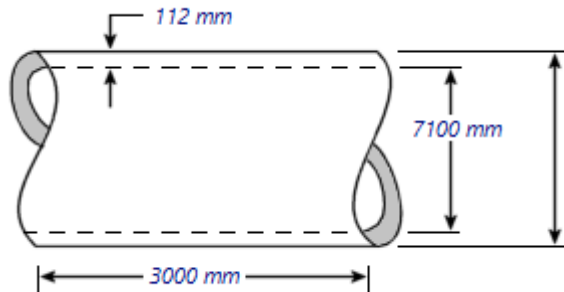
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e , or s _a		112.0000 mm	Spec. Nominal Thk.		112.000 mm
Surface Area		69027080.0 mm ²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		578555.4 N	Operating Weight		578555.4 N
			Comp. All. Ope. Stress		115.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.2814 + 6.0 + 0.0 = 111.2814 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0) - 0.013$$

$$= 4.594 - 0.013 = 4.581 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 112.0 + 7100.0) / 7100.0)$$

$$= 5.4358 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 7112.0 / (1.0 (7324.0 - 7112.0))$$

$$= 153.0844 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0))$
= 155.3552 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.553 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

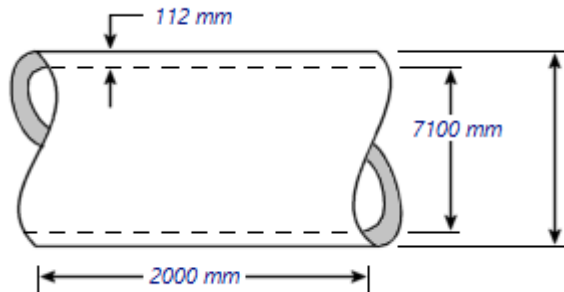
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e or s _a		112.0000 mm	Spec. Nominal Thk.		112.000 mm
Surface Area		46018052.0 mm ²	SG of Contents		0.000
Element Volume		79198.0 ltr	Weight of Contents		0.0 N
Empty Weight		385704.1 N	Operating Weight		385704.1 N
			Comp. All. Ope. Stress		115.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.2814 + 6.0 + 0.0 = 111.2814 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0) - 0.013$$

$$= 4.594 - 0.013 = 4.581 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 112.0 + 7100.0) / 7100.0)$$

$$= 5.4358 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 7112.0 / (1.0 (7324.0 - 7112.0))$$

$$= 153.0844 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0))$
= 155.3552 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.553 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

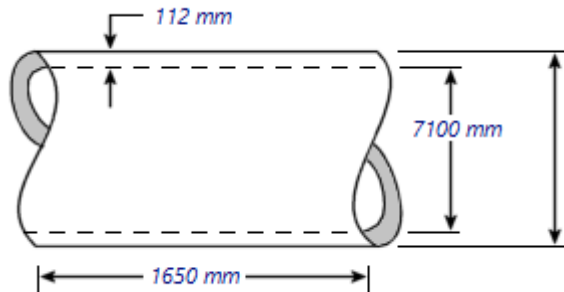
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1650.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		153.1 MPa
Spec. Min. Thk t _e , or s _a		112.0000 mm	Spec. Nominal Thk.		112.000 mm
Surface Area		37964896.0 mm ²	SG of Contents		0.000
Element Volume		65338.4 ltr	Weight of Contents		0.0 N
Empty Weight		318205.7 N	Operating Weight		318205.7 N
			Comp. All. Ope. Stress		115.0 MPa

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 105.2814 + 6.0 + 0.0 = 111.2814 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0) - 0.013$$

$$= 4.594 - 0.013 = 4.581 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 112.0 + 7100.0) / 7100.0)$$

$$= 5.4358 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 7112.0 / (1.0 (7324.0 - 7112.0))$$

$$= 153.0844 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 106.0 + 7112.0) / 7112.0))$
= 155.3552 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.553 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

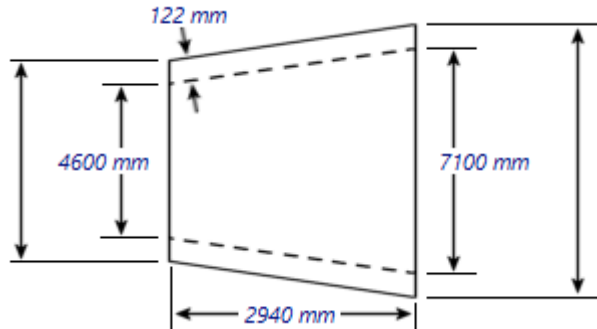
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: cone

Conical Head/Section Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2940.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		152.0 MPa
Spec. Min. Thk t _e , or s _a		122.0000 mm	Spec. Nominal Thk.		144.000 mm
Surface Area		61854092.0 mm ²	SG of Contents		0.000
Element Volume		80239.2 ltr	Weight of Contents		0.0 N
Empty Weight		659721.6 N	Operating Weight		659721.6 N
			Comp. All. Ope. Stress		117.4 MPa
			To End Diameter		4600.000 mm

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 / \cos(23.034) \cdot (\exp(4.563 / (156.4 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 114.4022 + 6.0 + 0.0 = 120.4022 \text{ mm}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 156.4 \cdot 1.0 \cdot \ln(2 \cdot 116.0 \cdot \cos(23.034) + 7112.0) / 7112.0$$

$$= 4.6261 \text{ MPa}$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 175.0 \cdot 1.0 \cdot \ln(2 \cdot 122.0 \cdot \cos(23.034) + 7100.0) / 7100.0$$

$$= 5.4489 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D) \cos(\alpha))$$

$$= 4.563 \cdot 7112.0 / (1.0(7344.0 - 7112.0) \cos(23.034))$$

$$= 152.007 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}))$$

$$= 4.563 / (1.0 \cdot \ln((2 \cdot 116.0 \cdot \cos(23.034) + 7112.0) / 7112.0))$$

$$= 154.277 \text{ MPa}$$



Percent Fiber Elongation per Table 6.1

Cones formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 3.027 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

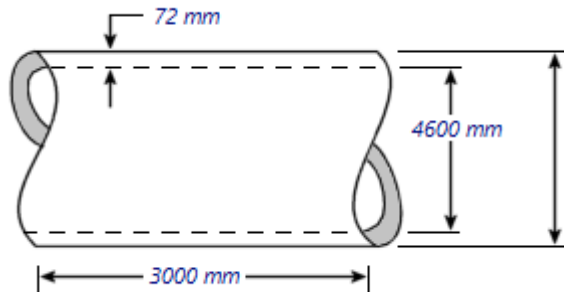
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t,e, or sa		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.563/(1.0 • Ln((2•69.0 + 4606.0)/4606.0))
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

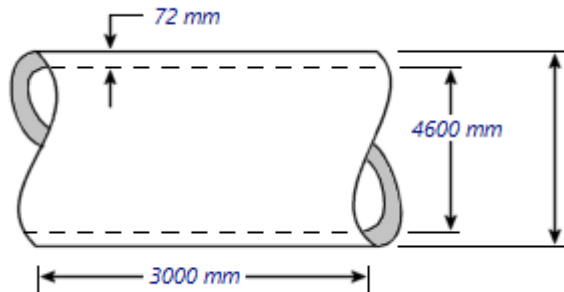
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-10

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm ²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.563/(1.0 • Ln((2•69.0 + 4606.0)/4606.0))
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

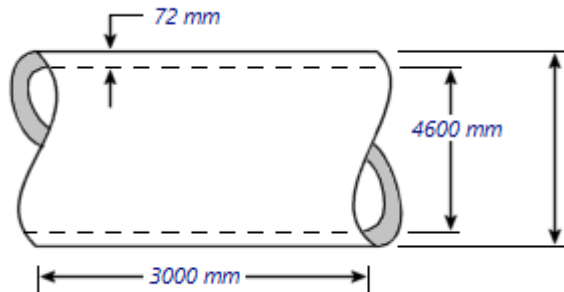
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-11

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm ²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0))$
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

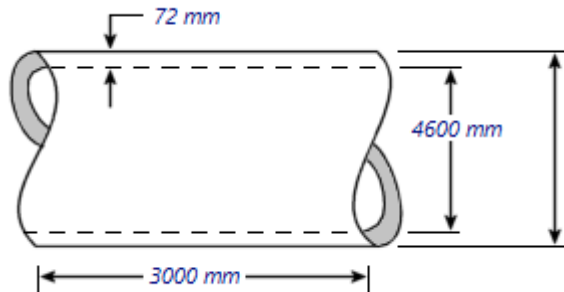
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-12

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk	t _e , or sa	72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm ²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.563/(1.0 • Ln((2•69.0 + 4606.0)/4606.0))
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

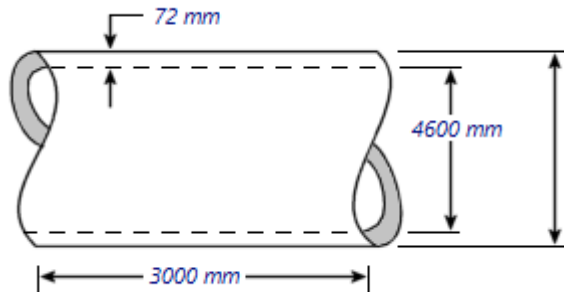
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-13

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm ²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0))$
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

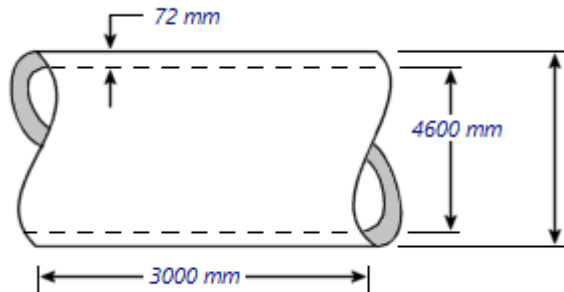
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-14

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		44711148.0 mm ²	SG of Contents		0.000
Element Volume		49866.0 ltr	Weight of Contents		0.0 N
Empty Weight		240938.2 N	Operating Weight		240938.2 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.563 / (1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0))$$
$$= 154.5773 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

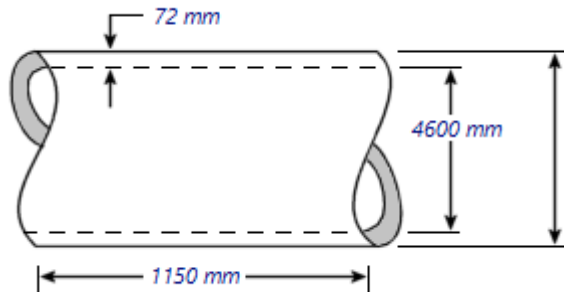
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-15

Cylindrical Shell Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1150.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4600.000 mm	Actual Stress		152.3 MPa
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.		72.000 mm
Surface Area		17139274.0 mm ²	SG of Contents		0.000
Element Volume		19115.3 ltr	Weight of Contents		0.0 N
Empty Weight		92359.7 N	Operating Weight		92359.7 N
			Comp. All. Ope. Stress		115.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 0.013$$

$$= 4.617 - 0.013 = 4.6041 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5.3943 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.563 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= $4.563 / (1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0))$
= 154.5773 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

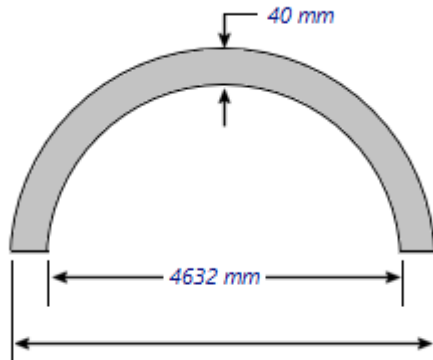
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: TOP dish

Spherical Head Design Information:					
Design Pressure	P	4.563 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	4632.000 mm	Actual Stress		141.9 MPa
Spec. Min. Thk t _e , or s _a		40.0000 mm	Spec. Nominal Thk.		46.000 mm
Surface Area		35054164.0 mm²	SG of Contents		0.000
Element Volume		26022.7 ltr	Weight of Contents		0.0 N
Empty Weight		120173.9 N	Operating Weight		120173.9 N
			Comp. All. Ope. Stress		103.3 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 57.900 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.950 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + ci + co$
 $= 4638.0/2(\exp(0.5 \cdot 4.563/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 34.0786 + 3.0 + 0.0 = 37.0786 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 37.0 + 4638.0) / 4638.0) - 0.013$
 $= 4.951 - 0.013 = 4.9384 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 40.0 + 4632.0) / 4632.0)$
 $= 5.9933 \text{ MPa}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4.563 \cdot 4638.0^2 / (1.0 (4712.0^2 - 4638.0^2)) \\ &= 141.871 \text{ MPa} \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4.563 / (1.0 \cdot \ln((2 \cdot 37.0 + 4638.0) / 4638.0)) \\ &= 144.140 \text{ MPa} \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.475 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.624 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 35054164 / 3.14159) \\ &= 6680.745 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 57.900 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.950 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 4.5796(1.117)$
 $= 6.391 \text{ MPa}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
BTM dish	176.6	247.0	0.715	6.46
shell-1	204.4	247.0	0.828	6.46
shell-2	204.4	247.0	0.828	6.46
shell-3	204.4	247.0	0.828	6.46
shell-4	208.0	247.0	0.842	6.46
shell-5	208.0	247.0	0.842	6.46
shell-6	208.0	247.0	0.842	6.46
shell-7	208.0	247.0	0.842	6.46
shell-8	208.0	247.0	0.842	6.46
cone	207.5	247.0	0.840	6.46
shell-9	208.8	247.0	0.845	6.44
shell-10	208.8	247.0	0.845	6.44
shell-11	208.8	247.0	0.845	6.44
shell-12	208.8	247.0	0.845	6.44
shell-13	208.8	247.0	0.845	6.44
shell-14	208.8	247.0	0.845	6.44
shell-15	208.8	247.0	0.845	6.44
TOP dish	188.0	247.0	0.761	6.44

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
C	Nozzle	165.00	147.60	1.118
A	Nozzle	165.00	147.60	1.118
MH-6	Nozzle	165.00	147.60	1.118
MH-5	Nozzle	165.00	147.60	1.118
MH-4	Nozzle	165.00	147.60	1.118
E	Nozzle	138.00	123.60	1.117
P	Nozzle	165.00	147.60	1.118
MH-3	Nozzle	165.00	147.60	1.118
MH-2	Nozzle	165.00	147.60	1.118
D	Nozzle	165.00	147.60	1.118
MH-1	Nozzle	165.00	147.60	1.118
B	Nozzle	165.00	147.60	1.118

Minimum 1.117

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
BTM dish	175.00	156.40	1.119
shell-1	175.00	156.40	1.119
shell-2	175.00	156.40	1.119
shell-3	175.00	156.40	1.119
shell-4	175.00	156.40	1.119
shell-5	175.00	156.40	1.119
shell-6	175.00	156.40	1.119
shell-7	175.00	156.40	1.119
shell-8	175.00	156.40	1.119

cone	175.00	156.40	1.119
shell-9	175.00	156.40	1.119
shell-10	175.00	156.40	1.119
shell-11	175.00	156.40	1.119
shell-12	175.00	156.40	1.119
shell-13	175.00	156.40	1.119
shell-14	175.00	156.40	1.119
shell-15	175.00	156.40	1.119
TOP dish	175.00	156.40	1.119

Minimum 1.119

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
C	33.27	237.50	0.140
A	24.04	237.50	0.101
MH-6	25.47	237.50	0.107
MH-5	25.47	237.50	0.107
MH-4	25.47	237.50	0.107
E	118.08	194.70	0.606
P	19.39	237.50	0.082
MH-3	31.13	237.50	0.131
MH-2	31.13	237.50	0.131
D	76.74	237.50	0.323
MH-1	31.13	237.50	0.131
B	52.20	237.50	0.220

Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 2, 2025****External Pressure Calculations:**

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	9000	...	...	No Calc	No Calc
20	30	9000	...	...	No Calc	No Calc
30	40	9000	...	...	No Calc	No Calc
40	50	9000	...	...	No Calc	No Calc
50	60	No Calc	7280	60	No Calc	No Calc
60	70	22841.3	7328	108	No Calc	No Calc
70	80	22841.3	7328	108	No Calc	No Calc
80	90	22841.3	7328	108	No Calc	No Calc
90	100	22841.3	7324	106	No Calc	No Calc
100	110	22841.3	7324	106	No Calc	No Calc
110	120	22841.3	7324	106	No Calc	No Calc
120	130	22841.3	7324	106	No Calc	No Calc
130	140	22841.3	7324	106	No Calc	No Calc
140	150	2940	7365.14	116	No Calc	No Calc
150	160	19922	4744	69	No Calc	No Calc
160	170	19922	4744	69	No Calc	No Calc
170	180	19922	4744	69	No Calc	No Calc
180	190	19922	4744	69	No Calc	No Calc
190	200	19922	4744	69	No Calc	No Calc
200	210	19922	4744	69	No Calc	No Calc
210	220	19922	4744	69	No Calc	No Calc
220	230	No Calc	4712	37	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	66	No Calc	...	2.77522
60	70	114	No Calc	...	1.67939
70	80	114	No Calc	...	1.67939
80	90	114	No Calc	...	1.67939
90	100	112	No Calc	...	1.61343
100	110	112	No Calc	...	1.61343
110	120	112	No Calc	...	1.61343
120	130	112	No Calc	...	1.61343
130	140	112	No Calc	...	1.61343
140	150	122	No Calc	...	3.56659
150	160	72	No Calc	...	1.24362
160	170	72	No Calc	...	1.24362
170	180	72	No Calc	...	1.24362
180	190	72	No Calc	...	1.24362
190	200	72	No Calc	...	1.24362
200	210	72	No Calc	...	1.24362
210	220	72	No Calc	...	1.24362
220	230	40	No Calc	...	2.59549

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners	Allowable Length Bet. Stiffeners	Ring Inertia Required	Ring Inertia Available
------	----	----------------------------------	-------------------------------------	--------------------------	---------------------------



		mm	mm	mm^4	mm^4
10	20	9000	No Calc	No Calc	No Calc
20	30	9000	No Calc	No Calc	No Calc
30	40	9000	No Calc	No Calc	No Calc
40	50	9000	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	22841.3	No Calc	No Calc	No Calc
70	80	22841.3	No Calc	No Calc	No Calc
80	90	22841.3	No Calc	No Calc	No Calc
90	100	22841.3	No Calc	No Calc	No Calc
100	110	22841.3	No Calc	No Calc	No Calc
110	120	22841.3	No Calc	No Calc	No Calc
120	130	22841.3	No Calc	No Calc	No Calc
130	140	22841.3	No Calc	No Calc	No Calc
140	150	2940	2940	No Calc	No Calc
150	160	19922	No Calc	No Calc	No Calc
160	170	19922	No Calc	No Calc	No Calc
170	180	19922	No Calc	No Calc	No Calc
180	190	19922	No Calc	No Calc	No Calc
190	200	19922	No Calc	No Calc	No Calc
200	210	19922	No Calc	No Calc	No Calc
210	220	19922	No Calc	No Calc	No Calc
220	230	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	19865	...	19304.4	...	...
20	30	17961.2	...	17119.9	...	...
30	40	17961.2	...	17119.9	...	...
40	50	5986.78	...	5706.33	...	...
50	60	44398.9	95631.3	40660.6	96113.8	...
60	70	60070.3	118797	56956	119199	...
70	80	60070.3	118797	56956	119199	...
80	90	60070.3	118797	56956	119199	...
90	100	59000.2	118797	55885.9	119199	...
100	110	59000.2	118797	55885.9	119199	...
110	120	59000.2	118797	55885.9	119199	...
120	130	39333.5	79198	37257.3	79466	...
130	140	32450.1	65338.4	30737.2	65559.4	...
140	150	67277.4	80239.2	64544.1	80591.9	...
150	160	24570.5	49866	23561.9	49996.2	...
160	170	24570.5	49866	23561.9	49996.2	...
170	180	24570.5	49866	23561.9	49996.2	...
180	190	24570.5	49866	23561.9	49996.2	...
190	200	24570.5	49866	23561.9	49996.2	...
200	210	24570.5	49866	23561.9	49996.2	...
210	220	9418.7	19115.3	9032.06	19165.2	...
220	230	12255.1	26022.7	11470.5	26123.9	...
Total		771542	1377523.38	732849	1382191.00	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	1805.02	2848.43	1999	2848.43	PLAT- SKIRT
20	Plat	1804.22	-2849.69	3000	2849.69	PLAT- SKIRT
30	Wght	800	1000	...	...	skirt manwy/ven
50	Pack	15889.5	...	-1775	...	Pack 1: Btm dis
50	Liqd	101300	...	-1775	...	Liquid: 40
50	Insl	2496.68	...	-1775	...	Ins: 50
50	Nozl	16823.8	...	-3791	...	C
60	Plat	7551.32	0.39645E-04	...	...	PLAT-1
60	Pack	6579.95	...	500.005	...	pack#5
60	Liqd	125848	...	1500	...	Liquid: 50
60	Insl	1968.4	...	1500	...	Ins: 50
60	Wght	260	...	0.0052334	...	VB
60	Wght	2770	...	0.0052334	...	btm dish SF wt
60	Wght	5249	-5343	0.0052334	...	C_PP_OVH
70	Plat	3646.11	-2850.32	1000	...	PLAT-2
70	Liqd	125848	...	1500	...	Liquid: 60
70	Insl	1968.4	...	1500	...	Ins: 50
80	Plat	3646.11	2850.32	2000	...	PLAT-3
80	Liqd	98580.6	...	1175	...	Liquid: 70
80	Insl	1968.4	...	1500	...	Ins: 50
80	Nozl	1204.56	-3750	2100	...	A
80	Wght	3300	...	2099.98	...	FP_A
80	Wght	250	...	2099.98	...	FP_A_Bolting ba
80	Wght	21250	-5267	2099.98	...	A_PP_OVH
90	Pack	8882.94	...	2324.97	...	pack#4
90	Insl	1967.34	...	1500	...	Ins: 50
90	Nozl	1787.64	1875	800.006	3247.59	MH-6
90	Wght	300	...	1649.97	...	B#4_btm TSR
90	Wght	4400	...	1649.97	...	B#4_btm supp de
100	Plat	3644.5	-2850.32	0.0011613	...	PLAT-4
100	Pack	17436.9	...	1325.02	...	pack#4
100	Insl	1967.34	...	1500	...	Ins: 50



100	Wght	940	...	2649.98	...	B#4_top HDG
100	Wght	5010	...	2649.98	...	B#4_top dist. d
100	Wght	300	...	2649.98	...	B#4_top dist. T
110	Plat	3644.5	2850.32	1000	...	PLAT-5
110	Pack	8224.94	...	2374.97	...	pack#3
110	Insl	1967.34	...	1500	...	Ins: 50
110	Nozl	1783.79	1875	1399.98	3247.59	MH-5
110	Wght	300	...	1749.97	...	B#3_btm TSR
110	Wght	4400	...	1749.97	...	B#3_btm supp de
110	Wght	3250	...	1699.98	...	Nozzle wt
110	Wght	10000	...	1699.98	...	10T margin
110	Wght	14300	...	1699.98	...	Cleat wt
110	Wght	27000	...	1699.98	...	Lifting wts
120	Pack	13159.4	...	1000.04	...	pack#3
120	Insl	1311.56	...	1000	...	Ins: 50
130	Plat	4370	-2850.32	...	...	PLAT-6
130	Pack	4934.98	...	375.025	...	pack#3
130	Insl	1082.04	...	825	...	Ins: 50
130	Nozl	1783.79	1875	599.977	3247.59	MH-4
130	Wght	940	...	749.975	...	B#3_top HDG
130	Wght	6000	...	749.975	...	B#3_top distri.
130	Wght	300	...	749.975	...	B#3_top dist. T
140	Plat	3652.52	2850.32	2350	...	PLAT-7
140	Insl	2062.16	...	1470	...	Ins: 50
140	Nozl	3184.02	-3335.98	973.78	...	E
140	Wght	56821	-5343	973.778	...	E_PP_OVH
140	Wght	3150	...	973.778	...	FP_E
140	Wght	250	...	973.778	...	FP_E_Bolting ba
140	Wght	1720	...	973.778	...	FP_E_insert plt
150	Pack	6873.36	...	1649.98	...	pack#2
150	Insl	1281.8	...	1500	...	Ins: 50
150	Nozl	29.9693	-2425.01	149.979	...	P
150	Wght	200	...	299.977	...	B#2_btm TSR
150	Wght	1800	...	299.977	...	B#2_btm supp de
160	Plat	2610.54	-2065.11	850.002	...	PLAT-8
160	Pack	7636.87	...	1500.04	...	pack#2
160	Insl	1281.8	...	1500	...	Ins: 50
170	Plat	2610.54	2065.11	1850	...	PLAT-9
170	Pack	381.871	...	75.0268	...	pack#2
170	Pack	1909.24	...	2624.97	...	pack#1
170	Insl	1281.8	...	1500	...	Ins: 50
170	Nozl	1543.71	-1767.77	1499.98	1767.77	MH-3
170	Wght	300	...	149.977	...	B#2_top HDG
170	Wght	1850	...	149.977	...	B#2_top distri.
170	Wght	200	...	149.977	...	B#2_top dis. TA
170	Wght	200	...	2249.98	...	B#1_btm TSR
170	Wght	1800	...	2249.98	...	B#1_btm supp de
180	Plat	2610.54	-2065.11	2850	...	PLAT-10
180	Pack	7637.04	...	1500.01	...	pack#1
180	Insl	1281.8	...	1500	...	Ins: 50
190	Pack	5345.97	...	1050.02	...	pack#1
190	Insl	1281.8	...	1500	...	Ins: 50
190	Wght	300	...	2099.98	...	B#1_top HDG
190	Wght	2260	...	2099.98	...	B#1_top distri.
190	Wght	200	...	2099.98	...	B#1_top dis. TA
200	Plat	2610.54	2065.11	2850	...	PLAT-11
200	Insl	1281.8	...	1500	...	Ins: 50
200	Nozl	1543.71	1250	999.976	2165.06	MH-2
200	Nozl	483.077	-2500	499.974	...	D
200	Tray	8292.41	...	2199.98	...	Tray#1-2-3
200	Wght	1000	...	699.976	...	FP_D
200	Wght	150	...	699.976	...	FP_D_Bolting ba
200	Wght	17147	-3779	499.979	...	D_PP_OVH
210	Insl	491.355	...	575	...	Ins: 50
210	Nozl	1546.59	1250	299.977	2165.06	MH-1
210	Wght	410	...	849.981	...	Demister dead
210	Wght	1400	...	849.981	...	Demister TA
210	Wght	1210	...	849.981	...	Tray Truss
210	Wght	900	2300	499.979	...	davit
210	Wght	17550	-3841	574.979	...	B_PP_OVH



220	Plat	11848	...	2500.01	...
220	Insl	1143.16	...	1150	...
220	Nozl	744	...	2496	...
220	Wght	1050	...	...	...
220	Wght	3000	-4000	2300	...
220	Wght	3000	-3758.77	2300	1368.08
220	Forc	...	...	2300	...
90	Pliq	14157.90	...	2324.975	...
100	Pliq	27791.44	...	1325.023	...
110	Pliq	13109.17	...	2374.974	...
120	Pliq	20973.88	...	1000.037	...
130	Pliq	7865.53	...	375.025	...
150	Pliq	11885.79	...	1649.977	...
160	Pliq	13206.09	...	1500.037	...
170	Pliq	660.35	...	75.027	...
170	Pliq	3301.57	...	2624.974	...
180	Pliq	13206.39	...	1500.014	...
190	Pliq	9244.55	...	1050.024	...

TOP PLT

Ins: 50

B

top dish SF wt

PSV-1

PSV-2

B_FxFz

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
200	Tliq	0.00	0.000	2199.977

Total Weight of Each Detail Type:

Platforms	56054.5
Packing	104892.9
Packing Liquid	135402.7
Liquid	451575.8
Insulation	28084.9
Nozzles	32458.7
Trays	7482.4
Tray Bolting Weight	810.0
Weights	229187.0

Sum of the Detail Weights 1045948.9 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	771542.3	771542.3	771542.3	771542.3	771542.3	771542.3
Nozzles	32458.7	32458.7	32458.7	32458.7	32458.7	32458.7
Tray Bar	810.0	810.0	810.0	810.0	810.0	810.0
Wld Weights	73550.0	73550.0	73550.0	73550.0	73550.0	73550.0
Platforms	...	...	56054.5	56054.5	56054.5	56054.5
Packing	...	...	104892.9	104892.9	104892.9	104892.9
Insulation	...	...	28084.9	28084.9	28084.9	28084.9
Trays	...	...	7482.4	7482.4	7482.4	7482.4
Empty Weights	...	...	...	155637.0	155637.0	...
Test Weights	...	155637.0	...	...	...	...
Ope Weights	...	...	...	...	...	155637.0
Ope. Liquid	...	...	...	...	...	451575.8
Packing Liq	...	...	...	...	...	135402.7
Test Liquid	...	1376682.5	...	...	...	...
Totals	878360.9	2410680.5	1074875.8	1230512.5	1230512.6	1817491.2

Field Installation Options:**Weight Summary:**

Fabricated Wt.	- Bare weight without removable internals	878360.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	2410680.5 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1074875.8 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1230512.5 kgm



Ope. Wt. no Liq - Fab. weight + internals. + details + weights 1230512.6 kgm
 Operating Wt. - Empty weight + operating liq. uncorroded 1817491.2 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating) 321033.4 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 588198.4 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (empty) 281814.6 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	45653628	45201232
20	30	68480432	67839552
30	40	68480432	67839552
40	50	22826814	22613184
50	60	83432760	80258120
60	70	69064776	66915920
70	80	69064776	66915920
80	90	69064776	66915920
90	100	69027080	66915920
100	110	69027080	66915920
110	120	69027080	66915920
120	130	46018052	44610616
130	140	37964896	36803760
140	150	61854092	58713192
150	160	44711148	43353980
160	170	44711148	43353980
170	180	44711148	43353980
180	190	44711148	43353980
190	200	44711148	43353980
200	210	44711148	43353980
210	220	17139274	16619026
220	230	35054164	33702104

Total 1169446784.0 1135819520.0 mm²
 Total 1169.4 1135.8 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	21670	21670	21670	50437696	21670
20	30	19765.4	19765.4	19765.4	50437632	19765.4
30	40	18761.2	18761.2	18761.2	7847978	18761.2
40	50	5986.78	5986.78	5986.78	...	5986.78
50	60	79608.9	180909	159292	...	79608.9
60	70	84449	210297	196594	275124672	84449
70	80	65684.8	191532	184409	101950920	65684.8
80	90	91689.4	190270	210414	1040330624	91689.4
90	100	76338.1	90496	186180	32881306	76338.1
100	110	88298.9	116090	189587	101906104	88298.9
110	120	133871	146980	244370	134716608	133871
120	130	53804.5	74778.3	119795	...	53804.5
130	140	51860.9	59726.5	112224	89381632	51860.9
140	150	138117	138117	218307	2.980E+09	138117
150	160	34755.6	46641.4	77717.8	712947	34755.6
160	170	36099.7	49305.8	78298.4	52886212	36099.7
170	180	36647.7	40609.6	84192.1	26115594	36647.7
180	190	36099.9	49306.3	78298.4	52886212	36099.9
190	200	33958.3	43202.8	78447.9	...	33958.3
200	210	57079	57079	106915	575703104	57079
210	220	32926.6	32926.6	52030.3	622014208	32926.6
220	230	33040.3	33040.3	59047.1	228339984	33040.3



Empty Support Force + the Sum of the Y forces 10798015.00 N
 Operating Support Force + the Sum of the Y forces 17881680.00 N
 Hydro Support Force + the Sum of the Y forces 24596932.00 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	1230512	1817490	2502301
20	30	1208842	1795820	2480631
30	40	1189077	1776055	2460866
40	50	1170316	1757294	2442105
50	60	1164329	1751307	2436118
60	70	1084720	1570398	2276826
70	80	1000271	1360102	2080232
80	90	934587	1168569	1895823
90	100	842897	978300	1685409
100	110	766559	887804	1499229
110	120	678260	771714	1309643
120	130	544390	624734	1065273
130	140	490585	549955	945478
140	150	438724	490229	833254
150	160	300607	352112	614947
160	170	265851	305470	537229
170	180	229752	256165	458930
180	190	193104	215555	374738
190	200	157004	166249	296440
200	210	123046	123046	217992
210	220	65966.9	65966.9	111077
220	230	33040.3	33040.3	59047.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	6.424E+09	6.424E+09	6.424E+09
20	30	6.374E+09	6.374E+09	6.374E+09
30	40	6.323E+09	6.323E+09	6.323E+09
40	50	6.315E+09	6.315E+09	6.315E+09
50	60	6.315E+09	6.315E+09	6.315E+09
60	70	6.315E+09	6.315E+09	6.315E+09
70	80	6.040E+09	6.040E+09	6.040E+09
80	90	5.938E+09	5.938E+09	5.938E+09
90	100	4.898E+09	4.898E+09	4.898E+09
100	110	4.865E+09	4.865E+09	4.865E+09
110	120	4.763E+09	4.763E+09	4.763E+09
120	130	4.628E+09	4.628E+09	4.628E+09
130	140	4.628E+09	4.628E+09	4.628E+09
140	150	4.539E+09	4.539E+09	4.539E+09
150	160	1558658304	1558658304	1558658304
160	170	1557945216	1557945216	1557945216
170	180	1505059072	1505059072	1505059072
180	190	1478943488	1478943488	1478943488
190	200	1426057216	1426057216	1426057216
200	210	1426057216	1426057216	1426057216
210	220	850354176	850354176	850354176
220	230	228339984	228339984	228339984

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
C	9.14	10.20	130	600	GR 1.1	...	...	...	...
A	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-6	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-5	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-4	9.14	10.20	130	600	GR 1.1	...	...	...	...
E	9.14	10.20	130	600	GR 1.1	...	...	...	...
P	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-3	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-2	9.14	10.20	130	600	GR 1.1	...	...	...	...
D	9.14	10.20	130	600	GR 1.1	2.354	9.143	9.143	9.143
MH-1	9.14	10.20	130	600	GR 1.1	...	...	...	...
B	9.14	10.20	130	600	GR 1.1	...	...	...	...
Min Rating	9.143	10.201, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
D	498.475	0.50	57275556	0	4.580	9.143	Pass

Evaluating Flange on Nozzle [D]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(57275556) + 4 \cdot \text{abs}(0) \cdot 498.475 \leq \pi \cdot 498.475^3 \cdot [(9.143 - 4.580) + .500 \cdot 9.143]$$

$$= 0.91641E+09 \leq 0.35555E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 916408896.0 / (\pi \cdot 498.475^3))$$

$$= \min(9.143, 11.360)$$

$$= 9.143 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.56923 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.46004 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	60646.4	251554880

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	60646.4	251554880

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
220	2300.00	42883.	-59369.	42883.	0.792E+08	0.792E+08	



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.165 \cdot 0.879) / (1 + 1.7 \cdot 3.4 \cdot 0.165)))$
 $= \min(0.85, 0.87)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	1.460 Hz
Natural Frequency of Vessel (Empty)	1.569 Hz
Natural Frequency of Vessel (Test)	1.157 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.610
Structure Height to Diameter ratio	8.721
Height to top of Structure	55175.992 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 0.0 = 1000.0$ mm
 $= 3.281$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.281 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.61 \cdot 187.705$$

$$= 1556.1 \text{ lbs. } [6921.5] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Btm skirt-1	1000.0	0.000	0.000	0.000	0.849	1.000	0.766
Btm skirt-2	3500.0	0.000	0.000	0.000	0.849	1.000	0.766
Btm skirt-3	6500.0	0.000	0.000	0.000	0.914	1.000	0.766
Top skirt	8500.0	0.000	0.000	0.000	0.967	1.000	0.766
BTM dish	9000.0	0.000	0.000	0.000	0.979	1.000	0.766
shell-1	10500.0	0.000	0.000	0.000	1.011	1.000	0.766
shell-2	13500.0	0.000	0.000	0.000	1.066	1.000	0.766
shell-3	16500.0	0.000	0.000	0.000	1.112	1.000	0.766
shell-4	19500.0	0.000	0.000	0.000	1.152	1.000	0.766
shell-5	22500.0	0.000	0.000	0.000	1.187	1.000	0.766
shell-6	25500.0	0.000	0.000	0.000	1.219	1.000	0.766
shell-7	28000.0	0.000	0.000	0.000	1.243	1.000	0.766
shell-8	29825.0	0.000	0.000	0.000	1.260	1.000	0.766
cone	32015.3	0.000	0.000	0.000	1.279	1.000	0.766
shell-9	35090.0	0.000	0.000	0.000	1.304	1.000	0.766
shell-10	38090.0	0.000	0.000	0.000	1.326	1.000	0.766
shell-11	41090.0	0.000	0.000	0.000	1.348	1.000	0.766
shell-12	44090.0	0.000	0.000	0.000	1.368	1.000	0.766
shell-13	47090.0	0.000	0.000	0.000	1.387	1.000	0.766
shell-14	50090.0	0.000	0.000	0.000	1.405	1.000	0.766
shell-15	52165.0	0.000	0.000	0.000	1.417	1.000	0.766
TOP dish	53773.8	0.000	0.000	0.000	1.426	1.000	0.766

Note:

The height/(avg. diameter) ratio [8.2], was less than or equal to 15,
 therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT- SKIRT	1499999.88	1999.00	0.77	1149.07	1.00
PLAT- SKIRT	1499999.88	5000.00	0.77	1149.07	1.00
PLAT-1	5844799.50	9000.00	0.77	4477.39	1.00
PLAT-2	2999999.75	13000.00	0.77	2298.14	1.00
PLAT-3	2999999.75	17000.00	0.77	2298.14	1.00
PLAT-4	2999999.75	21000.00	0.77	2298.14	1.00
PLAT-5	2999999.75	25000.00	0.77	2298.14	1.00
PLAT-6	10965440.00	29000.00	0.77	8400.03	1.00
PLAT-7	2999999.75	33000.00	0.77	2298.14	1.00
PLAT-8	2999999.75	37440.00	0.77	2298.14	1.00
PLAT-9	2999999.75	41440.00	0.77	2298.14	1.00
PLAT-10	2999999.75	45440.00	0.77	2298.14	1.00
PLAT-11	2999999.75	51440.00	0.77	2298.14	1.00
TOP PLT	18566400.00	55239.97	0.77	14222.72	1.00

Wind Loads on Masses/Equipment/Piping



ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
skirt manwy/v	0.00	5000.00	0.77	0.00
VB	0.00	9000.00	0.77	0.00
btm dish SF w	0.00	9000.00	0.77	0.00
CPPOVH	0.00	9000.00	0.77	0.00
FPA	0.00	17099.97	0.77	0.00
FPABolting	0.00	17099.97	0.77	0.00
APPOVH	0.00	17099.98	0.77	0.00
B#4btm TSR	0.00	19649.98	0.77	0.00
B#4btm supp	0.00	19649.98	0.77	0.00
B#4top HDG	0.00	23649.98	0.77	0.00
B#4top dist.	0.00	23649.98	0.77	0.00
B#4top dist.	0.00	23649.98	0.77	0.00
B#3btm TSR	0.00	25749.98	0.77	0.00
B#3btm supp	0.00	25749.98	0.77	0.00
Nozzle wt	0.00	25699.98	0.77	0.00
10T margin	0.00	25699.98	0.77	0.00
Cleat wt	0.00	25699.98	0.77	0.00
Lifting wts	0.00	25699.98	0.77	0.00
B#3top HDG	0.00	29749.98	0.77	0.00
B#3top distr	0.00	29749.98	0.77	0.00
B#3top dist.	0.00	29749.98	0.77	0.00
EPPOVH	0.00	31623.78	0.77	0.00
FPE	0.00	31623.78	0.77	0.00
FPEBolting	0.00	31623.78	0.77	0.00
FPEinsert p	0.00	31623.78	0.77	0.00
B#2btm TSR	0.00	33889.98	0.77	0.00
B#2btm supp	0.00	33889.98	0.77	0.00
B#2top HDG	0.00	39739.98	0.77	0.00
B#2top distr	0.00	39739.98	0.77	0.00
B#2top dis.	0.00	39739.98	0.77	0.00
B#1btm TSR	0.00	41839.98	0.77	0.00
B#1btm supp	0.00	41839.98	0.77	0.00
B#1top HDG	0.00	47689.98	0.77	0.00
B#1top distr	0.00	47689.98	0.77	0.00
B#1top dis.	0.00	47689.98	0.77	0.00
FPD	0.00	49289.97	0.77	0.00
FPDBolting	0.00	49289.97	0.77	0.00
DPPOVH	0.00	49089.97	0.77	0.00
Demister dead	0.00	52439.97	0.77	0.00
Demister TA	0.00	52439.97	0.77	0.00
Tray Truss	0.00	52439.97	0.77	0.00
davit	0.00	52089.97	0.77	0.00
BPPOVH	0.00	52164.97	0.77	0.00
top dish SF w	0.00	52739.96	0.77	0.00
PSV-1	0.00	55039.96	0.77	0.00
PSV-2	0.00	55039.96	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 1.46004 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1000	8719.2	17438400	0.76606	10140.3
20	30	3500	8719.2	26157600	0.76606	13226.1
30	40	6500	8719.2	26157600	0.76606	11420.4
40	50	8500	8719.2	8719200	0.76606	3806.81
50	60	8999.98	8928	272.125	0.76606	0.11881
60	70	10500	8985.6	26956800	0.76606	15990.9
70	80	13500	8985.6	26956800	0.76606	14203.7
80	90	16500	8985.6	26956800	0.76606	14373.9
90	100	19500	8980.8	26942400	0.76606	11763.1
100	110	22500	8980.8	26942400	0.76606	14122.5
110	120	25500	8980.8	26942400	0.76606	14241.5



120	130	28000	8980.8	17961598	0.76606	7842.05
130	140	29825	8980.8	14818319	0.76606	15454.1
140	150	32015.3	7504.8	22064114	0.76606	12238.9
150	160	35090	6031.92	18095760	0.76606	7900.63
160	170	38090	6031.92	18095760	0.76606	10385.4
170	180	41090	6031.92	18095760	0.76606	10450.1
180	190	44090	6031.92	18095760	0.76606	10506
190	200	47090	6031.92	18095760	0.76606	7900.63
200	210	50090	6031.92	18095760	0.76606	10496.7
210	220	52165	6031.92	6936709	0.76606	3028.57
220	230	53773.8	5992.56	11465329	0.76606	21077.3

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(180.62990.75)$
= 0.985 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/1.46)
= 0.685

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.985$, 0.685) per 12.8.2
= 0.685

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (1.25 / 1.25)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.685(1.25 / 1.25))$
= 0.152

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1518 \cdot 17822310$
= 2706219.000 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 17822310$



= 558907.6 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0925

The Natural Frequency for the Vessel (Ope...) is 1.46004 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1000	212496	1028.68	1005.38	499538
20	30	3500	193820	3687.22	3497.72	492874
30	40	6500	183972	6882.65	6433.31	486796
40	50	8500	58706.4	2944.19	2734.44	481027
50	60	8999.98	1773993	94699.9	38651.2	479186
60	70	10500	2062166	130274	48343.4	423553
70	80	13500	1878165	156136	49186.1	358884
80	90	16500	1865786	193126	85080.1	299984
90	100	19500	887403	110245	84679.8	241473
100	110	22500	1138381	165355	114132	213645
110	120	25500	1441284	240029	197800	177945
120	130	28000	733276	135255	87854.3	132746
130	140	29825	585677	115745	90591.9	109751
140	150	32120	1354375	290239	261157	91383.8
150	160	35090	457365	107954	72230.8	48910.6
160	170	38090	483492	124821	81898.1	34567.6
170	180	41090	398217	111683	90157.3	19405.3
180	190	44090	483497	146451	95755.7	6917.21
190	200	47090	423647	137891	96639.6	-8245.26
200	210	50090	559717	194895	173520	-21530.8
210	220	52165	322879	117525	104534	-39083.5
220	230	52740	323993	119352	106131	-49209



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	60646.4	60646.4	3.451E+09	3.451E+09
20	30	3500	60646.4	60646.4	3.330E+09	3.330E+09
30	40	6500	60646.4	60646.4	3.148E+09	3.148E+09
40	50	8500	60646.4	60646.4	2.966E+09	2.966E+09
50	60	8999.98	60646.4	60646.4	2.905E+09	2.905E+09
60	70	10500	60646.4	60646.4	2.905E+09	2.905E+09
70	80	13500	60646.4	60646.4	2.723E+09	2.723E+09
80	90	16500	60646.4	60646.4	2.541E+09	2.541E+09
90	100	19500	60646.4	60646.4	2.359E+09	2.359E+09
100	110	22500	60646.4	60646.4	2.177E+09	2.177E+09
110	120	25500	60646.4	60646.4	1995236864	1995236864
120	130	28000	60646.4	60646.4	1813223808	1813223808
130	140	29825	60646.4	60646.4	1691881728	1691881728
140	150	32120	60646.4	60646.4	1591774720	1591774720
150	160	35090	60646.4	60646.4	1413401984	1413401984
160	170	38090	60646.4	60646.4	1231389184	1231389184
170	180	41090	60646.4	60646.4	1049376064	1049376064
180	190	44090	60646.4	60646.4	867363136	867363136
190	200	47090	60646.4	60646.4	685350272	685350272
200	210	50090	60646.4	60646.4	503337440	503337440
210	220	52165	60646.4	60646.4	321324512	321324512
220	230	52740	60646.4	60646.4	251554896	251554896

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	240570	2706219	6.501E+09	84.58E+09
20	30	3500	230429	2705190	6.030E+09	79.16E+09
30	40	6500	217203	2701503	5.358E+09	71.05E+09
40	50	8500	205783	2694620	4.723E+09	62.95E+09
50	60	8999.98	201976	2691676	4.519E+09	60.26E+09
60	70	10500	201976	2596976	4.519E+09	60.26E+09
70	80	13500	185985	2466701	3.937E+09	52.66E+09
80	90	16500	171781	2310565	3.400E+09	45.49E+09
90	100	19500	157408	2117439	2.906E+09	38.85E+09
100	110	22500	145644	2007195	2.452E+09	32.66E+09
110	120	25500	131522	1841839	2035698944	26.88E+09
120	130	28000	117280	1601811	1662343936	21.71E+09
130	140	29825	109438	1466555	1435533056	18.64E+09
140	150	32120	93984.3	1350810	1267641472	16.32E+09
150	160	35090	81745.4	1060571	1009214080	12.77E+09
160	170	38090	73844.8	952618	775734208	9.752E+09
170	180	41090	63459.3	827797	569694528	7.081E+09
180	190	44090	53009.2	716114	394921088	4.764E+09
190	200	47090	42503.2	569663	251594448	2.834E+09
200	210	50090	34602.6	431772	135888896	1331734784
210	220	52165	24105.9	236877	47790548	328355936
220	230	52740	21077.3	119352	21800370	123445904

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	60646.4	0.0016265	0.0064677	0.64294E-05
20	30	60646.4	0.01984	0.040533	0.16189E-04
30	40	60646.4	0.068339	0.10305	0.25400E-04
40	50	60646.4	0.11612	0.12993	0.28349E-04
50	60	60646.4	0.12993	0.12993	0.28349E-04
60	70	60646.4	0.17347	0.21898	0.30987E-04
70	80	60646.4	0.26641	0.31569	0.33455E-04
80	90	60646.4	0.36675	0.41954	0.35752E-04
90	100	60646.4	0.474	0.53009	0.37920E-04
100	110	60646.4	0.58774	0.64688	0.39914E-04
110	120	60646.4	0.70746	0.7694	0.41734E-04
120	130	60646.4	0.81142	0.854	0.42851E-04
130	140	60646.4	0.88953	0.92542	0.43714E-04
140	150	60646.4	0.99055	1.05734	0.45978E-04
150	160	60646.4	1.12909	1.20616	0.53077E-04
160	170	60646.4	1.2882	1.37482	0.59198E-04
170	180	60646.4	1.46567	1.56037	0.64343E-04
180	190	60646.4	1.65857	1.7599	0.68510E-04
190	200	60646.4	1.86398	1.97046	0.71700E-04
200	210	60646.4	2.07896	2.18912	0.73914E-04
210	220	60646.4	2.23171	2.27448	0.74503E-04
220	230	60646.4	2.27448	2.27448	0.74503E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 263.700 Actual deflection : 2.274 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	240570	0.0030827	0.012179	0.12028E-04
20	30	230429	0.036991	0.074824	0.29393E-04
30	40	217203	0.12492	0.18655	0.44766E-04
40	50	205783	0.20954	0.23372	0.49537E-04
50	60	201976	0.23372	0.23372	0.49537E-04
60	70	201976	0.30962	0.38858	0.53608E-04
70	80	185985	0.47038	0.55483	0.57140E-04
80	90	171781	0.64173	0.73092	0.60175E-04
90	100	157408	0.82222	0.9155	0.62805E-04
100	110	145644	1.01058	1.10732	0.65006E-04
110	120	131522	1.20555	1.30515	0.66820E-04
120	130	117280	1.37223	1.43982	0.67833E-04
130	140	109438	1.49594	1.55236	0.68563E-04
140	150	93984.3	1.65385	1.75662	0.70321E-04
150	160	81745.4	1.86411	1.97528	0.75234E-04
160	170	73844.8	2.08966	2.20681	0.78931E-04
170	180	63459.3	2.32632	2.44781	0.81578E-04
180	190	53009.2	2.57095	2.6954	0.83347E-04
190	200	42503.2	2.8209	2.94719	0.84405E-04
200	210	34602.6	3.07404	3.20126	0.84897E-04
210	220	24105.9	3.25009	3.29894	0.84970E-04
220	230	21077.3	3.29894	3.29894	0.84970E-04



Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	228.53	1428.31
20	30	228.53	1428.31
30	40	228.53	1428.31
40	50	228.53	1428.31
50	60	234.003	1462.52
60	70	235.512	1471.95
70	80	235.512	1471.95
80	90	235.512	1471.95
90	100	235.387	1471.17
100	110	235.387	1471.17
110	120	235.387	1471.17
120	130	235.387	1471.17
130	140	235.387	1471.17
140	150	196.701	1229.38
150	160	158.096	988.103
160	170	158.096	988.103
170	180	158.096	988.103
180	190	158.096	988.103
190	200	158.096	988.103
200	210	158.096	988.103
210	220	158.096	988.103
220	230	157.065	981.655

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 263.700 Actual deflection : 3.299 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 263.700 Actual deflection : 5.573 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	817695	781529	1470701312	1406270848
20	30	772476	736290	1390134400	1325597312
30	40	772476	736290	1390134400	1325597312
40	50	772476	736290	1390134400	1325597312
50	60	1495784	1360933	2.673E+09	2.436E+09
60	70	2583634	2449689	4.588E+09	4.357E+09
70	80	2583634	2449689	4.588E+09	4.357E+09
80	90	2583634	2449689	4.588E+09	4.357E+09
90	100	2537606	2403662	4.506E+09	4.276E+09
100	110	2537606	2403662	4.506E+09	4.276E+09
110	120	2537606	2403662	4.506E+09	4.276E+09
120	130	2537606	2403662	4.506E+09	4.276E+09
130	140	2537606	2403662	4.506E+09	4.276E+09
140	150	3012205	2866646	5.350E+09	5.101E+09
150	160	1056782	1013401	1215878144	1167439744
160	170	1056782	1013401	1215878144	1167439744
170	180	1056782	1013401	1215878144	1167439744
180	190	1056782	1013401	1215878144	1167439744
190	200	1056782	1013401	1215878144	1167439744
200	210	1056782	1013401	1215878144	1167439744
210	220	1056782	1013401	1215878144	1167439744
220	230	587102	543418	679963968	630171904



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	113.099	246.996	-90.3134	-91.6078
20	30	113.099	246.996	-88.5705	-89.9765
30	40	113.099	246.996	-88.5705	-89.9765
40	50	86.02	247	-88.5705	-89.9765
50	60	156.4	247	-104.421	-106.479
60	70	156.4	247	-115.291	-116.098
70	80	156.4	247	-115.291	-116.098
80	90	156.4	247	-115.291	-116.098
90	100	156.4	247	-115.012	-115.845
100	110	156.4	247	-115.012	-115.845
110	120	156.4	247	-115.012	-115.845
120	130	156.4	247	-115.012	-115.845
130	140	156.4	247	-115.012	-115.845
140	150	156.4	247	-117.437	-118.102
150	160	156.4	247	-115.088	-115.733
160	170	156.4	247	-115.088	-115.733
170	180	156.4	247	-115.088	-115.733
180	190	156.4	247	-115.088	-115.733
190	200	156.4	247	-115.088	-115.733
200	210	156.4	247	-115.088	-115.733
210	220	156.4	247	-115.088	-115.733
220	230	156.4	247	-103.335	-105.071

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	135.007	...	171.469
60	70	74.0011	...	97.9428
70	80	74.0011	...	97.9428
80	90	74.0011	...	97.9428
90	100	75.4181	...	99.7193
100	110	75.4181	...	99.7193
110	120	75.4181	...	99.7193
120	130	75.4181	...	99.7193
130	140	75.4181	...	99.7193
140	150	68.736	...	91.2854
150	160	75.0295	...	100.512
160	170	75.0295	...	100.512
170	180	75.0295	...	100.512
180	190	75.0295	...	100.512
190	200	75.0295	...	100.512
200	210	75.0295	...	100.512
210	220	75.0295	...	100.512
220	230	141.871	...	183.447

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-15.4408	-22.8064	...	4.56664	4.56664
20	30	-16.1009	-23.919	...	4.80653	4.80653
30	40	-15.8376	-23.6557	...	4.76849	4.76849
40	50	-15.5877	-23.4059	...	4.76257	4.76257
50	60	-8.39011	-9.36581	...	2.59121	2.59121
60	70	-4.34246	-4.88451	...	1.44883	1.44883
70	80	-4.00438	-4.54644	...	1.38571	1.38571
80	90	-3.74143	-4.28348	...	1.36232	1.36232
90	100	-3.43898	-3.99142	...	1.14518	1.14518
100	110	-3.12753	-3.6222	...	1.1375	1.1375
110	120	-2.76727	-3.14856	...	1.11367	1.11367
120	130	-2.22108	-2.54888	...	1.08217	1.08217
130	140	-2.00156	-2.24379	...	1.08217	1.08217
140	150	-1.50088	-1.67708	...	0.88958	0.88958
150	160	-2.90902	-3.40744	...	1.33468	1.33468
160	170	-2.57268	-2.95608	...	1.33407	1.33407
170	180	-2.22334	-2.47894	...	1.28878	1.28878
180	190	-1.8687	-2.08596	...	1.26642	1.26642
190	200	-1.51935	-1.60881	...	1.22113	1.22113
200	210	-1.19073	-1.19073	...	1.22113	1.22113
210	220	-0.63837	-0.63837	...	0.72816	0.72816
220	230	-0.59626	-0.59626	...	0.36223	0.36223

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:



From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	4.6214	60.1243	...	...
20	30	...	4.54731	59.7008	...	...
30	40	...	4.04074	53.5822	...	...
40	50	...	3.56206	47.4757	...	...
50	60	...	1.85435	24.725	...	...
60	70	...	1.03683	13.8245	...	...
70	80	...	0.90326	12.0813	...	...
80	90	...	0.7801	10.4366	...	...
90	100	...	0.67955	9.08303	...	...
100	110	...	0.57322	7.63586	...	...
110	120	...	0.47597	6.28538	...	...
120	130	...	0.38868	5.07713	...	...
130	140	...	0.33565	4.35942	...	...
140	150	...	0.24844	3.19844	...	...
150	160	...	0.86419	10.9379	...	...
160	170	...	0.66426	8.35103	...	...
170	180	...	0.48783	6.06325	...	...
180	190	...	0.33817	4.07937	...	...
190	200	...	0.21544	2.42718	...	...
200	210	...	0.11636	1.14037	...	...
210	220	...	0.040923	0.28117	...	...
220	230	...	0.034583	0.19583	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	45.3705
20	30	...	...	...	45.2777
30	40	...	...	...	41.0017
40	50	...	...	...	36.7369
50	60	...	...	...	19.2162
60	70	...	...	...	10.7444
70	80	...	...	...	9.49432
80	90	...	...	...	8.27786
90	100	...	...	...	7.24384
100	110	...	...	...	6.11074
110	120	...	...	...	5.0474
120	130	...	...	...	4.0935
130	140	...	...	...	3.52438
140	150	...	...	...	2.58952
150	160	...	...	...	8.91776
160	170	...	...	...	6.90099
170	180	...	...	...	5.08228
180	190	...	...	...	3.48465
190	200	...	...	...	2.12592
200	210	...	...	...	1.0144
210	220	...	...	...	0.25004
220	230	...	...	...	0.17414

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.075972	0.63924
20	30	-0.08064	0.66946
30	40	-0.08064	0.6612
40	50	-0.08064	0.65337
50	60	-0.043628	0.35213



60	70	-0.024238	0.17292
70	80	-0.024238	0.14651
80	90	-0.024238	0.12247
90	100	-0.024702	0.10047
100	110	-0.024702	0.08889
110	120	-0.024702	0.074037
120	130	-0.024702	0.055231
130	140	-0.024702	0.045664
140	150	-0.020712	0.031881
150	160	-0.058589	0.048268
160	170	-0.058589	0.034113
170	180	-0.058589	0.01915
180	190	-0.058589	0.0068263
190	200	-0.058589	-0.0081369
200	210	-0.058589	-0.021248
210	220	-0.058589	-0.03857
220	230	-0.10926	-0.090562

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	2.45346	2.45346	2.34598	2.34598
20	30	2.51127	2.51127	2.39468	2.39468
30	40	2.37401	2.37401	2.26379	2.26379
40	50	2.23674	2.23674	2.1329	2.1329
50	60	1.19207	1.19207	1.08639	1.08639
60	70	0.66652	0.66652	0.633	0.633
70	80	0.62477	0.62477	0.59335	0.59335
80	90	0.58301	0.58301	0.55369	0.55369
90	100	0.55163	0.55163	0.52337	0.52337
100	110	0.50907	0.50907	0.48299	0.48299
110	120	0.46651	0.46651	0.44261	0.44261
120	130	0.42396	0.42396	0.40224	0.40224
130	140	0.39558	0.39558	0.37532	0.37532
140	150	0.31197	0.31197	0.29742	0.29742
150	160	1.2103	1.2103	1.16208	1.16208
160	170	1.05444	1.05444	1.01243	1.01243
170	180	0.89858	0.89858	0.86278	0.86278
180	190	0.74272	0.74272	0.71314	0.71314
190	200	0.58687	0.58687	0.56349	0.56349
200	210	0.43101	0.43101	0.41384	0.41384
210	220	0.27515	0.27515	0.26419	0.26419
220	230	0.39906	0.39906	0.36983	0.36983

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...



160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.16	113.10	-27.16	101.11	0.2401	0.2686	0.1641
20	28.05	113.10	-28.05	99.01	0.2480	0.2833	0.1746
30	27.10	113.10	-27.10	99.01	0.2396	0.2737	0.1718
40	26.23	86.02	-26.23	95.06	0.3049	0.2759	0.1759
50	14.07	156.40	-14.07	72.29	0.0900	0.1947	0.3328
60	7.52	156.40	-7.52	161.75	0.0481	0.0465	0.0297
70	6.94	156.40	-6.94	161.75	0.0444	0.0429	0.0274
80	6.49	156.40	-6.49	161.75	0.0415	0.0401	0.0256
90	5.84	156.40	-5.84	160.93	0.0373	0.0363	0.0237
100	5.37	156.40	-5.37	160.93	0.0343	0.0334	0.0215
110	4.85	156.40	-4.85	160.93	0.0310	0.0301	0.0191
120	4.14	156.40	-4.14	160.93	0.0265	0.0257	0.0153
130	3.84	156.40	-3.84	160.93	0.0246	0.0239	0.0138
150	6.38	156.40	-6.38	161.40	0.0408	0.0395	0.0223
160	5.68	156.40	-5.68	161.40	0.0363	0.0352	0.0198
170	4.96	156.40	-4.96	161.40	0.0317	0.0307	0.0172
180	4.27	156.40	-4.27	161.40	0.0273	0.0265	0.0145



190	3.60	156.40	-3.60	161.40	0.0230	0.0223	0.0125
200	3.02	156.40	-3.02	161.40	0.0193	0.0187	0.0110
210	1.74	156.40	-1.74	161.40	0.0111	0.0108	0.0065
220	1.50	156.40	-1.50	70.96	0.0096	0.0212	0.0194

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	67.19	113.10	-67.19	101.11	0.5941	0.6646	0.4838
20	68.03	113.10	-68.03	99.01	0.6015	0.6870	0.4927
30	63.32	113.10	-63.32	99.01	0.5599	0.6395	0.4510
40	58.67	86.02	-58.67	95.06	0.6821	0.6172	0.4256
50	31.04	156.40	-31.04	138.04	0.1984	0.2248	0.4093
60	17.03	156.40	-17.03	161.75	0.1089	0.1053	0.0795
70	15.36	156.40	-15.36	161.75	0.0982	0.0950	0.0711
80	13.84	156.40	-13.84	161.75	0.0885	0.0856	0.0632
90	12.28	156.40	-12.28	160.93	0.0785	0.0763	0.0555
100	10.80	156.40	-10.80	160.93	0.0690	0.0671	0.0482
110	9.32	156.40	-9.32	160.93	0.0596	0.0579	0.0412
120	7.77	156.40	-7.77	160.93	0.0497	0.0483	0.0348
130	6.96	156.40	-6.96	160.93	0.0445	0.0432	0.0311
150	14.32	156.40	-14.32	161.40	0.0916	0.0887	0.0710
160	11.83	156.40	-11.83	161.40	0.0756	0.0733	0.0575
170	9.47	156.40	-9.47	161.40	0.0606	0.0587	0.0450
180	7.36	156.40	-7.36	161.40	0.0470	0.0456	0.0480
190	5.46	156.40	-5.46	161.40	0.0349	0.0338	0.0510
200	3.88	156.40	-3.88	161.40	0.0248	0.0240	0.0553
210	1.93	156.40	-1.93	161.40	0.0123	0.0120	0.0578
220	1.62	156.40	-1.62	135.84	0.0104	0.0119	0.0115

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.52	113.10	-34.52	101.11	0.3053	0.3415	0.2421
20	35.86	113.10	-35.86	99.01	0.3171	0.3622	0.2590
30	34.92	113.10	-34.92	99.01	0.3088	0.3527	0.2561
40	34.05	86.02	-34.05	95.06	0.3958	0.3582	0.2636
50	15.05	156.40	-15.05	72.29	0.0962	0.2082	0.4410
60	8.06	156.40	-8.06	161.75	0.0515	0.0498	0.0334
70	7.48	156.40	-7.48	161.75	0.0479	0.0463	0.0311
80	7.03	156.40	-7.03	161.75	0.0450	0.0435	0.0293
90	6.39	156.40	-6.39	160.93	0.0409	0.0397	0.0274
100	5.87	156.40	-5.87	160.93	0.0375	0.0365	0.0249
110	5.23	156.40	-5.23	160.93	0.0334	0.0325	0.0217
120	4.47	156.40	-4.47	160.93	0.0286	0.0278	0.0176
130	4.08	156.40	-4.08	160.93	0.0261	0.0254	0.0155
150	6.88	156.40	-6.88	161.40	0.0440	0.0426	0.0261
160	6.07	156.40	-6.07	161.40	0.0388	0.0376	0.0227
170	5.21	156.40	-5.21	161.40	0.0333	0.0323	0.0191
180	4.49	156.40	-4.49	161.40	0.0287	0.0278	0.0161
190	3.69	156.40	-3.69	161.40	0.0236	0.0229	0.0125
200	3.02	156.40	-3.02	161.40	0.0193	0.0187	0.0110
210	1.74	156.40	-1.74	161.40	0.0111	0.0108	0.0065
220	1.50	156.40	-1.50	70.96	0.0096	0.0212	0.0194

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	89.31	113.10	-89.31	101.11	0.7897	0.8833	0.6200
20	90.27	113.10	-90.27	99.01	0.7981	0.9117	0.6278
30	83.72	113.10	-83.72	99.01	0.7402	0.8455	0.5689
40	77.23	86.02	-77.23	95.06	0.8978	0.8124	0.5301
50	37.52	156.40	-37.52	138.10	0.2399	0.2717	0.5435
60	20.65	156.40	-20.65	161.75	0.1320	0.1277	0.0985
70	18.49	156.40	-18.49	161.75	0.1182	0.1143	0.0871
80	16.54	156.40	-16.54	161.75	0.1058	0.1023	0.0765
90	14.67	156.40	-14.67	160.93	0.0938	0.0912	0.0669



100	12.82	156.40	-12.82	160.93	0.0819	0.0796	0.0576
110	10.94	156.40	-10.94	160.93	0.0699	0.0680	0.0488
120	9.08	156.40	-9.08	160.93	0.0580	0.0564	0.0409
130	8.04	156.40	-8.04	160.93	0.0514	0.0499	0.0363
150	16.84	156.40	-16.84	161.40	0.1077	0.1044	0.0835
160	13.66	156.40	-13.66	161.40	0.0873	0.0846	0.0665
170	10.71	156.40	-10.71	161.40	0.0685	0.0664	0.0511
180	8.17	156.40	-8.17	161.40	0.0522	0.0506	0.0480
190	5.85	156.40	-5.85	161.40	0.0374	0.0363	0.0510
200	4.00	156.40	-4.00	161.40	0.0256	0.0248	0.0553
210	1.96	156.40	-1.96	161.40	0.0125	0.0122	0.0578
220	1.64	156.40	-1.64	135.84	0.0105	0.0121	0.0117

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.61	0.0000	-0.0000	0.0023
20	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0030
30	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0034
40	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0037
50	0.00	247.00	0.00	106.48	0.0000	-0.0000	...
60	0.00	247.00	0.00	116.10	0.0000	-0.0000	...
70	0.00	247.00	0.00	116.10	0.0000	-0.0000	0.0010
80	0.00	247.00	0.00	116.10	0.0000	-0.0000	0.0012
90	0.00	247.00	0.00	115.85	0.0000	-0.0000	0.0013
100	0.00	247.00	0.00	115.85	0.0000	-0.0000	0.0014
110	0.00	247.00	0.00	115.85	0.0000	-0.0000	0.0015
120	0.00	247.00	0.00	115.85	0.0000	-0.0000	0.0016
130	0.00	247.00	0.00	115.85	0.0000	-0.0000	0.0017
150	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0047
160	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0049
170	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0051
180	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0053
190	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0055
200	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0057
210	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0058
220	0.00	247.00	0.00	105.07	0.0000	-0.0000	...

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.61	0.0000	-0.0000	0.0020
20	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0022
30	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0022
40	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0023
50	0.00	247.00	0.00	106.48	0.0000	-0.0000	...
60	0.00	247.00	0.00	116.10	0.0000	-0.0000	...
70	0.00	247.00	0.00	116.10	0.0000	-0.0000	...
80	0.00	247.00	0.00	116.10	0.0000	-0.0000	...
90	0.00	247.00	0.00	115.85	0.0000	-0.0000	...
100	0.00	247.00	0.00	115.85	0.0000	-0.0000	...
110	0.00	247.00	0.00	115.85	0.0000	-0.0000	...
120	0.00	247.00	0.00	115.85	0.0000	-0.0000	...
130	0.00	247.00	0.00	115.85	0.0000	-0.0000	...
150	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
160	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
170	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
180	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
190	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
200	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
210	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
220	0.00	247.00	0.00	105.07	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.52	113.10	-34.52	101.11	0.3053	0.3415	0.2421



20	35.86	113.10	-35.86	99.01	0.3171	0.3622	0.2590
30	34.92	113.10	-34.92	99.01	0.3088	0.3527	0.2561
40	34.05	86.02	-34.05	95.06	0.3958	0.3582	0.2636
50	135.10	156.40		72.29	0.8638		0.4410
60	133.02	156.40		161.75	0.8505		0.0334
70	132.05	156.40		161.75	0.8443		0.0311
80	131.10	156.40		161.75	0.8382		0.0293
90	132.79	156.40		160.93	0.8490		0.0274
100	132.76	156.40		160.93	0.8488		0.0249
110	132.73	156.40		160.93	0.8486		0.0217
120	132.69	156.40		160.93	0.8484		0.0176
130	132.68	156.40		160.93	0.8483		0.0155
150	132.14	156.40		161.40	0.8449		0.0261
160	132.10	156.40		161.40	0.8446		0.0227
170	132.05	156.40		161.40	0.8443		0.0191
180	132.02	156.40		161.40	0.8441		0.0161
190	131.99	156.40		161.40	0.8439		0.0125
200	131.97	156.40		161.40	0.8438		0.0110
210	131.93	156.40		161.40	0.8436		0.0065
220	141.92	156.40		103.33	0.9074		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	89.31	113.10	-89.31	101.11	0.7897	0.8833	0.6200
20	90.27	113.10	-90.27	99.01	0.7981	0.9117	0.6278
30	83.72	113.10	-83.72	99.01	0.7402	0.8455	0.5689
40	77.23	86.02	-77.23	95.06	0.8978	0.8124	0.5301
50	147.22	156.40		138.10	0.9413		0.5435
60	134.61	156.40		161.75	0.8607		0.0985
70	133.30	156.40		161.75	0.8523		0.0871
80	132.06	156.40		161.75	0.8444		0.0765
90	133.51	156.40		160.93	0.8537		0.0669
100	133.31	156.40		160.93	0.8523		0.0576
110	133.12	156.40		160.93	0.8512		0.0488
120	132.97	156.40		160.93	0.8502		0.0409
130	132.89	156.40		160.93	0.8497		0.0363
150	133.12	156.40		161.40	0.8511		0.0835
160	132.73	156.40		161.40	0.8486		0.0665
170	132.43	156.40		161.40	0.8467		0.0511
180	132.23	156.40		161.40	0.8455		0.0480
190	132.09	156.40		161.40	0.8445		0.0510
200	132.00	156.40		161.40	0.8440		0.0553
210	131.94	156.40		161.40	0.8436		0.0578
220	142.01	156.40		103.33	0.9080		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.52	113.10	-34.52	101.28	0.3053	0.3409	0.2413
20	35.86	113.10	-35.86	99.21	0.3171	0.3615	0.2581
30	34.92	113.10	-34.92	99.21	0.3088	0.3520	0.2553
40	34.05	86.02	-34.05	95.35	0.3958	0.3571	0.2623
50	15.05	156.40	-15.05	72.90	0.0962	0.2064	0.4373
60	8.06	156.40	-8.06	163.05	0.0515	0.0494	0.0330
70	7.48	156.40	-7.48	163.05	0.0479	0.0459	0.0307
80	7.03	156.40	-7.03	163.05	0.0450	0.0431	0.0290
90	6.39	156.40	-6.39	162.26	0.0409	0.0394	0.0271
100	5.87	156.40	-5.87	162.26	0.0375	0.0362	0.0246
110	5.23	156.40	-5.23	162.26	0.0334	0.0322	0.0214
120	4.47	156.40	-4.47	162.26	0.0286	0.0275	0.0174
130	4.08	156.40	-4.08	162.26	0.0261	0.0252	0.0153
150	6.88	156.40	-6.88	162.70	0.0440	0.0423	0.0257
160	6.07	156.40	-6.07	162.70	0.0388	0.0373	0.0224
170	5.21	156.40	-5.21	162.70	0.0333	0.0320	0.0188
180	4.49	156.40	-4.49	162.70	0.0287	0.0276	0.0159
190	3.69	156.40	-3.69	162.70	0.0236	0.0227	0.0124
200	3.02	156.40	-3.02	162.70	0.0193	0.0185	0.0109



210	1.74	156.40	-1.74	162.70	0.0111	0.0107	0.0064
220	1.50	156.40	-1.50	71.59	0.0096	0.0210	0.0192

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	89.31	113.10	-89.31	101.28	0.7897	0.8818	0.6192
20	90.27	113.10	-90.27	99.21	0.7981	0.9099	0.6270
30	83.72	113.10	-83.72	99.21	0.7402	0.8439	0.5681
40	77.23	86.02	-77.23	95.35	0.8978	0.8099	0.5289
50	37.52	156.40	-37.52	139.10	0.2399	0.2698	0.5396
60	20.65	156.40	-20.65	163.05	0.1320	0.1267	0.0977
70	18.49	156.40	-18.49	163.05	0.1182	0.1134	0.0864
80	16.54	156.40	-16.54	163.05	0.1058	0.1015	0.0759
90	14.67	156.40	-14.67	162.26	0.0938	0.0904	0.0664
100	12.82	156.40	-12.82	162.26	0.0819	0.0790	0.0572
110	10.94	156.40	-10.94	162.26	0.0699	0.0674	0.0485
120	9.08	156.40	-9.08	162.26	0.0580	0.0559	0.0406
130	8.04	156.40	-8.04	162.26	0.0514	0.0495	0.0360
150	16.84	156.40	-16.84	162.70	0.1077	0.1035	0.0828
160	13.66	156.40	-13.66	162.70	0.0873	0.0840	0.0660
170	10.71	156.40	-10.71	162.70	0.0685	0.0658	0.0507
180	8.17	156.40	-8.17	162.70	0.0522	0.0502	0.0478
190	5.85	156.40	-5.85	162.70	0.0374	0.0360	0.0508
200	4.00	156.40	-4.00	162.70	0.0256	0.0246	0.0551
210	1.96	156.40	-1.96	162.70	0.0125	0.0121	0.0576
220	1.64	156.40	-1.64	136.91	0.0105	0.0120	0.0117

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.61	0.0000	-0.0000	0.0023
20	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0030
30	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0034
40	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0037
50	174.07	247.00		106.48	0.7047		...
60	177.08	247.00		116.10	0.7169		...
70	177.08	247.00		116.10	0.7169		0.0010
80	177.08	247.00		116.10	0.7169		0.0012
90	180.19	247.00		115.85	0.7295		0.0013
100	180.19	247.00		115.85	0.7295		0.0014
110	180.19	247.00		115.85	0.7295		0.0015
120	180.19	247.00		115.85	0.7295		0.0016
130	180.19	247.00		115.85	0.7295		0.0017
150	180.88	247.00		115.73	0.7323		0.0047
160	180.88	247.00		115.73	0.7323		0.0049
170	180.88	247.00		115.73	0.7323		0.0051
180	180.88	247.00		115.73	0.7323		0.0053
190	180.88	247.00		115.73	0.7323		0.0055
200	180.88	247.00		115.73	0.7323		0.0057
210	180.88	247.00		115.73	0.7323		0.0058
220	185.74	247.00		105.07	0.7520		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.61	0.0000	-0.0000	0.0020
20	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0022
30	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0022
40	0.00	247.00	0.00	89.98	0.0000	-0.0000	0.0023
50	174.07	247.00		106.48	0.7047		...
60	177.08	247.00		116.10	0.7169		...
70	177.08	247.00		116.10	0.7169		...
80	177.08	247.00		116.10	0.7169		...
90	180.19	247.00		115.85	0.7295		...
100	180.19	247.00		115.85	0.7295		...
110	180.19	247.00		115.85	0.7295		...



120	180.19	247.00	115.85	0.7295	...
130	180.19	247.00	115.85	0.7295	...
150	180.88	247.00	115.73	0.7323	0.0012
160	180.88	247.00	115.73	0.7323	0.0012
170	180.88	247.00	115.73	0.7323	0.0012
180	180.88	247.00	115.73	0.7323	0.0012
190	180.88	247.00	115.73	0.7323	0.0012
200	180.88	247.00	115.73	0.7323	0.0012
210	180.88	247.00	115.73	0.7323	0.0012
220	185.74	247.00	105.07	0.7520	...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.01	113.10	-20.01	101.11	0.1769	0.1979	0.1633
20	20.91	113.10	-20.91	99.01	0.1849	0.2112	0.1737
30	20.61	113.10	-20.61	99.01	0.1822	0.2081	0.1709
40	20.35	86.02	-20.35	95.06	0.2366	0.2141	0.1750
50	134.19	156.40		72.29	0.8580		0.2515
60	132.86	156.40		161.75	0.8495		0.0295
70	131.91	156.40		161.75	0.8434		0.0272
80	130.98	156.40		161.75	0.8375		0.0254
90	132.70	156.40		160.93	0.8485		0.0235
100	132.68	156.40		160.93	0.8484		0.0214
110	132.67	156.40		160.93	0.8483		0.0189
120	132.65	156.40		160.93	0.8481		0.0152
130	132.64	156.40		160.93	0.8481		0.0137
150	132.01	156.40		161.40	0.8441		0.0219
160	132.00	156.40		161.40	0.8440		0.0194
170	131.98	156.40		161.40	0.8439		0.0167
180	131.97	156.40		161.40	0.8438		0.0141
190	131.96	156.40		161.40	0.8437		0.0114
200	131.95	156.40		161.40	0.8437		0.0090
210	131.93	156.40		161.40	0.8435		0.0048
220	141.75	156.40		103.33	0.9064		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	21.80	113.10	-21.80	101.11	0.1927	0.2156	0.2306
20	22.80	113.10	-22.80	99.01	0.2016	0.2303	0.2460
30	22.55	113.10	-22.55	99.01	0.1994	0.2277	0.2433
40	22.31	86.02	-22.31	95.06	0.2594	0.2347	0.2504
50	132.99	156.40		72.29	0.8503		0.3086
60	132.79	156.40		161.75	0.8490		0.0315
70	131.86	156.40		161.75	0.8431		0.0293
80	130.94	156.40		161.75	0.8372		0.0276
90	132.67	156.40		160.93	0.8482		0.0258
100	132.65	156.40		160.93	0.8482		0.0234
110	132.64	156.40		160.93	0.8481		0.0204
120	132.62	156.40		160.93	0.8480		0.0165
130	132.61	156.40		160.93	0.8479		0.0145
150	131.97	156.40		161.40	0.8438		0.0246
160	131.96	156.40		161.40	0.8437		0.0213
170	131.95	156.40		161.40	0.8437		0.0179
180	131.94	156.40		161.40	0.8436		0.0151
190	131.93	156.40		161.40	0.8435		0.0116
200	131.92	156.40		161.40	0.8435		0.0086
210	131.91	156.40		161.40	0.8434		0.0046
220	141.60	156.40		103.33	0.9053		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.37	113.10	-27.37	101.11	0.2420	0.2707	0.2413
20	28.73	113.10	-28.73	99.01	0.2540	0.2901	0.2581
30	28.42	113.10	-28.42	99.01	0.2513	0.2871	0.2553



40	28.17	86.02	-28.17	95.06	0.3275	0.2963	0.2627
50	133.75	156.40		72.29	0.8552		0.3596
60	132.89	156.40		161.75	0.8497		0.0332
70	131.95	156.40		161.75	0.8436		0.0309
80	131.01	156.40		161.75	0.8377		0.0291
90	132.72	156.40		160.93	0.8486		0.0273
100	132.71	156.40		160.93	0.8485		0.0247
110	132.68	156.40		160.93	0.8484		0.0215
120	132.66	156.40		160.93	0.8482		0.0174
130	132.65	156.40		160.93	0.8481		0.0153
150	132.03	156.40		161.40	0.8442		0.0256
160	132.01	156.40		161.40	0.8441		0.0222
170	131.99	156.40		161.40	0.8439		0.0187
180	131.98	156.40		161.40	0.8438		0.0157
190	131.96	156.40		161.40	0.8437		0.0121
200	131.95	156.40		161.40	0.8437		0.0090
210	131.93	156.40		161.40	0.8435		0.0048
220	141.75	156.40		103.33	0.9064		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.01	113.10	-20.01	101.11	0.1769	0.1979	0.1633
20	20.91	113.10	-20.91	99.01	0.1849	0.2112	0.1737
30	20.61	113.10	-20.61	99.01	0.1822	0.2081	0.1709
40	20.35	86.02	-20.35	95.06	0.2366	0.2141	0.1750
50	134.19	156.40		72.29	0.8580		0.2515
60	132.86	156.40		161.75	0.8495		0.0295
70	131.91	156.40		161.75	0.8434		0.0272
80	130.98	156.40		161.75	0.8375		0.0254
90	132.70	156.40		160.93	0.8485		0.0235
100	132.68	156.40		160.93	0.8484		0.0214
110	132.67	156.40		160.93	0.8483		0.0189
120	132.65	156.40		160.93	0.8481		0.0152
130	132.64	156.40		160.93	0.8481		0.0137
150	132.01	156.40		161.40	0.8441		0.0219
160	132.00	156.40		161.40	0.8440		0.0194
170	131.98	156.40		161.40	0.8439		0.0167
180	131.97	156.40		161.40	0.8438		0.0141
190	131.96	156.40		161.40	0.8437		0.0114
200	131.95	156.40		161.40	0.8437		0.0090
210	131.93	156.40		161.40	0.8435		0.0048
220	141.75	156.40		103.33	0.9064		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.37	113.10	-27.37	101.11	0.2420	0.2707	0.2413
20	28.73	113.10	-28.73	99.01	0.2540	0.2901	0.2581
30	28.42	113.10	-28.42	99.01	0.2513	0.2871	0.2553
40	28.17	86.02	-28.17	95.06	0.3275	0.2963	0.2627
50	11.96	156.40	-11.96	72.29	0.0765	0.1654	0.3596
60	6.33	156.40	-6.33	161.75	0.0405	0.0392	0.0332
70	5.93	156.40	-5.93	161.75	0.0379	0.0367	0.0309
80	5.65	156.40	-5.65	161.75	0.0361	0.0349	0.0291
90	5.14	156.40	-5.14	160.93	0.0328	0.0319	0.0273
100	4.76	156.40	-4.76	160.93	0.0304	0.0296	0.0247
110	4.26	156.40	-4.26	160.93	0.0273	0.0265	0.0215
120	3.63	156.40	-3.63	160.93	0.0232	0.0226	0.0174
130	3.33	156.40	-3.33	160.93	0.0213	0.0207	0.0153
150	4.74	156.40	-4.74	161.40	0.0303	0.0294	0.0256
160	4.29	156.40	-4.29	161.40	0.0274	0.0266	0.0222
170	3.77	156.40	-3.77	161.40	0.0241	0.0233	0.0187
180	3.35	156.40	-3.35	161.40	0.0214	0.0208	0.0157
190	2.83	156.40	-2.83	161.40	0.0181	0.0175	0.0121
200	2.41	156.40	-2.41	161.40	0.0154	0.0149	0.0090
210	1.37	156.40	-1.37	161.40	0.0087	0.0085	0.0048
220	0.96	156.40	-0.96	70.96	0.0061	0.0135	0.0141



Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	29.19	113.10	-29.19	101.11	0.2581	0.2887	0.2345
20	30.57	113.10	-30.57	99.01	0.2703	0.3087	0.2509
30	30.14	113.10	-30.14	99.01	0.2665	0.3044	0.2481
40	29.75	86.02	-29.75	95.06	0.3459	0.3130	0.2554
50	134.44	156.40		72.29	0.8596		0.3815
60	132.93	156.40		161.75	0.8499		0.0320
70	131.98	156.40		161.75	0.8438		0.0299
80	131.04	156.40		161.75	0.8378		0.0283
90	132.74	156.40		160.93	0.8488		0.0266
100	132.72	156.40		160.93	0.8486		0.0241
110	132.70	156.40		160.93	0.8485		0.0210
120	132.67	156.40		160.93	0.8483		0.0170
130	132.66	156.40		160.93	0.8482		0.0150
150	132.09	156.40		161.40	0.8446		0.0253
160	132.06	156.40		161.40	0.8444		0.0220
170	132.03	156.40		161.40	0.8442		0.0185
180	132.00	156.40		161.40	0.8440		0.0156
190	131.98	156.40		161.40	0.8439		0.0122
200	131.96	156.40		161.40	0.8437		0.0091
210	131.93	156.40		161.40	0.8436		0.0051
220	141.91	156.40		103.33	0.9073		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	29.19	113.10	-29.19	101.28	0.2581	0.2882	0.2338
20	30.57	113.10	-30.57	99.21	0.2703	0.3081	0.2501
30	30.14	113.10	-30.14	99.21	0.2665	0.3038	0.2473
40	29.75	86.02	-29.75	95.35	0.3459	0.3120	0.2541
50	12.80	156.40	-12.80	72.90	0.0818	0.1756	0.3784
60	6.83	156.40	-6.83	163.05	0.0437	0.0419	0.0317
70	6.41	156.40	-6.41	163.05	0.0410	0.0393	0.0296
80	6.11	156.40	-6.11	163.05	0.0390	0.0375	0.0280
90	5.59	156.40	-5.59	162.26	0.0357	0.0344	0.0263
100	5.18	156.40	-5.18	162.26	0.0331	0.0319	0.0239
110	4.65	156.40	-4.65	162.26	0.0298	0.0287	0.0208
120	4.00	156.40	-4.00	162.26	0.0256	0.0247	0.0169
130	3.68	156.40	-3.68	162.26	0.0235	0.0227	0.0149
150	5.90	156.40	-5.90	162.70	0.0378	0.0363	0.0249
160	5.31	156.40	-5.31	162.70	0.0340	0.0326	0.0217
170	4.65	156.40	-4.65	162.70	0.0297	0.0286	0.0183
180	4.09	156.40	-4.09	162.70	0.0261	0.0251	0.0154
190	3.42	156.40	-3.42	162.70	0.0219	0.0211	0.0120
200	2.86	156.40	-2.86	162.70	0.0183	0.0176	0.0090
210	1.68	156.40	-1.68	162.70	0.0107	0.0103	0.0050
220	1.45	156.40	-1.45	71.59	0.0093	0.0202	0.0184

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.37	113.10	-27.37	101.11	0.2420	0.2707	0.2413
20	28.73	113.10	-28.73	99.01	0.2540	0.2901	0.2581
30	28.42	113.10	-28.42	99.01	0.2513	0.2871	0.2553
40	28.17	86.02	-28.17	95.06	0.3275	0.2963	0.2627
50	133.75	156.40		72.29	0.8552		0.3596
60	132.89	156.40		161.75	0.8497		0.0332
70	131.95	156.40		161.75	0.8436		0.0309
80	131.01	156.40		161.75	0.8377		0.0291
90	132.72	156.40		160.93	0.8486		0.0273
100	132.71	156.40		160.93	0.8485		0.0247
110	132.68	156.40		160.93	0.8484		0.0215
120	132.66	156.40		160.93	0.8482		0.0174
130	132.65	156.40		160.93	0.8481		0.0153



150	132.03	156.40	161.40	0.8442	0.0256
160	132.01	156.40	161.40	0.8441	0.0222
170	131.99	156.40	161.40	0.8439	0.0187
180	131.98	156.40	161.40	0.8438	0.0157
190	131.96	156.40	161.40	0.8437	0.0121
200	131.95	156.40	161.40	0.8437	0.0090
210	131.93	156.40	161.40	0.8435	0.0048
220	141.75	156.40	103.33	0.9064	...

Absolute Maximum of the all of the Stress Ratio's 0.9413

Governing Element: BTM dish
 Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Btm skirt-1	...	-22.2	4.6	62.6	45.0	+ Tens
Btm skirt-1	...	-22.2	-4.6	-62.6	-89.3	- Comp
Btm skirt-2	...	-23.2	4.8	62.2	43.8	+ Tens
Btm skirt-2	...	-23.2	-4.8	-62.2	-90.3	- Comp
Btm skirt-3	...	-23.0	4.8	56.0	37.7	+ Tens
Btm skirt-3	...	-23.0	-4.8	-56.0	-83.7	- Comp
Top skirt	...	-22.8	4.8	49.7	31.7	+ Tens
Top skirt	...	-22.8	-4.8	-49.7	-77.2	- Comp
BTM dish	135.0	-9.0	2.6	25.9	154.5	+ Tens
BTM dish	135.0	-9.0	-2.6	-25.9	...	- Comp
shell-1	74.0	-4.7	1.4	14.5	85.2	+ Tens
shell-1	74.0	-4.7	-1.4	-14.5	...	- Comp
shell-2	74.0	-4.4	1.4	12.7	83.7	+ Tens
shell-2	74.0	-4.4	-1.4	-12.7	...	- Comp
shell-3	74.0	-4.2	1.4	11.0	82.2	+ Tens
shell-3	74.0	-4.2	-1.4	-11.0	...	- Comp
shell-4	75.4	-3.9	1.1	9.6	82.3	+ Tens
shell-4	75.4	-3.9	-1.1	-9.6	...	- Comp
shell-5	75.4	-3.5	1.1	8.1	81.2	+ Tens
shell-5	75.4	-3.5	-1.1	-8.1	...	- Comp
shell-6	75.4	-3.1	1.1	6.8	80.2	+ Tens
shell-6	75.4	-3.1	-1.1	-6.8	...	- Comp
shell-7	75.4	-2.5	1.1	5.5	79.5	+ Tens
shell-7	75.4	-2.5	-1.1	-5.5	...	- Comp
shell-8	75.4	-2.2	1.1	4.8	79.1	+ Tens
shell-8	75.4	-2.2	-1.1	-4.8	...	- Comp
cone	68.7	-1.6	0.9	3.5	71.5	+ Tens
cone	68.7	-1.6	-0.9	-3.5	...	- Comp
shell-9	75.0	-3.4	1.3	12.1	85.2	+ Tens
shell-9	75.0	-3.4	-1.3	-12.1	...	- Comp
shell-10	75.0	-2.9	1.3	9.4	82.8	+ Tens
shell-10	75.0	-2.9	-1.3	-9.4	...	- Comp
shell-11	75.0	-2.5	1.3	7.0	80.8	+ Tens
shell-11	75.0	-2.5	-1.3	-7.0	...	- Comp
shell-12	75.0	-2.1	1.3	4.8	79.0	+ Tens



shell-12	75.0	-2.1	-1.3	-4.8	...	- Comp
shell-13	75.0	-1.6	1.2	3.0	77.6	+ Tens
shell-13	75.0	-1.6	-1.2	-3.0	...	- Comp
shell-14	75.0	-1.2	1.2	1.6	76.6	+ Tens
shell-14	75.0	-1.2	-1.2	-1.6	...	- Comp
shell-15	75.0	-0.7	0.7	0.6	75.6	+ Tens
shell-15	75.0	-0.7	-0.7	-0.6	...	- Comp
TOP dish	141.9	-0.7	0.4	0.6	142.1	+ Tens
TOP dish	141.9	-0.7	-0.4	-0.6	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, MPa
- Fba - Allowable all comp mem stress due to bending moment, MPa
- Fva - Allowable shear stress when only shear is present, MPa
- Fha - Allowable hoop comp memb stress under ext pressure alone, MPa
- Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, MPa
- Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, MPa
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15, MPa
- Fbha - Allowable axial comp from bending w/hoop compression, MPa
- Fhba - Allowable hoop comp w/long compression due to moment, MPa

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220						70	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						138	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895



170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220						135	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220						70	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						138	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220						135	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	102	109	74		6895	102	6895	6895	6895
20	0.418	9000	100	108	72		6895	100	6895	6895	6895
30	0.418	9000	100	108	72		6895	100	6895	6895	6895
40	0.410	9000	96	104	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
70	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
80	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
90	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
100	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
110	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
120	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
130	0.513	22841	164	164	95	53	6895	164	6895	6895	6895



150	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
160	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
170	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
180	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
190	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
200	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
210	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	102	109	74		6895	102	6895	6895	6895
20	0.418	9000	100	108	72		6895	100	6895	6895	6895
30	0.418	9000	100	108	72		6895	100	6895	6895	6895
40	0.410	9000	96	104	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
70	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
80	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
90	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
100	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
110	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
120	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
130	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
150	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
160	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
170	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
180	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
190	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
200	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
210	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						138	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895



120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.416	9000	101	108	73		6895	101	6895	6895	6895
20	0.413	9000	99	106	71		6895	99	6895	6895	6895
30	0.413	9000	99	106	71		6895	99	6895	6895	6895
40	0.405	9000	95	102	68		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
70	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
80	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
90	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
100	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
110	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
120	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
130	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
150	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
160	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
170	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
180	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
190	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
200	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
210	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.416	9000	101	108	73		6895	101	6895	6895	6895
20	0.413	9000	99	106	71		6895	99	6895	6895	6895
30	0.413	9000	99	106	71		6895	99	6895	6895	6895
40	0.405	9000	95	102	68		6895	95	6895	6895	6895
50						139	6895		6895	6895	6895
60	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
70	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
80	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
90	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
100	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
110	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
120	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
130	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
150	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
160	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
170	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
180	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
190	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
200	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
210	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
220						136	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	102	109	74		6895	102	6895	6895	6895
20	0.418	9000	100	108	72		6895	100	6895	6895	6895
30	0.418	9000	100	108	72		6895	100	6895	6895	6895
40	0.410	9000	96	104	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
70	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
80	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
90	0.513	22841	164	164	95	53	6895	164	6895	6895	6895



100	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
110	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
120	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
130	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
150	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
160	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
170	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
180	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
190	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
200	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
210	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	102	109	74		6895	102	6895	6895	6895
20	0.418	9000	100	108	72		6895	100	6895	6895	6895
30	0.418	9000	100	108	72		6895	100	6895	6895	6895
40	0.410	9000	96	104	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
70	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
80	0.514	22841	164	164	96	54	6895	164	6895	6895	6895
90	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
100	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
110	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
120	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
130	0.513	22841	164	164	95	53	6895	164	6895	6895	6895
150	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
160	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
170	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
180	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
190	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
200	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
210	0.789	19922	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895



80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895



60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220						70	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895
40	0.413	9000	95	102	67		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.416	9000	101	108	73		6895	101	6895	6895	6895
20	0.413	9000	99	106	71		6895	99	6895	6895	6895
30	0.413	9000	99	106	71		6895	99	6895	6895	6895
40	0.405	9000	95	102	68		6895	95	6895	6895	6895
50						72	6895		6895	6895	6895
60	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
70	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
80	0.508	22841	163	163	94	52	6895	163	6895	6895	6895
90	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
100	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
110	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
120	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
130	0.508	22841	162	162	94	51	6895	162	6895	6895	6895
150	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
160	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
170	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
180	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
190	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
200	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
210	0.782	19922	162	162	90	41	6895	162	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.421	9000	101	108	72		6895	101	6895	6895	6895
20	0.418	9000	99	106	70		6895	99	6895	6895	6895
30	0.418	9000	99	106	70		6895	99	6895	6895	6895



40	0.413	9000	95	102	67	6895	95	6895	6895	6895
50						72	6895	6895	6895	6895
60	0.517	22841	161	161	94	51	6895	161	6895	6895
70	0.517	22841	161	161	94	51	6895	161	6895	6895
80	0.517	22841	161	161	94	51	6895	161	6895	6895
90	0.516	22841	160	161	93	50	6895	160	6895	6895
100	0.516	22841	160	161	93	50	6895	160	6895	6895
110	0.516	22841	160	161	93	50	6895	160	6895	6895
120	0.516	22841	160	161	93	50	6895	160	6895	6895
130	0.516	22841	160	161	93	50	6895	160	6895	6895
150	0.794	19922	161	161	90	40	6895	161	6895	6895
160	0.794	19922	161	161	90	40	6895	161	6895	6895
170	0.794	19922	161	161	90	40	6895	161	6895	6895
180	0.794	19922	161	161	90	40	6895	161	6895	6895
190	0.794	19922	161	161	90	40	6895	161	6895	6895
200	0.794	19922	161	161	90	40	6895	161	6895	6895
210	0.794	19922	161	161	90	40	6895	161	6895	6895
220						6895		6895	6895	6895

Cone Junction Stresses and related Calculations are shown in the Conical Sections report.

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

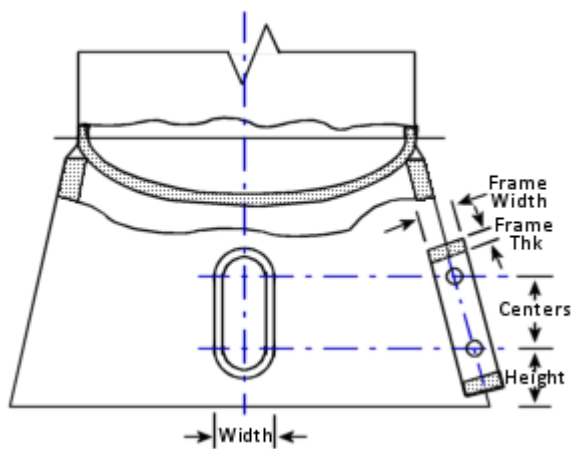
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Top skirt

Total vessel (empty) weight	Fg	12788405.000 N
Head Weight below section 2-2	DeltaFg	780644.562 N
Weight of vessel contents (liquid)	Ff	4428151.500 N
Vessel Gross Weight	Fg + Ff	17216556.000 N
Skirt Mean Diameter	Dz	7233.600 mm
Skirt Analysis Thickness (Corroded)	ez	32.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-59369.422 N
Moment at section 1-1	M1	63164923904.000 N-mm
Applied Moment - Bottom of Skirt	Mb	65919758336.000 N-mm
Weight at Bottom of Skirt	W	17232022.000 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	7220.000 mm
Head Corroded Thickness	eb	60.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	4815795519488.000 mm ⁴
Moment of Inertia	Iyy =	4815795519488.000 mm ⁴
Section Modulus	W =	1331508480.000 mm ³
Section Metal Area	A =	736290.875 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(Ixx, Iyy)}{\text{Area}}}$$

$$= \sqrt{\frac{\min(0.48158E+13, 0.48158E+13)}{736291}}$$

$$= 2557.464 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 2557.464 / 1000.0$$

$$= 2.557$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 0.63165E+11 N-mm



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -0.63165E+11 / 53260E+10 - 17232022 / 736291$$

$$= -35.262 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +.63165E+11 / 53260E+10 - 17232022 / 736291$$

$$= -11.55 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= -59369.422 - 12788405 - 4428152 + 4 \cdot 0.63165E+11 / 7233.6$$

$$= 17638544 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= -59369.422 - 12788405 - 4428152 - 4 \cdot 0.63165E+11 / 7233.6$$

$$= -52190392.0 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (17638544 + 780645 + 4428152) / (\pi \cdot 7220.0 \cdot 60.0) + 4.55 \cdot 7220.0 / (4 \cdot 60.0)$$

$$= 153.68 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.5E+08 + 780645 + 4428152) / (\pi \cdot 7220.0 \cdot 60.0) + 4.55 \cdot 7220.0 / (4 \cdot 60.0)$$

$$= 102.36 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (4428152 + 780645) / (\pi \cdot 7220.0 \cdot 60.0) + 4.55 \cdot 7220.0 / (4 \cdot 60.0)$$

$$= 140.71 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 17638544 / (\pi \cdot 7233.6 \cdot 32.4)$$

$$= 23.96 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -0.5E+08 / (\pi \cdot 7233.6 \cdot 32.4)$$

$$= -70.89 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(7220.0 + 60.0 - 7233.6 + 32.4) / (3580.0 + 60.0)$$

$$= 0.9892$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.989) = 8.438 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(60.0^2 + 32.4^2 + 2 \cdot 60.0 \cdot 32.4 \cdot 0.989)}$$

$$= 46.086 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 46.08 \cdot 17638544.0 = 813.2194 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 46.08 \cdot -52190392.0 = -2.407 \times 10^6 \text{ N-mm} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{60.0}{32.4} \right)^2 \\ &= 0.435 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.435 \cdot 6 \cdot 0.81322 \times 10^9 / (\pi \cdot 7220.0 \cdot 60.0^2) \\ &= 25.96 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.435 \cdot 6 \cdot -0.2 \times 10^9 / (\pi \cdot 7220.0 \cdot 60.0^2) \\ &= -76.80 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.435 \cdot 6 \cdot 0.81322 \times 10^9 / (\pi \cdot 7233.6 \cdot 32.4^2) \\ &= 88.85 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.435 \cdot 6 \cdot -0.2 \times 10^9 / (\pi \cdot 7233.6 \cdot 32.4^2) \\ &= -262.89 \text{ MPa} \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 2.362/284.252 = 0.0083 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 0.0083 = 1.0388 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ MPa (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 153.676 - 25.957 + 0.0 \\ &= 127.72 \text{ MPa} \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 153.676 + 25.957 - 0.0 \\ &= 179.63 \text{ MPa} \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 102.363 - -76.802 + 0.0 \\ &= 179.17 \text{ MPa} \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 102.363 + -76.802 - 0.0 \\ &= 25.56 \text{ MPa} \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 140.715 + 25.957 + 0.0 \\ &= 166.67 \text{ MPa} \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 140.715 - 25.957 - 0.0 \\ &= 114.76 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qitot}]: \\ &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 140.715 + -76.802 + 0.0 \\ &= 63.91 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qotot}]: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 140.715 - -76.802 - 0.0 \\ &= 217.52 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pitot}]: \\ &= \sigma_{3pm} - \sigma_{3pb} \\ &= 23.958 - 88.847 \\ &= -64.89 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3potot}]: \\ &= \sigma_{3pm} + \sigma_{3pb} \\ &= 23.958 + 88.847 \\ &= 112.80 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qitot}]: \\ &= \sigma_{3qm} - \sigma_{3qb} \\ &= -70.889 - -262.887 \\ &= 192.00 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qotot}]: \\ &= \sigma_{3qm} + \sigma_{3qb} \\ &= -70.889 + -262.887 \\ &= -333.78 \text{ MPa} \end{aligned}$$

Allowable stresses:

$$\begin{aligned} \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (153.676/156.4)^2 \\ &= 368.53 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (102.363/156.4)^2 \\ &= 424.54 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (140.715/156.4)^2 \\ &= 384.80 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (140.715/156.4)^2 \\ &= 384.80 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (23.958/156.4)^2 \\ &= 466.75 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\ &= 156.4(3 - (1/1.5)) \cdot (-70.889/156.4)^2 \\ &= 447.78 \text{ MPa} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	127.72	368.53	Pass	46
Stot1po	179.63	368.53	Pass	47
Stot1qi	179.17	424.54	Pass	48
Stot1qo	25.56	424.54	Pass	49
Stot2pi	166.67	384.80	Pass	50
Stot2po	114.76	384.80	Pass	51
Stot2qi	63.91	384.80	Pass	52
Stot2qo	217.52	384.80	Pass	53
Stot3pi	-64.89	466.75	Pass	54
Stot3po	112.80	466.75	Pass	55
Stot3qi	192.00	447.78	Pass	56
Stot3qo	-333.78	447.78	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Packing is installed in the Shop.
 Trays are installed in the Shop.
 Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	30652.270 mm
Center of Gravity of the Packing	25668.088 mm
Center of Gravity of the Liquid	11840.254 mm
Center of Gravity of the Insulation	27793.260 mm
Center of Gravity of the Nozzles	19307.525 mm
Center of Gravity of the Trays	50789.973 mm
Center of Gravity of the Added Weights (Operating)	31507.803 mm
Center of Gravity of the Added Weights (Empty)	31507.803 mm
Center of Gravity of Bare Shell New and Cold	24158.139 mm
Center of Gravity of Bare Shell Corroded	24217.113 mm
Vessel CG in the Operating Condition	22243.266 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	26086.070 mm
Vessel CG in the Test Condition	25879.850 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	52740.00 mm
Total vessel weight (No Liquid)	Twt 12066400.00 N
Impact weight multiplication factor	Imp 1.20
Design lifting weight, DWT = Imp * Twt	14479680.00 N
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	40000.00 mm
Design Reaction force at the tailing lug	5036731.50 N
Design Reaction force at the lifting lug	9442948.00 N
CG distance during field lifting operation	26086.07 mm
CG distance from tailing lug	26086.07 mm
CG distance from the nearer lifting lug/trunnion	13913.93 mm
Compressive Allowable Increase Factor	1.20

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	28.47	8800.00	107.97
Shear	6.58	39590.03	122.50 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	28934.4	0.0	4040172.8	4.9
19200.00	56963547136.0	12.6	99639.1	0.0
29990.01	37050060800.0	8.2	4252206.0	1.7
41990.03	16260086784.0	13.4	2034548.2	1.9
52740.05	432086848.0	0.6	64798.6	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.2637 at elevation 8800.0 mm - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	7266.0000	mm
Skirt Thickness	STHK	36.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.6000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	52.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	6914.0005	mm
Outside Diameter of Basing	DOU, D5	7673.9995	mm
Nominal Diameter of Bolts	BND, d(b0)	76.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	3131.0000	mm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	165.00	MPa
Number of Bolts	RN, nb	52	
Diameter of Bolt Circle	DC, DBC	7469.9995	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	41.4	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	25.0000	mm
Width of Gussets at Top Plate	TWDT	194.0000	mm
Width of Gussets at Base Plate	BWDT	194.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	450.0000	mm
Distance between Gussets	RG, b6	140.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	70.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	64.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	204.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	102.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	12125773.0	N
Operating Weight of Vessel	ROW	17881680.0	N
Earthquake Moment on Basing	EQMOM	90426662912.0	N-mm
Wind Moment on Basing	WIMOM	12349510656.0	N-mm
User Moment at the Base [Earthquake Case]		3451340800.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		3451340800.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.93878E+11/7470.0) - 17322772) / 52$$

$$= 633196.1 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.15801E+11/7470.0) - 12125773) / 52$$

$$= -70543.0 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7470.0) - 17881680) / 52$$

$$= -343878.5 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 3131.0000 mm²
 Area Available in all the Bolts Abss * RN : 162811.9844 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.279	0.418	1.451	2.488	0.442	0.780	6282092.5	23604864.0	62854.9
4	0.348	0.383	1.636	2.337	0.427	0.783	6595125.5	23917896.0	70267.3
6	0.401	0.409	1.766	2.222	0.415	0.784	6840114.5	24162886.0	76628.2
8	0.405	0.403	1.777	2.213	0.414	0.784	6860946.5	24183718.0	77194.6
10	0.404	0.404	1.774	2.215	0.415	0.784	6855731.0	24178502.0	77052.4
12	0.405	0.404	1.776	2.213	0.415	0.784	6859640.5	24182412.0	77159.0

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 6859641 / (6.938 \cdot 7470.0 \cdot 2.213)$$

$$= 119.607 \text{ MPa, Should be } \leq 165.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 52 \cdot 3131.0 / (\pi \cdot 7470.0)$$

$$= 6.938 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 8.262[(2 \cdot 0.4046 \cdot 7470.0 + 380.0) / (2 \cdot 0.4046 \cdot 7470.0)]$$

$$= 8.781 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 204.0, b = 225.651, l/b = 0.904051

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(38128.496, 54842.07) = 54842.07 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 54842.07 / 172.9)} + 1.6$$

$$= 45.225 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 204.0, b = 140.0, b/l = 0.686275

As b/l (0.686) is less than 1, inverting b/l = 1.457.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 516562 / (4 \cdot \pi) [1.3(\ln((2 \cdot 204.0 \cdot \sin(\pi \cdot 102.0 / 204.0)) / (\pi \cdot 58.737))) + 1] - [(0.7 - 0.076) \cdot 516562 / (4 \cdot \pi)]$$

$$= 57842.7656 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$\begin{aligned} &= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}} \\ &= \sqrt{(6 \cdot \text{Abs}(57842.77) / 174.66) + 1.6} \\ &= 46.1777 \text{ mm} \end{aligned}$$

Required Gusset Plate Thickness [tg]:

$$\begin{aligned} &= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca} \\ &= 516562 / (124.11 \cdot 194.0) + 1.6 \\ &= 23.056 \text{ (not less than } 9.525 + 1.6 \text{) mm} \end{aligned}$$

Bolt spacing [m]:

$$\begin{aligned} &= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts} \\ &= \pi \cdot 7470.0 / 52 = 451.302 \text{ mm} \end{aligned}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$\begin{aligned} &= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca} \\ &= 1.76 (516562 \cdot 102.0 / (451.3 \cdot 386.0 \cdot 242))^{2/3} \cdot 3633.0^{1/3} + \text{Ca} \\ &= 31.364 + 1.6 = 32.964 \text{ mm} \end{aligned}$$

Shear Stress in a Single Bolt [tb]:

$$\begin{aligned} &= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 2766866 / (3131.0 \cdot 52) \\ &= 17.0 \text{ MPa. Must be less than } 82.8 \text{ MPa.} \end{aligned}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	45.2247	mm
Actual Basing Thickness as entered by user	52.0000	mm
Required Thickness of Chair Cap	46.1777	mm
Actual Top Ring Thickness as entered by user	64.0000	mm
Required Gusset thickness, + CA	23.0561	mm
Actual Gusset Thickness as entered by user	25.0000	mm
Required Thickness of Skirt for Local Stress	32.9644	mm
Given Thickness of Skirt	36.0000	mm
Required Gusset Height to meet local stress	284.0509	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$\begin{aligned} &= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA}) \\ &= 89.312 \cdot (36.0 - 1.6) \\ &= 3072.06 \text{ N/mm} \end{aligned}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$\begin{aligned} &= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 3072 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 22.391 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta})) \\ &= 516562 / (122.0 + 2 \cdot (334.0 + 64.0)) \\ &= 562.703 \text{ N/mm} \end{aligned}$$

Horizontal Plate Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2) \\ &= 516562 \cdot 102.0 / (122.0 \cdot (398.0) + 0.6667 \cdot (398.0)^2) \\ &= 341.785 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{Wv^2 + Wh^2} \\ &= \sqrt{562.7^2 + 341.78^2} \\ &= 658.371 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 658.37 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 4.799 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 516562 / (2 \cdot 194.0)$$

$$= 1331.345 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 516562 \cdot 102.0 / (2 \cdot 398.0 \cdot 194.0)$$

$$= 341.199 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{1331.34^2 + 341.2^2}$$

$$= 1374.371 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1374.37 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 10.017 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	22.3909	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	10.0172	mm
Required Gusset to Top Plate Double Fillet Weld Size	10.0172	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

cone

Design Internal Pressure, Small End	4.563	MPa
Design Internal Pressure, Large End	4.563	MPa
Temperature for Internal Pressure	130.0	°C
Design External Pressure	0.000	MPa
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material		SA-516 70	+IT
Cone Allowable Stress at Temperature	Ss	156.40	MPa
Cone Allowable Stress At Ambient		175.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	122.0000	mm
Corrosion Allowance for Cone	Cac	6.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	4600.000	mm
Diameter of Large End of Cone	Dl	7100.000	mm
Half Apex Angle for Cone		23.03	degrees
Axial Length of Cone	Lc	2940.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	+IT
Operating Allowable Stress	Ss	156.40	MPa
Ambient Allowable Stress		175.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	72.0000	mm
Corrosion Allowance	Cas	3.0000	mm
Axial Length of Attached Element	Ls	19922.000	mm

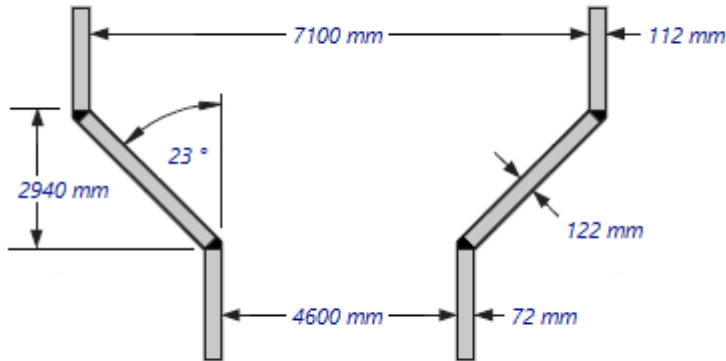
LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	+IT
Operating Allowable Stress	Sl	156.40	MPa
Ambient Allowable Stress		175.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	112.0000	mm
Corrosion Allowance	Cal	6.0000	mm
Axial Length of Attached Element	Ll	22841.336	mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -6.000mm, must be > Tl, 106.000mm, see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.203E+06 MPa at 0 °C
Elastic Modulus of Small Cylinder Material	0.204E+06 MPa at 0 °C
Elastic Modulus of Large Cylinder Material	0.203E+06 MPa at 0 °C
Elastic Modulus of Large End Reinforcement	0.203E+06 MPa at 0 °C
Elastic Modulus of Small End Reinforcement	0.204E+06 MPa at 0 °C

Elastic Modulus Data from ASME Section II Part D at 130 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 130 °C
Elastic Modulus of Small Cylinder Material	0.196E+06 MPa at 130 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 130 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 130 °C
Elastic Modulus of Small End Reinforcement	0.196E+06 MPa at 130 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	-2898841.25	N
Axial Force on Large End of Cone	-4210743.50	N
Moment on Small End of Cone	15745519616.00	N-mm
Moment on Large End of Cone	22450384896.00	N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Maximum Centroid Reinforcement Distance Large End	25.4000	mm
Maximum Centroid Reinforcement Distance Small End	25.4000	mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 202721(116.0/7324.0)$
 $= 1284.3 \text{ MPa}$

Factor of Safety [FS]:
 Since $Fic [234.4] \geq Fy[234.4]$, $FS = 1.667$

Allowable Stress per 4.3.11 Step 6 [Fha]:
 = Fic/FS
 = 234.4/1.667
 = 140.612 MPa

Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:
 $= 2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}}$
 $= 2 \cdot \sqrt{(3556.52 \cdot 116.0/\cos(23.03)) + 1.4 \cdot \sqrt{(2303.26 \cdot 116.0/\cos(23.03))}}$
 = 2093.402 mm

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	80.104	234.600	Pass
Sigma sm-	69.448	234.600	Pass
Sigma thetam+	33.605	234.600	Pass
Sigma thetam-	49.027	234.600	Pass
Sigma sm+ + Sigma	-137.368	469.200	Pass
Sigma sm+ - Sigma	297.575	469.200	Pass
Sigma sm- + Sigma	-117.908	469.200	Pass
Sigma sm- - Sigma	256.803	469.200	Pass
Sigma thetam+ +	-31.636	469.200	Pass
Sigma thetam+ -	98.847	469.200	Pass
Sigma thetam- +	-7.179	469.200	Pass
Sigma thetam- -	105.234	469.200	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	73.108	234.600	Pass
Sigma sm-	63.378	234.600	Pass
Sigma thetam+	27.561	234.600	Pass
Sigma thetam-	43.425	234.600	Pass
Sigma sm+ + Sigma	-108.485	469.200	Pass
Sigma sm+ - Sigma	254.700	469.200	Pass
Sigma sm- + Sigma	-93.068	469.200	Pass
Sigma sm- - Sigma	219.823	469.200	Pass
Sigma thetam+ +	-26.917	469.200	Pass
Sigma thetam+ -	82.038	469.200	Pass
Sigma thetam- +	-3.508	469.200	Pass
Sigma thetam- -	90.359	469.200	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= -4210743.500 / (2 \cdot 3.141 \cdot 3556.52) + 0.22450E+11 / (3.142 \cdot 3556.52^2)$
 = 376.3069 N/mm

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= -4210743.500 / (2 \cdot 3.142 \cdot 3556.52) - 0.22450E+11 / (3.141 \cdot 3556.52^2)$
 = -753.17 N/mm

Thickness Ratio [n]:
 $= tc/tL = 116.0/106.0 = 1.0943$

Curve Fitting Geometry [H and B]:
 $H = \sqrt{0.5 \cdot D/tL} = \sqrt{0.5 \cdot 7113.04/106.0} = 5.7924$
 $B = \tan(23.034) = 0.4252$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.0943) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.083635 C2 = 3.667384 C3 = 0.782362 C4 = -0.227969
 C5 = -0.079526 C6 = 0.053210 C7 = 0.026400 C8 = -0.015506
 C9 = 0.001684 C10 = -0.008491 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(5.792) + C3 \cdot \ln(0.425) + C4 \cdot (\ln(5.792))^2 + C5 \cdot (\ln(0.425))^2 + C6 \cdot \ln(5.792) \cdot \ln(0.425) + C7 \cdot (\ln(5.792))^3 + C8 \cdot (\ln(0.425))^3 + C9 \cdot \ln(5.792) \cdot (\ln(0.425))^2 + C10 \cdot (\ln(5.792))^2 \cdot \ln(0.425) \\ X = 2.025, MsN1 = -\exp(X) = -7.5764$$

Table 4.3.3 [Qn1]:

C1 = -1.956502 C2 = 2.365935 C3 = 0.570438 C4 = -0.104062
 C5 = -0.100300 C6 = 0.159315 C7 = 0.010307 C8 = -0.018730
 C9 = 0.005358 C10 = -0.023129 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(5.792) + C3 \cdot \ln(0.425) + C4 \cdot (\ln(5.792))^2 + C5 \cdot (\ln(0.425))^2 + C6 \cdot \ln(5.792) \cdot \ln(0.425) + C7 \cdot (\ln(5.792))^3 + C8 \cdot (\ln(0.425))^3 + C9 \cdot \ln(5.792) \cdot (\ln(0.425))^2 + C10 \cdot (\ln(5.792))^2 \cdot \ln(0.425) \\ X = 1.2132, Qn = -\exp(X) = -3.3643$$

Table 4.3.4 MsN2:

C1 = -5.708597 C2 = 0.004941 C3 = 0.474625 C4 = -0.213099
 C5 = 2.243076 C6 = 0.000024 C7 = 0.002828 C8 = -0.001820
 C9 = -0.214342 C10 = 0.000045 C11 = -0.106228

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(5.792^2) + C5 \cdot \ln(0.402) + C7 \cdot (\ln(5.792^2))^2 + C9 \cdot (\ln(0.402))^2 + C11 \cdot \ln(5.792^2) \cdot \ln(0.402) = -0.2498 \\ X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(5.792) + C4 \cdot \ln(0.402) + C6 \cdot (\ln(5.792^2))^2 + C8 \cdot (\ln(0.402))^2 + C10 \cdot \ln(5.792^2) \cdot \ln(0.4) = 0.3318 \\ MsN2 = \exp(X1/X2) = \exp(-0.25/0.332) = -0.471$$

Table 4.3.4 QN2:

C1 = -4.906900 C2 = 0.008542 C3 = -0.022868 C4 = -0.197015
 C5 = 2.048010 C6 = -0.000127 C7 = -0.000075 C8 = -0.004217
 C9 = -0.210561 C10 = -0.000970 C11 = 0.005642

Compute using Table 4.3.4 [Qn2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(5.792^2) + C5 \cdot \ln(0.402) + C7 \cdot (\ln(5.792^2))^2 + C9 \cdot (\ln(0.402))^2 + C11 \cdot \ln(5.792^2) \cdot \ln(0.402) = -0.2498 \\ X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(5.792) + C4 \cdot \ln(0.402) + C6 \cdot (\ln(5.792^2))^2 + C8 \cdot (\ln(0.402))^2 + C10 \cdot \ln(5.792^2) \cdot \ln(0.402) = 0.3318 \\ Qn2 = \exp(X3/X4) = \exp(-0.573/0.358) = -0.2017$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = 4.563 \cdot 106.0^2 \cdot -7.576 = -388428.7812 \text{ N}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 376.307 \cdot 106.0 \cdot -0.471 = -18788.0703 \text{ N}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = -753.17 \cdot 106.0 \cdot -0.471 = 37603.9102 \text{ N}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = -388428.781 + -18788.07 = -407216.8750 \text{ N}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = -388428.781 + 37603.91 = -350824.8750 \text{ N}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = 4.563 \cdot 106.0 \cdot -3.364 = -1627.1838 \text{ N/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 376.307 \cdot -0.035 = -75.9047 \text{ N/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = -753.17 \cdot -0.035 = 151.9216 \text{ N/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = -1627.184 + -75.905 = -1703.0884 \text{ N/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = -1627.184 + 151.922 = -1475.2622 \text{ N/mm}$$

Compute [βcy]:

$$= 3 \cdot (1 - \nu^2) / (RL^2 \cdot tL^2) \cdot 0.25$$

$$= (3 \cdot (1 - 0.3^2) / (3556.52^2 \cdot 106.0^2)) \cdot 0.25 = 0.0021 \text{ 1/mm}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = 4.563 \cdot 3556.52/2 + 376.307 = 8490.2871 \text{ N/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = 4.563 \cdot 3556.52/2 + -753.17 = 7360.8105 \text{ N/mm}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q)$$

$$= 4.563 \cdot 3556.52 + 2 \cdot 0.0021 \cdot 3556.52 \cdot (-407216.875 \cdot 0.002 + -1703.088) = 3561.8489 \text{ N/mm}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q)$$

$$= 4.563 \cdot 3556.52 + 2 \cdot 0.0021 \cdot 3556.52 \cdot (-350824.875 \cdot 0.002 + -1475.262) = 5196.4443 \text{ N/mm}$$

Note: Kpc has been set to 1.0

σSm with XL+ [σsm+]:

$$= Ns / tL$$

$$= 8490.287/106.0$$

$$= 80.104 \text{ MPa}$$

σSm with XL- [σsm-]:

$$= Ns / tL$$

$$= 7360.811/106.0$$

$$= 69.448 \text{ MPa}$$

σSb with XL+ [σsb+]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot -407216.875 / (106.0^2 \cdot 1.0)$$

$$= -217.471 \text{ MPa}$$

σSb with XL- [σsb-]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot -350824.875 / (106.0^2 \cdot 1.0)$$

$$= -187.356 \text{ MPa}$$

σThetam with XL+ [σthetam+]:

$$= Ntheta / tL$$

$$= 3561.849/106.0 \text{ MPa}$$

$$= 33.605 \text{ MPa}$$

σThetam with XL- [σthetam-]:

$$= Ntheta / tL$$

$$= 5196.444/106.0$$

$$= 49.027 \text{ MPa}$$

σThetab with XL+ [σthetab+]:

$$= 6 \cdot \nu \cdot Ms / (tL^2 \cdot Kpc)$$



$$= 6 \cdot 0.3 \cdot -407216.875 / (106.0^2 \cdot 1.0)$$

$$= -65.241 \text{ MPa}$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot -350824.875 / (106.0^2 \cdot 1.0)$$

$$= -56.207 \text{ MPa}$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= Msp = Msp = -388428.7812 \text{ N}$$

McsX with XL+ [McsX+]:

$$= MsX = -18788.0703 \text{ N}$$

McsX with XL- [McsX-]:

$$= MsX = 37603.9102 \text{ N}$$

Mcs with XL+ [Mcs+]:

$$= McsP + MsX = -388428.781 + -18788.07$$

$$= -407216.875 \text{ N}$$

Mcs with XL- [Mcs-]:

$$= McsP + MsX = -388428.781 + 37603.91$$

$$= -350824.875 \text{ N}$$

Compute with XL+ [McsX+]:

$$= MsX = -18788.0703 \text{ N}$$

Compute with XL- [McsX-]:

$$= MsX = 37603.9102 \text{ N}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= -1703.088 \cdot \cos(23.034) + 8490.287 \cdot \sin(23.034)$$

$$= 1754.712 \text{ N /mm}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) - Ns \cdot \sin(a)$$

$$= -1475.262 \cdot \cos(23.034) + 7360.811 \cdot \sin(23.034)$$

$$= 1522.441 \text{ N /mm}$$

Compute [Rc]:

$$= RL / \cos(a) = 3556.52 / \cos(23.034) = 3864.6287 \text{ mm}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (3864.629^2 \cdot 116.0^2))^{1/4}$$

$$= 0.002 \text{ 1/mm}$$

Compute with XL +ve [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 8490.287 \cdot \cos(23.034) - -1703.088 \cdot \sin(23.034)$$

$$= 8479.769 \text{ N /mm}$$

Compute with XL -ve [Ncs-]:

$$= Ns \cdot \cos(a) + Q \cdot \sin(a)$$

$$= 7360.811 \cdot \cos(23.034) - -1475.262 \cdot \sin(23.034)$$

$$= 7351.198 \text{ N /mm}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 4.563 \cdot 3556.52 / \cos(23.034) + 2 \cdot 0.002 \cdot 3864.629 \cdot$$

$$(-407216.875 \cdot 0.002 - 1754.712)$$

$$= 3196.758 \text{ N/mm}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (Mcs \cdot \beta_{co} - Qc)$$

$$= 4.563 \cdot 3556.52 / \cos(23.034) + 2 \cdot 0.002 \cdot 3864.629 \cdot$$



$$(-350824.875 \cdot 0.002 - 1522.441) \\ = 5036.892 \text{ N/mm}$$

Note: $K_{cpc} = 1.0$

$$\text{Compute with XL +ve } [\sigma_{sm+}]: \\ = N_{cs}/tC = 8479.769/116.0 = 73.1077 \text{ MPa}$$

$$\text{Compute with XL -ve } [\sigma_{sm-}]: \\ = N_{cs}/tC = 7351.198/116.0 = 63.3778 \text{ MPa}$$

$$\text{Compute with XL +ve } [\sigma_{sb+}] \\ = 6 \cdot M_{cs}/(tC^2 \cdot K_{cpc}) \\ = 6 \cdot -407216.875/(116.0^2 \cdot 1.0) \\ = -181.592 \text{ MPa}$$

$$\text{Compute with XL -ve } [\sigma_{sb-}] \\ = 6 \cdot M_{cs}/(tC^2 \cdot K_{cpc}) \\ = 6 \cdot -350824.875/(116.0^2 \cdot 1.0) \\ = -156.445 \text{ MPa}$$

$$\text{Compute with XL +ve } [\sigma_{\theta_{etam+}}]: \\ = N_{C\theta_{etam}}/tC = 3196.758/116.0 = 27.56 \text{ MPa}$$

$$\text{Compute with XL -ve } [\sigma_{\theta_{etam-}}]: \\ = N_{C\theta_{etam}}/tC = 5036.892/116.0 = 43.43 \text{ MPa}$$

$$\text{Compute with XL +ve } [\sigma_{\theta_{etab+}}]: \\ = 6 \cdot v \cdot M_{cs}/(tC^2 \cdot K_{cpc}) \\ = 6 \cdot 0.3 \cdot -407216.875/(116.0^2 \cdot 1.0) \\ = -54.478 \text{ MPa}$$

$$\text{Compute with XL -ve } [\sigma_{\theta_{etab-}}]: \\ = 6 \cdot v \cdot M_{cs} / (tC^2 \cdot K_{cpc}) \\ = 6 \cdot 0.3 \cdot -350824.875/(116.0^2 \cdot 1.0) \\ = -46.934 \text{ MPa}$$

Calculations for the Small End Cone/Shell Junction

$$\text{Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:} \\ = 0.4 \cdot E(tc/D) \\ = 0.4 \cdot 204100(116.0/4744.0) \\ = 1996.3 \text{ MPa}$$

$$\text{Factor of Safety [FS]:} \\ \text{Since } F_{ic} [234.4] \geq F_y [234.4], FS = 1.667$$

$$\text{Allowable Stress per 4.3.11 Step 6 [Fha]:} \\ = F_{ic}/FS \\ = 234.4/1.667 \\ = 140.612 \text{ MPa}$$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	86.949	234.600	Pass
Sigma sm-	59.568	234.600	Pass
Sigma thetam+	226.032	234.600	Pass
Sigma thetam-	193.828	234.600	Pass
Sigma sm+ + Sigma	300.634	469.200	Pass
Sigma sm- + Sigma	196.412	469.200	Pass
Sigma sm+ - Sigma	-126.736	469.200	Pass
Sigma sm- - Sigma	-77.277	469.200	Pass
Sigma thetam+ +	290.137	469.200	Pass
Sigma thetam- +	234.881	469.200	Pass
Sigma thetam+ -	161.926	469.200	Pass



Sigma thetam- - 152.774 469.200 Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	50.596	234.600	Pass
Sigma sm-	34.439	234.600	Pass
Sigma thetam+	214.237	234.600	Pass
Sigma thetam-	185.908	234.600	Pass
Sigma sm+ + Sigma	126.202	469.200	Pass
Sigma sm- + Sigma	82.857	469.200	Pass
Sigma sm+ - Sigma	-25.010	469.200	Pass
Sigma sm- - Sigma	-13.979	469.200	Pass
Sigma thetam+ +	236.919	469.200	Pass
Sigma thetam- +	200.434	469.200	Pass
Sigma thetam+ -	191.555	469.200	Pass
Sigma thetam- -	171.383	469.200	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= tc/ts = 116.0/69.0 = 1.6812$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(Rs/ts)} = \sqrt{(2303.26/69.0)} = 5.7776$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(23.034) = 0.4252$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= Fs / (2 \cdot \pi \cdot Rs) + Ms / (\pi \cdot Rs^2)$
 $= -2898841.250 / (2 \cdot 3.142 \cdot 2303.26) + 0.15739E+11 / (3.141 \cdot 2303.26^2)$
 $= 744.2574 \text{ N /mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= Fs / (2 \cdot \pi \cdot Rs) - Ms / (\pi \cdot Rs^2) + XS'$
 $= -2898841.250 / (2 \cdot 3.142 \cdot 2303.26) - 0.15739E+11 / (3.141 \cdot 2303.26^2) + 0.0$
 $= -1144.9216 \text{ N /mm}$

From Table 4.3.5 MsN:

C1 = -13.349358 C2 = 2.599535 C3 = 5.670165 C4 = -0.117561
 C5 = -1.340457 C6 = -0.242701 C7 = 0.006510 C8 = 0.136643
 C9 = 0.031180 C10 = 0.002018 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:

$= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) +$
 $C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-13.349 + 2.6 \cdot \ln(5.778^2) + 5.67 \cdot \ln(23.034) +$
 $-0.118 \cdot (\ln(5.778^2))^2 + -1.34 \cdot \ln(23.034) +$
 $-0.243 \cdot \ln(5.778^2) \cdot \ln(23.034) + 0.007 \cdot \ln(5.778^2)^3$
 $0.137 \cdot \ln(23.034)^3 + 0.031 \cdot \ln(5.778^2) \cdot \ln(23.034)^2 +$
 $0.002 \cdot \ln(5.778^2)^2 \cdot \ln(23.034)] = 6.6989$

Compute from Table 4.3.2 [Msp]:

$= p \cdot tL^2 \cdot MsN = 4.563 \cdot 69.0^2 \cdot 6.6989 = 145525.9062 \text{ N}$

From table 4.3.6 MsN:

C1 = 0.004842 C2 = 0.000162 C3 = -0.000986 C4 = 0.122961
 C5 = 0.010282 C6 = -0.000007 C7 = -0.000007 C8 = 0.006624
 C9 = 0.002192 C10 = 0.000148 C11 = 0.199294

Compute from Table 4.3.6 [MsN]:

$= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$
 $(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
 $= (0.00484 + -0.00099 \cdot 5.778 + 0.01028 \cdot 23.034 +$
 $-0.00001 \cdot 5.778^2 + 0.00219 \cdot 23.034^2 +$
 $0.19929 \cdot 5.778 \cdot 23.034) / (1 + 0.00016 \cdot 5.778 +$
 $0.12296 \cdot 23.034 + -0.00001 \cdot 5.778^2 + 0.00662 \cdot 23.034^2 +$
 $0.00015 \cdot 5.778 \cdot 23.034)$



$$= 0.4677$$

Compute from Table 4.3.2 [Msx+]:

$$= X_s \cdot t_s \cdot M_s N = 744.257 \cdot 69.0 \cdot 0.468 = 24018.7422 \text{ N}$$

Compute from Table 4.3.2 [Msx-]:

$$= X_s \cdot t_s \cdot M_s N = -1144.922 \cdot 69.0 \cdot 0.468 = -36949.0117 \text{ N}$$

Compute from Table 4.3.2 [Ms+]:

$$= M_s + M_{sx+} = 145525.906 + 24018.742 = 169544.6406 \text{ N}$$

Compute from Table 4.3.2 [Ms-]:

$$= M_s + M_{sx-} = 145525.906 + -36949.012 = 108576.8984 \text{ N}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.489468 & C2 &= -0.000019 & C3 &= 0.007388 & C4 &= 0.002729 \\ C5 &= -0.002439 & C6 &= 0.000000 & C7 &= -0.000004 & C8 &= -0.000117 \\ C9 &= -0.000094 & C10 &= 0.000001 & C11 &= -0.003924 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned} &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\ &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\ &= (0.48947 + 0.00739 \cdot 5.778^2 + -0.00244 \cdot 23.034 \\ &\quad - 0.5.778^4 + -0.00009 \cdot 23.034^2 + -0.00392 \cdot 5.778^2 \\ &\quad \cdot 23.034) / (1 + -0.00002 \cdot 5.778^2 + 0.00273 \cdot 23.034 + \\ &\quad 0.5.778^4 + -0.00012 \cdot 23.034^2 + 0.000001 \cdot 5.778^2 \cdot 23.034) \\ &= -2.391 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot t_s \cdot Q_n = 4.563 \cdot 69.0 \cdot -2.391 = -752.6733 \text{ N/mm}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.416618 & C2 &= 0.017824 & C3 &= -0.337192 & C4 &= -0.003178 \\ C5 &= -0.115365 & C6 &= 0.002180 & C7 &= 0.000183 & C8 &= -0.015006 \\ C9 &= 0.000788 & C10 &= 0.000084 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.41662 + 0.01782 \cdot \ln(5.778) + -0.33719 \cdot \ln(0.425) + \\ &\quad -0.00318 \cdot \ln(5.778)^2 + -0.11536 \cdot (\ln(0.425))^2 + \\ &\quad 0.00218 \cdot \ln(5.778) \cdot \ln(0.425) + 0.00018 \cdot \ln(5.778)^3 + \\ &\quad -0.01501 \cdot \ln(0.425)^3 + 0.00079 \cdot \ln(5.778) \cdot \ln(0.425)^2 + \\ &\quad 0.00008 \cdot \ln(5.778)^2 \cdot \ln(0.425) \\ &= -0.1832 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= X_{s+} \cdot Q_n = 744.257 \cdot -0.183 = -136.3778 \text{ N/mm}$$

Compute with Xs- [Qx-]:

$$= X_{s-} \cdot Q_n = -1144.922 \cdot -0.183 = 209.7956 \text{ N/mm}$$

Compute with Xs+ [Q+]:

$$= Q_p + Q_{x+} = -752.673 + -136.378 = -889.0511 \text{ N/mm}$$

Compute with Xs- [Q-]:

$$= Q_p + Q_{x-} = -752.673 + 209.796 = -542.8777 \text{ N/mm}$$

Compute angle [βcy]:

$$\begin{aligned} &= (3 \cdot (1 - \nu^2) / (R_s^2 \cdot t_s^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (2303.26^2 \cdot 69.0^2))^{0.25} \\ &= 0.0032 \text{ 1/mm} \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P \cdot R_s / 2 + X_{s+} = 4.563 \cdot 2303.26 / 2 + 744.257 = 5999.0034 \text{ N/mm}$$

Compute with Xs- [Ns-]:

$$= P \cdot R_s / 2 + X_{s-} = 4.563 \cdot 2303.26 / 2 + -1144.922 = 4109.8242 \text{ N/mm}$$

Compute with Xs+ [NTheta+]:

$$= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q)$$



$$= 4.563 \cdot 2303.26 + 2 \cdot 0.003 \cdot 2303.26 \cdot (-169544.641 \cdot 0.003 - -1703.088) = 15594.8545 \text{ N/mm}$$

Compute with Xs- [NTheta-]:

$$= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q) \\ = 4.563 \cdot 2303.26 + 2 \cdot 0.003 \cdot 2303.26 \cdot (-108576.898 \cdot 0.003 - -1475.262) = 13372.9629 \text{ N/mm}$$

Compute Stress with Xs+ [σsm+]:

$$= N_s(+)/t_s = 5999.003/69.0 = 86.9495 \text{ MPa}$$

Compute Stress with Xs- [σsm-]:

$$= N_s(-)/t_s = 4109.824/69.0 = 59.5677 \text{ MPa}$$

Compute stress with Xs+ [σsb+]:

$$= 6 \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot 169544.641/(69.0^2 \cdot 1.0) = 213.685 \text{ MPa}$$

Compute stress with Xs- [σsb-]:

$$= 6 \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot 108576.898/(69.0^2 \cdot 1.0) = 136.8445 \text{ MPa}$$

Compute stress with Xs+ [σthetam+]:

$$= N_{\theta} (+)/t_s = 15594.855/69.0 = 226.0316 \text{ MPa}$$

Compute stress with Xs- [σthetam-]:

$$= N_{\theta} (-)/t_s = 13372.963/69.0 = 193.8275 \text{ MPa}$$

Compute stress with Xs+ [σthetab+]:

$$= 6 \cdot v \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 169544.641/(69.0^2 \cdot 1.0) = 64.1055 \text{ MPa}$$

Compute stress with Xs- [σthetab-]:

$$= 6 \cdot v \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 108576.898/(69.0^2 \cdot 1.0) = 41.0534 \text{ MPa}$$

Small End Cone Without a Flare

Compute with Xs+ [Qc+]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = -889.1 \cdot \cos(23.03) + 5999.0 \cdot \sin(23.03) \\ = 1529.0776 \text{ N/mm}$$

Compute with Xs- [Qc-]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = -542.9 \cdot \cos(23.03) + 4109.8 \cdot \sin(23.03) \\ = 1108.4674 \text{ N/mm}$$

Compute effective radius [Rc]:

$$= R_s/\cos(a) = 2303.26/\cos(23.034) = 2502.7964 \text{ mm}$$

Compute the angle [βco]:

$$= (3 \cdot (1 - v^2)/(R_c^2 \cdot t_c^2))^{0.25} = (3 \cdot (1 - 0.3^2)/(2502.796^2 \cdot 116.0^2))^{0.25} \\ = 0.0024 \text{ 1/mm}$$

Compute with Ms+ [Ncs+]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 5999.003 \cdot \cos(23.03) - 889.051 \cdot \sin(23.03) = 5868.5928 \text{ N/mm}$$

Compute with Ms- [Ncs-]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 4109.824 \cdot \cos(23.03) - 542.878 \cdot \sin(23.03) = 3994.5806 \text{ N/mm}$$

Compute with Ms+ [Nctheta+]:

$$= P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c) \\ = 4.563 \cdot 2303.26/\cos(23.03) + 2 \cdot 0.0024 \cdot 2502.796 \cdot (-169544.641 \cdot 0.0024 + 1529.078) = 24849.3535 \text{ N/mm}$$

Compute with Ms- [Nctheta-]:

$$= P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c) \\ = 4.563 \cdot 2303.26/\cos(23.03) + 2 \cdot 0.0024 \cdot (-108576.898 \cdot 0.0024 + 1108.467) = 21563.502 \text{ N/mm}$$

Compute the stress with Ms+ [σsm+]:

$$= N_{cs}/t_c = 5868.593/116.0 = 50.5956 \text{ MPa}$$



Compute the stress with Ms- [σ_{sm}]:
= $Ncs/tc = 3994.581/116.0 = 34.439$ MPa

Compute the stress with Ms+ [σ_{sb}]:
= $6 \cdot Mcs/(tc^2 \cdot Kcpc) = 6 \cdot 169544.641/(116.0^2 \cdot 1.0) = 75.606$ MPa

Compute the stress with Ms- [σ_{sb}]:
= $6 \cdot Mcs/(tc^2 \cdot Kcpc) = 6 \cdot 108576.898/(116.0^2 \cdot 1.0) = 48.4183$ MPa

Compute stress with Ms+ [$\sigma_{\theta am}$]:
= $Nc\theta eta/tc = 24849.354/116.0 = 214.237$ MPa

Compute stress with Ms- [$\sigma_{\theta am}$]:
= $Nc\theta eta/tc = 21563.502/116.0 = 185.908$ MPa

Compute stress with Ms+ [$\sigma_{\theta ab}$]:
= $6 \cdot v \cdot Mcs/(tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 169544.641/(116.0^2 \cdot 1.0) = 22.6818$ MPa

Compute stress with Ms- [$\sigma_{\theta ab}$]:
= $6 \cdot v \cdot Mcs/(tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 108576.898/(116.0^2 \cdot 1.0) = 14.5255$ MPa

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Input, Nozzle Desc: C

From: 50

Pressure for Reinforcement Calculations	P	4.7162	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Hemispherical Head	D	7148.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

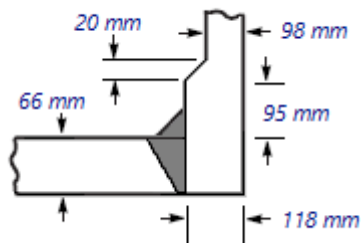
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	98.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	95.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		118.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		135.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4500.0	mm
Centerline Distance to Tangent		5500.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	3.858	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 926.4 \cdot (\exp(4.716/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 15.0395 + 6.0 + 0.0 = 21.0395 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 926.4) / 926.4) - 0.0$
 $= 26.741 - 0.0 = 26.7406 \text{ MPa}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	60.387 mm
Nozzle neck minimum required thickness :	14.340 mm

Minimum nozzle neck thickness 97.993 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	14.4,	Allowable :	147.6 MPa	Passed
Expansion :	0.0,	Allowable :	376.4 MPa	Passed
Occasional :	10.8,	Allowable :	196.3 MPa	Passed
Shear :	2.1,	Allowable :	103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 118.0 - 6.0$
 $= 112.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 98.0 - 6.0$
 $= 92.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7148.0/2 + 6.52$

= 3580.000 mm

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(3580.0 \cdot 60.0)}, 2 \cdot 463.2) \\ &= 463.465 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min((t + t_e)/t_n)^{0.35}, 1) \\ &= \min((60.0 + 0.0)/112.0)^{0.35}, 1) \\ &= \min(0.804, 1) \\ &= 0.804 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(60.0 + 0.0 + 0.804 \cdot \sqrt{(463.2 \cdot 112.0)}, 500.0 + 60.0) \\ &= 243.071 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / LR) \\ &= 60.0 + ((0.0 \cdot 1.0) / 463.465) \\ &= 60.000 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 60.0 \cdot 463.465 \\ &= 27807.908 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A_{2A} + A_{2C} \\ &= 17360.0 + 8302.547 \\ &= 25662.549 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\ &= 27807.908 + 0.944(25662.549 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 52076.527 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 112.0 / \ln[1 + 112.0/463.2] \\ &= 517.180 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/\text{Reff}] \\ &= 60.0 / \ln[1 + 60.0/3580.0] \\ &= 3609.922 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 4.716 \cdot 517.18 \cdot 243.071 \\ &= 592835.2 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 4.716 \cdot 3609.922 \cdot (463.465 + 112.0) / 2$$
$$= 4898301.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$
$$= (4.716 \cdot 3609.922 \cdot 463.2) / 2$$
$$= 3942711.5 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (4898301 + 3942712 + 592835) / 4.716$$
$$= 2000461.5 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$
$$= 234.6 / (2 \cdot 2000462 / 52076.527 - 3609.922 / (2 \cdot 60.0))$$
$$= 5.0 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$
$$= 2 \cdot 156.4 [60.0 / 3609.922]$$
$$= 5.2 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(5.019, 5.199)$$
$$= 5.019 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (592835 + 4898301 + 3942712) / 52076.527$$
$$= 181.169 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$
$$= 4.716 \cdot 3609.922 / (2 \cdot 60.0)$$
$$= 141.9 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 181.169 - 141.877, 141.877)$$
$$= 220.5 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	220.46	MPa
Maximum Allowable Working Pressure	Pmax	5.02	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (463.2 + 112.0) / 463.2$$
$$= 1.242 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$
$$= \pi / 2 \cdot (463.2 + 112.0)$$
$$= 903.522 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 112.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(3942712 \cdot 1.24, 1.5 \cdot 147.6(29774.686 + 0.0), \pi / 4 \cdot 4.7 \cdot 463.2^2 \cdot 1.24^2)$$
$$= 1225423.750 \text{ N}$$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 1225424 / (903.522(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
 $= 52.850 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
 The computed stress ratio is : 0.146

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (13.431 \cdot 1.0) / (98.0 - 6.0 - 0.0)$
 $= 0.146$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 46.5902 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 612.5900 mm

Input Echo, WRC107/537 Item	1,	Description: C	:
Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	7148.000	mm
Vessel Thickness	Tv	66.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	118.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa
Design Internal Pressure	Dp	4.716	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	80968.7	N
Longitudinal Shear (SUS)	(V1) V1	58477.4	N
Circumferential Shear (SUS)	(Vc) V2	58477.4	N
Circumferential Moment (SUS)	(Mc) M1	162007328.0	N-mm

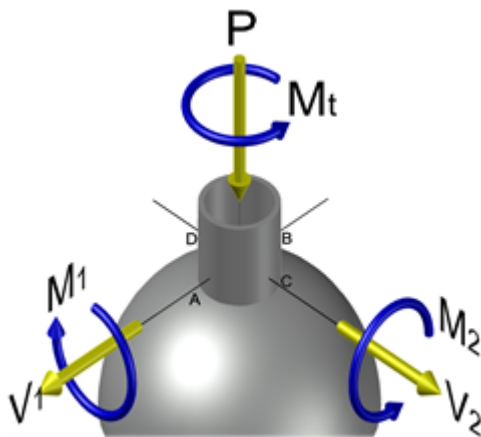


Longitudinal Moment (SUS) (M1) M2 162007328.0 N-mm
 Torsional Moment (SUS) Mt 186309504.0 N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 1150.4 + 2 \cdot 1.65 \cdot \sqrt{(3610.0 (66.0 - 6.0))}$$

$$= 2686.230 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	80968.7	N
Circumferential Shear	(VC) V2	58477.4	N
Longitudinal Shear	(VL) V1	58477.4	N
Circumferential Moment	(MC) M1	162007328.0	N-mm
Longitudinal Moment	(ML) M2	162007328.0	N-mm
Torsional Moment	MT	186309504.0	N-mm

Dimensionless Param: U = 1.24 TAU = 5.00 (4.64) RHO = 0.54

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.03280	(A,B,C,D)
M(x) / P	SP 1	0.03195	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.03584	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.04984	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.03584	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.04984	(A,B,C,D)
N(y) * T / P	SP 1	0.05140	(A,B,C,D)
M(y) / P	SP 1	0.00925	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.05028	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.01630	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.05028	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.01630	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad. Bend. P		-4.3	4.3	-4.3	4.3	-4.3	4.3	-4.3	4.3
Rad. Memb. MC		0.0	0.0	0.0	0.0	-3.5	-3.5	3.5	3.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-28.9	28.9	28.9	-28.9
Rad. Memb. ML		-3.5	-3.5	3.5	3.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-28.9	28.9	28.9	-28.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-37.4	29.0	27.3	-21.9	-37.4	29.0	27.3	-21.9
Tang. Memb. P		-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P		-1.2	1.2	-1.2	1.2	-1.2	1.2	-1.2	1.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-4.9	-4.9	4.9	4.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-9.5	9.5	9.5	-9.5
Tang. Memb. ML		-4.9	-4.9	4.9	4.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-9.5	9.5	9.5	-9.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-16.7	4.7	11.9	-4.5	-16.7	4.7	11.9	-4.5
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear		2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
Str. Int.		37.6	29.2	27.4	21.9	37.5	29.1	27.6	22.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		141.9	141.9	141.9	141.9	141.9	141.9	141.9	141.9
Rad. Pl (SUS)		-4.2	-4.2	2.7	2.7	-4.2	-4.2	2.7	2.7
Rad. Q (SUS)		-33.2	33.2	24.6	-24.6	-33.2	33.2	24.6	-24.6
Long. Pm (SUS)		141.9	141.9	141.9	141.9	141.9	141.9	141.9	141.9
Long. Pl (SUS)		-6.0	-6.0	3.7	3.7	-6.0	-6.0	3.7	3.7
Long. Q (SUS)		-10.7	10.7	8.2	-8.2	-10.7	10.7	8.2	-8.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		141.9	141.9	141.9	141.9	141.9	141.9	141.9	141.9
Pm+Pl (SUS)		137.8	137.8	145.8	145.8	137.8	137.8	145.8	145.8
Pm+Pl+Q (Total)		125.4	171.1	169.3	137.4	125.2	170.9	169.5	137.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	141.88	156.40	Passed
Pm+Pl (SUS)	145.82	234.60	Passed
Pm+Pl+Q (TOTAL)	171.06	469.20	Passed



The $Pm+PI+Q$ allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The $Pm+PI+Q$ allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: A

From: 80

Pressure for Reinforcement Calculations	P	4.5952	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	7100.00	mm
Shell Finished (Minimum) Thickness	t	114.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8100.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

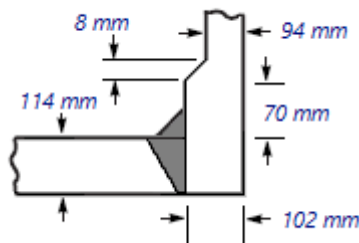
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	94.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	8.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		102.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.701	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4.595/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.8282 + 6.0 + 0.0 = 15.8282 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 88.0 + 621.6) / 621.6) - 0.0$$

$$= 36.798 - 0.0 = 36.7982 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 435.74 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1026.52 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 112.036 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 94.005 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 10.6,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 380.1 MPa	Passed
Occasional	: 7.1,	Allowable	: 196.3 MPa	Passed
Shear	: 2.2,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 102.0 - 6.0$$

$$= 96.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 94.0 - 6.0$$

$$= 88.000 \text{ mm}$$



Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 7112.0/108.0 \\ &= 65.852 \text{ must be less than } 400. \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 7100.0/2 + 6.52 \\ &= 3556.000 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(3556.0 \cdot 108.0)}, 2 \cdot 310.8) \\ &= 619.716 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 108.0, 0.0] + \sqrt{(310.8 \cdot 96.0)} \\ &= 172.733 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 400.0 \\ &= 400.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(108.0 + 0.0) \\ &= 864.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[172.733, 400.0, 864.0] + 108.0 \\ &= 280.733 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 108.0 + ((0.0 \cdot 1.0) / 619.716) \\ &= 108.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 310.8 + 96.0) / (\sqrt{(7112.0 + 108.0) \cdot 108.0})) \\ &= 0.813 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 108.0 \cdot 619.716 \cdot \max((0.813/5)^{0.85}, 1) \\ &= 66929.344 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A_{2A} + A_{2C} \\ &= 17088.002 + 9072.534 \\ &= 26160.535 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 66929.344 + 0.944(26160.535 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 91667.930 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 96.0 / \ln[1 + 96.0/310.8]$$
$$= 356.649 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 108.0 / \ln[1 + 108.0/3556.0]$$
$$= 3609.733 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 4.595 \cdot 356.649 \cdot 280.733$$
$$= 460046.0 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 4.595 \cdot 3609.733 (619.716 + 96.0)$$
$$= 11870850.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 4.595 \cdot 3609.733 \cdot 310.8$$
$$= 5154921.5 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$
$$= (11870850 + 5154922 + 460046.0) / 4.595$$
$$= 3805571.8 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$
$$= 234.6 / (2 \cdot 3805572 / 91667.93 - 3609.733 / 108.0)$$
$$= 4.7 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$
$$= 156.4 [108.0 / 3609.733]$$
$$= 4.7 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(4.729, 4.679)$$
$$= 4.679 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (460045.969 + 11870850 + 5154922) / 91667.93$$
$$= 190.768 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 4.595 \cdot 3609.733 / 108.0$$
$$= 153.6 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 190.768 - 153.587, 153.587)$$
$$= 227.9 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	227.95	MPa
Maximum Allowable Working Pressure	Pmax	4.68	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (310.8 + 96.0) / 310.8 \\ &= 1.309 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (310.8 + 96.0) \\ &= 639.000 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 96.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot ky^2) \\ &= \min(5154922 \cdot 1.31, 1.5 \cdot 147.6(26160.535 + 0.0), \pi/4 \cdot 4.6 \cdot 310.8^2 \cdot 1.31^2) \\ &= 597197.188 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 597197 / (639.0(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0)) \\ &= 36.418 < \text{ or } = \text{ to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 128.036 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -12 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.100

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (8.777 \cdot 1.0) / (94.0 - 6.0 - 0.0) \\ &= 0.100 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-48 °C
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The Drop for this Nozzle is : 23.3850 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.3850 mm

Input Echo, WRC297 Item 1, Description: A :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7100.000	mm
Shell Thickness	114.0000	mm
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		6.0000	mm
Nozzle Diameter		609.600	mm
Nozzle Thickness		102.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	4.60	MPa
Radial Load	P	53979.15	N
Circumferential Shear	Vc	38984.99	N
Longitudinal Shear	Vl	38984.99	N
Circumferential Moment	Mc	0.720E+08	N-mm
Longitudinal Moment	Ml	0.720E+08	N-mm
Torsional Moment	Mt	0.828E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.976) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 813.6 + 2 \cdot 1.65 \cdot \sqrt{(3610.0(114.0 - 6.0))}$
 $= 2874.132 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7220.000 mm
Nozzle Outside Diameter	(d) =	813.600 mm
Vessel Thickness used	(T) =	108.000 mm
Nozzle Thickness used	(t) =	96.000 mm
T / t	=	1.125
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.921$$

Nr/P	=	0.135
Mr/P	=	0.190
M0/P	=	0.068
N0/P	=	0.128
MrD/Mc	=	0.274
NrDL/Mc	=	0.152
M0d/Mc	=	0.115
N0DL/Mc	=	-0.052
MrD/Ml	=	0.202
NrDL/Ml	=	0.122
M0D/Ml	=	0.084
N0DL/Ml	=	0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	149	153	149	153

Outplane Stress Summary	141	157	151	151
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-5	5	-5	5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	74	74	74	74

Inplane Stress Summary	58	87	78	70
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	1	1	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	141	157	151	151
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-5	5	5	-5
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	74	74	74	74

Outplane Stress Summary	67	81	76	70
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-5	5	-5	5
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-12	12	12	-12
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	149	153	149	153
-------------------------	-----	-----	-----	-----

Inplane Stress Summary	129	169	156	146
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	129	169	156	146
------------------------	-----	-----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20

Hoop Stress Summary	14	19	15	21
---------------------	----	----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-7	17	13	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	21	20	15	21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----

(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	15	20	14	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-10	21	16	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	25	21	17	20

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	149	153	149	153	149	153	149	153
Circ. Pl (SUS)	0	0	0	0	-1	-1	1	1
Circ. Q (SUS)	-4	4	2	-2	-17	17	7	-7
Long. Pm (SUS)	74	74	74	74	74	74	74	74
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-14	14	4	-4	-6	6	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	149.0	153.0	149.0	153.0	149.0	153.0	149.0	153.0
Pm+Pl (SUS)	149.0	153.0	149.0	153.0	148.0	152.0	150.0	154.0
Pm+Pl+Q (Total)	145.0	157.0	151.0	151.0	131.0	169.0	157.0	147.0



Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	153.00	156.40	Passed
Pm+Pl (SUS)	154.00	234.60	Passed
Pm+Pl+Q (TOTAL)	169.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	20	15	20	15	20	15	20
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-12	12	4	-4	-15	15	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0
Pm+Pl (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0
Pm+Pl+Q (Total)	22.0	20.0	15.0	20.0	25.0	20.0	16.0	20.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	147.60	Passed
Pm+Pl (SUS)	20.00	221.40	Passed
Pm+Pl+Q (TOTAL)	25.00	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-6

From: 90

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	7100.00	mm
Shell Finished (Minimum) Thickness	t	112.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9800.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

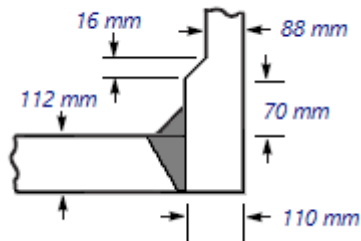
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.8226 + 6.0 + 0.0 = 15.8226 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$$

$$= 34.561 - 0.0 = 34.5607 \text{ MPa}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 6.0$$

$$= 104.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 88.0 - 6.0$$

$$= 82.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/106.0$$

$$= 67.094 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7100.0/2 + 6.52$$

$$= 3556.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3556.0 \cdot 106.0}, 2 \cdot 310.8)$$

$$= 613.951 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 106.0, 0.0] + \sqrt{310.8 \cdot 104.0}$$

$$= 179.787 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$



$$= 400.0$$
$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(106.0 + 0.0)$$
$$= 848.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[179.787, 400.0, 848.0] + 106.0$$
$$= 285.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 106.0 + ((0.0 \cdot 1.0) / 613.951)$$
$$= 106.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (Di + teff) \cdot teff }))$$
$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{ (7112.0 + 106.0) \cdot 106.0 }))$$
$$= 0.830$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)0.85, 1)$$
$$= 106.0 \cdot 613.951 \cdot \max((0.83/5) 0.85, 1)$$
$$= 65078.824 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$
$$= 18304.0 + 9178.496$$
$$= 27482.494 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$
$$= 65078.824 + 0.944(27482.494 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 91064.984 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/Rn]$$
$$= 104.0 / \ln[1 + 104.0/310.8]$$
$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$
$$= 106.0 / \ln[1 + 106.0/3556.0]$$
$$= 3608.742 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$
$$= 4.593 \cdot 360.302 \cdot 285.787$$
$$= 472855.8 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$
$$= 4.593 \cdot 3608.742(613.951 + 104.0)$$
$$= 11897925.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.593 \cdot 3608.742 \cdot 310.8 \\ &= 5150594.5 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (11897925 + 5150595 + 472855.8) / 4.593 \\ &= 3815467.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 234.6 / (2 \cdot 3815467 / 91064.984 - 3608.742 / 106.0) \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 156.4 [106.0 / 3608.742] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.715, 4.594) \\ &= 4.594 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (472855.781 + 11897925 + 5150595) / 91064.984 \\ &= 192.422 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.593 \cdot 3608.742 / 106.0 \\ &= 156.4 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 192.422 - 156.353, 156.353) \\ &= 228.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	228.49	MPa
Maximum Allowable Working Pressure	Pmax	4.59	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (310.8 + 104.0) / 310.8 \\ &= 1.335 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (310.8 + 104.0) \\ &= 651.566 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(5150595 \cdot 1.33, 1.5 \cdot 147.6(27482.494 + 0.0), \pi / 4 \cdot 4.6 \cdot 310.8^2 \cdot 1.33^2) \\ &= 620565.625 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= 620566/(651.566(0.49•7.071 + 0.6•37.0 + 0.49•0.0))
= 37.113 < or = to 156.400 Weld Stress Passed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 230.269 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.107

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (8.772 • 1.0)/(88.0 - 6.0 - 0.0)
= 0.107

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 24.3167 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 536.3169 mm

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Input, Nozzle Desc: MH-5
From: 110

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	7100.00	mm
Shell Finished (Minimum) Thickness	t	112.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16399.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

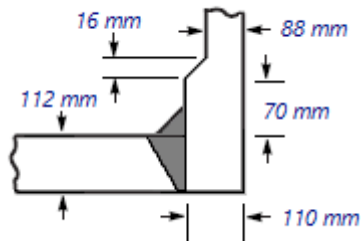
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.8226 + 6.0 + 0.0 = 15.8226 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$$

$$= 34.561 - 0.0 = 34.5607 \text{ MPa}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 6.0$$

$$= 104.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 88.0 - 6.0$$

$$= 82.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/106.0$$

$$= 67.094 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7100.0/2 + 6.52$$

$$= 3556.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3556.0 \cdot 106.0}, 2 \cdot 310.8)$$

$$= 613.951 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 106.0, 0.0] + \sqrt{310.8 \cdot 104.0}$$

$$= 179.787 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$



$$= 400.0$$
$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(106.0 + 0.0)$$
$$= 848.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[179.787, 400.0, 848.0] + 106.0$$
$$= 285.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 106.0 + ((0.0 \cdot 1.0) / 613.951)$$
$$= 106.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (Di + teff) \cdot teff }))$$
$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{ (7112.0 + 106.0) \cdot 106.0 }))$$
$$= 0.830$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)0.85, 1)$$
$$= 106.0 \cdot 613.951 \cdot \max((0.83/5)0.85, 1)$$
$$= 65078.824 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$
$$= 18304.0 + 9178.496$$
$$= 27482.494 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$
$$= 65078.824 + 0.944(27482.494 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 91064.984 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/Rn]$$
$$= 104.0 / \ln[1 + 104.0/310.8]$$
$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$
$$= 106.0 / \ln[1 + 106.0/3556.0]$$
$$= 3608.742 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$
$$= 4.593 \cdot 360.302 \cdot 285.787$$
$$= 472855.8 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$
$$= 4.593 \cdot 3608.742(613.951 + 104.0)$$
$$= 11897925.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.593 \cdot 3608.742 \cdot 310.8 \\ &= 5150594.5 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (11897925 + 5150595 + 472855.8) / 4.593 \\ &= 3815467.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 234.6 / (2 \cdot 3815467 / 91064.984 - 3608.742 / 106.0) \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / (t / Rxs) \\ &= 156.4 [106.0 / 3608.742] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.715, 4.594) \\ &= 4.594 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (472855.781 + 11897925 + 5150595) / 91064.984 \\ &= 192.422 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.593 \cdot 3608.742 / 106.0 \\ &= 156.4 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 192.422 - 156.353, 156.353) \\ &= 228.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	228.49	MPa
Maximum Allowable Working Pressure	Pmax	4.59	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (310.8 + 104.0) / 310.8 \\ &= 1.335 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (310.8 + 104.0) \\ &= 651.566 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(5150595 \cdot 1.33, 1.5 \cdot 147.6(27482.494 + 0.0), \pi / 4 \cdot 4.6 \cdot 310.8^2 \cdot 1.33^2) \\ &= 620565.625 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= $620566 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 37.113 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.107

The computed thickness ratio [sr]:
= $(tr \cdot E) / (tnom - ci - cext)$
= $(8.772 \cdot 1.0) / (88.0 - 6.0 - 0.0)$
= 0.107

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 24.3167 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 536.3169 mm

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Input, Nozzle Desc: MH-4

From: 130

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	7100.00	mm
Shell Finished (Minimum) Thickness	t	112.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		20599.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

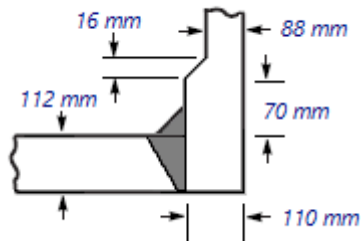
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.8226 + 6.0 + 0.0 = 15.8226 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$$

$$= 34.561 - 0.0 = 34.5607 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 930.94 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1028.75 mm

This nozzle or its diameter limit intersects weld seam # [8] on this element. The distance between the nozzle and seam is, 336.80 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1028.75 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 6.0$$

$$= 104.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 88.0 - 6.0$$

$$= 82.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/106.0$$

$$= 67.094 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7100.0/2 + 6.52$$

$$= 3556.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3556.0 \cdot 106.0}, 2 \cdot 310.8)$$

$$= 613.951 \text{ mm}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 106.0, 0.0] + \sqrt{(310.8 \cdot 104.0)} \\ &= 179.787 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 400.0 \\ &= 400.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(106.0 + 0.0) \\ &= 848.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[179.787, 400.0, 848.0] + 106.0 \\ &= 285.787 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 106.0 + ((0.0 \cdot 1.0) / 613.951) \\ &= 106.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{(7112.0 + 106.0) \cdot 106.0})) \\ &= 0.830 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 106.0 \cdot 613.951 \cdot \max((0.83/5)^{0.85}, 1) \\ &= 65078.824 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A_{2A} + A_{2C} \\ &= 18304.0 + 9178.496 \\ &= 27482.494 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 65078.824 + 0.944(27482.494 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 91064.984 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 104.0 / \ln[1 + 104.0/310.8] \\ &= 360.302 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 106.0 / \ln[1 + 106.0/3556.0] \\ &= 3608.742 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$



$$= 4.593 \cdot 360.302 \cdot 285.787$$
$$= 472855.8 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$
$$= 4.593 \cdot 3608.742(613.951 + 104.0)$$
$$= 11897925.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 4.593 \cdot 3608.742 \cdot 310.8$$
$$= 5150594.5 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (11897925 + 5150595 + 472855.8)/4.593$$
$$= 3815467.2 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 234.6 / (2 \cdot 3815467/91064.984 - 3608.742/106.0)$$
$$= 4.7 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 156.4[106.0/3608.742]$$
$$= 4.6 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(4.715, 4.594)$$
$$= 4.594 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (472855.781 + 11897925 + 5150595)/91064.984$$
$$= 192.422 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$
$$= 4.593 \cdot 3608.742/106.0$$
$$= 156.4 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$
$$= \max(2 \cdot 192.422 - 156.353, 156.353)$$
$$= 228.5 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	228.49	MPa
Maximum Allowable Working Pressure	Pmax	4.59	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (310.8 + 104.0)/310.8$$
$$= 1.335 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (310.8 + 104.0)$$
$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$



Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(5150595 \cdot 1.33, 1.5 \cdot 147.6(27482.494 + 0.0), \pi/4 \cdot 4.6 \cdot 310.8^2 \cdot 1.33^2)$
= 620565.625 N

Weld Stress Value [τ]:
= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $620566 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 37.113 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.107

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(8.772 \cdot 1.0) / (88.0 - 6.0 - 0.0)$
= 0.107

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 24.3167 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 536.3169 mm

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Input, Nozzle Desc: E

From: 140

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cone at Nozzle Location	D	6271.96	mm
Cone Half Apex Angle	Alpha	23.03	Degrees
Shell Finished (Minimum) Thickness	t	136.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		22623.78	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

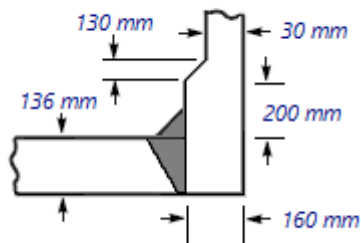
Material		SA-765 I	+IT
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	123.60	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	130.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		66.9663	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	42.000	in.
Actual Thickness Used in Calculation	1.181	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1078.8 \cdot (\exp(4.593/(123.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 20.4194 + 6.0 + 0.0 = 26.4194 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 123.6 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 1078.8) / 1078.8) - 0.0$$

$$= 5.381 - 0.0 = 5.3806 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 770.01 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1332.56 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 107.766 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 29.997 mm , is greater than required thickness 14.340 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[123.6/156.4, 1]$$

$$= 0.790$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 160.0 - 6.0$$

$$= 154.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 30.0 - 6.0$$

$$= 24.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6284.995/130.0$$

$$= 48.346 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6271.956/2 + 6.52$$

$$= 3142.498 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(3142.498 \cdot 130.0)}, 2 \cdot 539.4) \\ &= 639.159 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 130.0, 0.0] + \sqrt{(539.4 \cdot 154.0)} \\ &= 288.215 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 800.0 \\ &= 800.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(130.0 + 0.0) \\ &= 1040.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t / \cos(\alpha) \text{ for set-in nozzles} \\ &= \min[288.215, 800.0, 1040.0] + 130.0 / \cos(23.034) \\ &= 429.477 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot \text{frp}) / LR) \\ &= 130.0 + ((0.0 \cdot 1.0) / 639.159) \\ &= 130.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))) \\ &= \min(12, (2 \cdot 539.4 + 154.0) / (\sqrt{(6285.0 + 130.0) \cdot 130.0})) \\ &= 1.350 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 52554.375 + 9694.132 \\ &= 62248.508 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 154.0 / \ln[1 + 154.0 / 539.4] \\ &= 613.180 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 4.593 \cdot 613.18 \cdot 429.477 \\ &= 1209339.4 \text{ N} \end{aligned}$$

Effective Radius for Nozzles in Cones (perp to axis) [Rnc]:

$$\begin{aligned} &= R_n / \cos(\alpha) \\ &= 539.4 / \cos(23.034) \\ &= 586.129 \text{ mm} \end{aligned}$$

Area Contributed by the Vessel Wall, Cone, perpendicular [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda / 5)^{0.85}, 1) \\ &= 130.0 \cdot 639.159 \cdot \max((1.35/5)^{0.85}, 1) \\ &= 83090.719 \text{ mm}^2 \end{aligned}$$



The total area contributed by A1 through A5 [AT]:
= $A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
= $83090.719 + 0.79(62248.508 + 0.0) + 50.0 + 0.0 + 0.0 + 1.0(0.0)$
= 132334.547 mm^2

Shell Radius for Force Calculation [Rxs]:
= $\text{teff} / \ln[1 + \text{teff} \cdot \cos[\alpha] / (\text{reff} + \text{Lc} \cdot \sin[\alpha])]$
= $130.0 / \ln[1 + 130.0 \cdot \cos[23.034] / (3142.498 + 1392.63 \cdot \sin[23.034])]$
= 4071.500 mm

Force acting on the Nozzle on the Cone [fS]:
= $P / \cos(\alpha) [\text{Reff} + \text{Rnc} \cdot \sin[\alpha] + \text{Lt} \cdot \sin[\alpha/2] (\text{Lt})]$
= $4.593 / 0.92 [4071.5 + 586.13 \cdot 0.391 + 806.5 \cdot 0.39/2] (806.5)$
= 14204800.0 N

Discontinuity Force from Pressure [fY]:
= $P / \cos(\alpha) (\text{Reff} + \text{Rnc} \cdot \sin[\alpha/2]) \text{Rnc}$
= $4.593 / 0.92 \cdot (4071.5 + 586.129 \cdot 0.39/2) \cdot 586.129$
= 9526566.0 N

Cone Length Term [Lc]:
= $\text{Lt} + \text{Rnc}$
= $806.501 + 586.129 + 0.0$
= 1392.630 mm

Area Resisting Internal Pressure [Ap]:
= $(fN + fS + fY) / P$
= $(1209339 + 14204800 + 9526566) / 4.593$
= 5431106.0 mm^2

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $\text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$
= $234.6 / (2 \cdot 5431106 / 132334.547 - 4071.5 / 130.0)$
= 4.6 MPa

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S[t / \text{Rxs}]$
= $156.4 [130.0 / 4071.5]$
= 5.0 MPa

Maximum Allowable Working Pressure [Pmax]:
= $\min(\text{Pmax1}, \text{Pmax2})$
= $\min(4.622, 4.994)$
= 4.622 MPa

Average Primary Membrane Stress [σ_{Avg}]:
= $(fN + fS + fY) / \text{AT}$
= $(1209339 + 14204800 + 9526566) / 132334.547$
= 188.483 MPa

General Primary Membrane Stress [σ_{Circ}]:
= $P \cdot \text{Rxs} / \text{teff}$
= $4.593 \cdot 4071.5 / 130.0$
= 143.8 MPa

Maximum Local Primary Membrane Stress [PL]:
= $\max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$
= $\max(2 \cdot 188.483 - 143.836, 143.836)$
= 233.1 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	233.13	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
= $(\text{Rnc} + \text{tn}) / \text{Rnc}$



= (586.129 + 154.0)/586.129
= 1.263 For set-in Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

= $\pi/2 \cdot \sqrt{((R_{nc}+T_n)^2 + (R_n+T_n)^2)/2}$
= $\pi/2 \cdot \sqrt{((586.129+154.0)^2 + (539.4+154.0)^2)/2}$
= 1126.489 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be >= min(0.7 * 154.000, 6.000) Passed

Weld Load Value [fwelds]:

= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot R_n^2 \cdot k_y^2$)
= min(9526566•1.26, 1.5•123.6(62248.508+0.0), $\pi/4 \cdot 4.6 \cdot 539.4^2 \cdot 1.26^2$)
= 1673248.750 N

Weld Stress Value [τ]:

= fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
= 1673249/(1126.489(0.49•7.071 + 0.6•10.0 + 0.49•0.0))
= 156.949 > 156.400 Weld Stress Failed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -58 °C

The Drop for this Nozzle is : 77.6199 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1013.6199 mm

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Input, Nozzle Desc: P

From: 150

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		24739.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

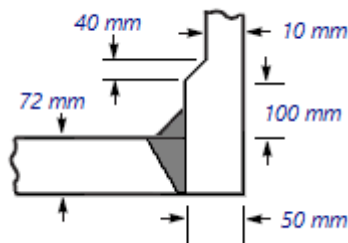
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: P

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.8976 + 3.0 + 0.0 = 3.8976 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 32.52 - 0.0 = 32.5196 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.520 mm
 Min. required shell thickness, internal p + c : 71.634 mm

Nozzle neck minimum required thickness : 7.520 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 7.520 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	8.3,	Allowable :	147.6 MPa	Passed
Expansion :	0.0,	Allowable :	382.5 MPa	Passed
Occasional :	8.3,	Allowable :	196.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 3.0$$

$$= 47.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.26$$



$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 28.4) \\ &= 56.800 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 69.0, 0.0] + \sqrt{28.4 \cdot 47.0} \\ &= 36.535 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 250.0 \\ &= 250.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(69.0 + 0.0) \\ &= 552.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[36.535, 250.0, 552.0] + 69.0 \\ &= 105.535 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot \text{frp}) / LR) \\ &= 69.0 + ((0.0 \cdot 1.0) / 56.8) \\ &= 69.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}})) \\ &= \min(12, (2 \cdot 28.4 + 47.0) / (\sqrt{(4606.0 + 69.0) \cdot 69.0})) \\ &= 0.183 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 69.0 \cdot 56.8 \cdot \max((0.183/5)^{0.85}, 1) \\ &= 3919.200 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 47.0 \cdot 105.535 \\ &= 4960.142 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 3919.2 + 0.944(4960.142 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 8650.255 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 47.0 / \ln[1 + 47.0/28.4] \\ &= 48.135 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= \text{teff} / \ln[1 + \text{teff}/\text{Reff}] \\ &= 69.0 / \ln[1 + 69.0/2303.0] \\ &= 2337.334 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:



$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.593 \cdot 48.135 \cdot 105.535 \\ &= 23328.1 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.593 \cdot 2337.334 (56.8 + 47.0) \\ &= 1114137.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.593 \cdot 2337.334 \cdot 28.4 \\ &= 304831.3 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (1114137 + 304831.3 + 23328.1) / 4.593 \\ &= 314075.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 234.6 / (2 \cdot 314075.531 / 8650.255 - 2337.334 / 69.0) \\ &= 6.1 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / R_{xs} \\ &= 156.4 [69.0 / 2337.334] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.055, 4.617) \\ &= 4.617 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (23328.064 + 1114137 + 304831.312) / 8650.255 \\ &= 166.749 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.593 \cdot 2337.334 / 69.0 \\ &= 155.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 166.749 - 155.571, 155.571) \\ &= 177.9 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	177.93	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (28.4 + 47.0) / 28.4 \\ &= 2.655 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi / 2 \cdot (R_n + t_n) \\ &= \pi / 2 \cdot (28.4 + 47.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 47.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 = $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(304831 \cdot 2.65, 1.5 \cdot 147.6(4960.142 + 0.0), \pi/4 \cdot 4.6 \cdot 28.4^2 \cdot 2.65^2)$
 = 20504.680 N

Weld Stress Value [τ]:
 = $\text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 = $20504.68 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 10.0 + 0.49 \cdot 0.0))$
 = 18.293 $\leq$ or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 1.2362 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 323.2362 mm

Input Echo, WRC297 Item 1,	Description: P	:
Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4600.000	mm
Shell Thickness	72.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	MPa
Vessel Hot S.I. Allowable	Smh 156.40	MPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 165.00	MPa
Nozzle Hot S.I. Allowable	SNmh 147.60	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.59	MPa
Radial Load	P	0.00	N
Circumferential Shear	Vc	0.00	N
Longitudinal Shear	Vl	0.00	N
Circumferential Moment	Mc	0.00	N-mm
Longitudinal Moment	Ml	0.00	N-mm
Torsional Moment	Mt	0.00	N-mm

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No



*Warning - The ratio, Dn/Tv, (2.094) is < 5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (3.016) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 150.8 + 2 \cdot 1.65 \cdot \sqrt{(2337.5(72.0 - 3.0))}$
 $= 1476.100 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 4675.000 mm
 Nozzle Outside Diameter (d) = 150.800 mm
 Vessel Thickness used (T) = 69.000 mm
 Nozzle Thickness used (t) = 47.000 mm
 T / t = 1.468
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.266$$

Nr/P = 0.240
 Mr/P = 0.318
 M0/P = 0.148
 N0/P = 0.191
 MrD/Mc = 0.265
 NrDL/Mc = 0.169
 M0d/Mc = 0.149
 N0DL/Mc = -0.126
 MrD/Ml = 0.259
 NrDL/Ml = 0.172
 M0D/Ml = 0.145
 N0DL/Ml = -0.084

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	151	155	151	155
Outplane Stress Summary	151	155	151	155

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	75	75	75	75
Inplane Stress Summary	75	75	75	75

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress		151	155	151	155

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)		0	0	0	0
Outplane Bending (P)		0	0	0	0
Outplane Membrane (Mc)		0	0	0	0
Outplane Bending (Mc)		0	0	0	0
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		75	75	75	75
Outplane Stress Summary		75	75	75	75

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		0	0	0	0
Inplane Membrane (Mc)		0	0	0	0
Inplane Bending (Mc)		0	0	0	0
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		151	155	151	155
Inplane Stress Summary		151	155	151	155

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		151	155	151	155

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)		0	0	0	0
Hoop Bending (P)		0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	7	1	7

Hoop Stress Summary	1	7	1	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1	7	1	7
------------------------	---	---	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	7	1	7

Hoop Stress Summary	1	7	1	7
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	1	7	1	7

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	151	155	151	155	151	155	151	155
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	75	75	75	75	75	75	75	75
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	151.0	155.0	151.0	155.0	151.0	155.0	151.0	155.0
Pm+Pl (SUS)	151.0	155.0	151.0	155.0	151.0	155.0	151.0	155.0
Pm+Pl+Q (Total)	151.0	155.0	151.0	155.0	151.0	155.0	151.0	155.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	155.00	156.40	Passed
Pm+Pl (SUS)	155.00	234.60	Passed
Pm+Pl+Q (TOTAL)	155.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	7	1	7	1	7	1	7
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0



Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	7.0	1.0	7.0	1.0	7.0	1.0	7.0
Pm+P1 (SUS)	1.0	7.0	1.0	7.0	1.0	7.0	1.0	7.0
Pm+P1+Q (Total)	1.0	7.0	1.0	7.0	1.0	7.0	1.0	7.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	147.60	Passed
Pm+P1 (SUS)	7.00	221.40	Passed
Pm+P1+Q (TOTAL)	7.00	442.80	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-3

From: 170

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		32089.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

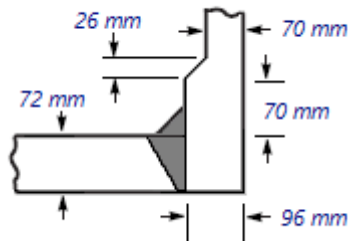
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		96.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 9.7278 + 3.0 + 0.0 = 12.7278 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$$

$$= 29.069 - 0.0 = 29.0687 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 565.56 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 799.43 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 96.0 - 3.0$$

$$= 93.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 70.0 - 3.0$$

$$= 67.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.26$$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 307.8)$$

$$= 398.631 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 69.0, 0.0] + \sqrt{307.8 \cdot 93.0}$$



= 169.190 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 400.0
= 400.000 mm

Thickness Limit Candidate [LH3]:

= $8(t + t_e)$
= $8(69.0 + 0.0)$
= 552.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[169.19, 400.0, 552.0] + 69.0$
= 238.190 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
= $69.0 + ((0.0 \cdot 1.0) / 398.631)$
= 69.000 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 307.8 + 93.0) / (\sqrt{(4606.0 + 69.0) \cdot 69.0}))$
= 1.248

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $69.0 \cdot 398.631 \cdot \max((1.248/5)^{0.85}, 1)$
= 27505.570 mm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= A2A + A2C
= 12927.001 + 6983.758
= 19910.760 mm²

Area Contributed by the Outside Fillet Weld [A41]:

= $0.5 \cdot Leg^2$
= $0.5 \cdot 10.0^2$
= 50.000 mm²

The total area contributed by A1 through A43 [AT]:

= $A1 + frn(A2 + A3) + A41 + A42 + A43$
= $27505.57 + 0.944(19910.76 + 0.0) + 50.0 + 0.0 + 0.0$
= 46346.031 mm²

Nozzle Radius for Force Calculation [Rxn]:

= $tn / \ln[1 + tn/Rn]$
= $93.0 / \ln[1 + 93.0/307.8]$
= 352.256 mm

Shell Radius for Force Calculation [Rxs]:

= $teff / \ln[1 + teff/Reff]$
= $69.0 / \ln[1 + 69.0/2303.0]$
= 2337.334 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
= $1.5 \cdot 156.4 \cdot 1.0$
= 234.6 MPa

Force acting on the Nozzle [fN]:

= $P \cdot Rxn \cdot LH$
= $4.593 \cdot 352.256 \cdot 238.19$
= 385303.9 N

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.593 \cdot 2337.334(398.631 + 93.0) \\ &= 5276925.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.593 \cdot 2337.334 \cdot 307.8 \\ &= 3303770.8 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (5276925 + 3303771 + 385303.9)/4.593 \\ &= 1952442.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 1952443/46346.031 - 2337.334/69.0) \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[69.0/2337.334] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.657, 4.617) \\ &= 4.617 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (385303.938 + 5276925 + 3303771)/46346.031 \\ &= 193.474 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.593 \cdot 2337.334/69.0 \\ &= 155.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 193.474 - 155.571, 155.571) \\ &= 231.4 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	231.38	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (307.8 + 93.0)/307.8 \\ &= 1.302 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (307.8 + 93.0) \\ &= 629.575 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(3303771 \cdot 1.3, 1.5 \cdot 147.6(19910.76 + 0.0), \pi/4 \cdot 4.6 \cdot 307.8^2 \cdot 1.3^2) \end{aligned}$$



= 579382.938 N

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
= $579383 / (629.575(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 35.861 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:

= $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
= $(8.688 \cdot 1.0) / (70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 35.1911 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 507.1911 mm

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Input, Nozzle Desc: MH-2

From: 200

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		40589.97	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

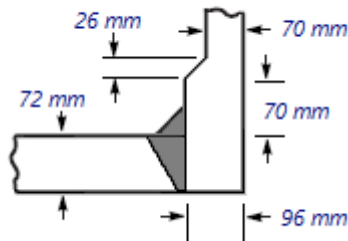
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		96.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 9.7278 + 3.0 + 0.0 = 12.7278 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$$

$$= 29.069 - 0.0 = 29.0687 \text{ MPa}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 96.0 - 3.0$$

$$= 93.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 70.0 - 3.0$$

$$= 67.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.26$$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 307.8)$$

$$= 398.631 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 69.0, 0.0] + \sqrt{307.8 \cdot 93.0}$$

$$= 169.190 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$



$$= 400.0$$
$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(69.0 + 0.0)$$
$$= 552.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$
$$= \min[169.19, 400.0, 552.0] + 69.0$$
$$= 238.190 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 69.0 + ((0.0 \cdot 1.0) / 398.631)$$
$$= 69.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$
$$= \min(12, (2 \cdot 307.8 + 93.0) / (\sqrt{(4606.0 + 69.0) \cdot 69.0}))$$
$$= 1.248$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 69.0 \cdot 398.631 \cdot \max((1.248/5)^{0.85}, 1)$$
$$= 27505.570 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$
$$= 12927.001 + 6983.758$$
$$= 19910.760 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$
$$= 27505.57 + 0.944(19910.76 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 46346.031 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 93.0 / \ln[1 + 93.0/307.8]$$
$$= 352.256 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$
$$= 69.0 / \ln[1 + 69.0/2303.0]$$
$$= 2337.334 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$
$$= 4.593 \cdot 352.256 \cdot 238.19$$
$$= 385303.9 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$
$$= 4.593 \cdot 2337.334(398.631 + 93.0)$$
$$= 5276925.0 \text{ N}$$



Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.593 \cdot 2337.334 \cdot 307.8 \\ &= 3303770.8 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (5276925 + 3303771 + 385303.9) / 4.593 \\ &= 1952442.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 234.6 / (2 \cdot 1952443 / 46346.031 - 2337.334 / 69.0) \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 156.4 [69.0 / 2337.334] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.657, 4.617) \\ &= 4.617 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (385303.938 + 5276925 + 3303771) / 46346.031 \\ &= 193.474 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.593 \cdot 2337.334 / 69.0 \\ &= 155.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 193.474 - 155.571, 155.571) \\ &= 231.4 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	231.38	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (307.8 + 93.0) / 307.8 \\ &= 1.302 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (307.8 + 93.0) \\ &= 629.575 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(3303771 \cdot 1.3, 1.5 \cdot 147.6(19910.76 + 0.0), \pi / 4 \cdot 4.6 \cdot 307.8^2 \cdot 1.3^2) \\ &= 579382.938 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= $579383 / (629.575(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 35.861 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(8.688 \cdot 1.0) / (70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 35.1911 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 507.1911 mm

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Input, Nozzle Desc: D

From: 200

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		40089.97	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

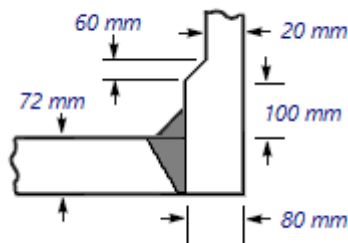
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	18.000	in.
Actual Thickness Used in Calculation	0.787	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 463.2 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 7.3195 + 3.0 + 0.0 = 10.3195 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 17.0 + 463.2) / 463.2) - 0.0$$

$$= 10.455 - 0.0 = 10.455 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 71.634 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 19.990 mm, is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51.0,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 339.8 MPa	Passed
Occasional	: 30.2,	Allowable	: 196.3 MPa	Passed
Shear	: 11.5,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 3.0$$

$$= 77.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.0 - 3.0$$

$$= 17.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$



$$= 4600.0/2 + 3.26$$
$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$
$$= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 231.6)$$
$$= 398.631 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$
$$= \min[1.5 \cdot 69.0, 0.0] + \sqrt{231.6 \cdot 77.0}$$
$$= 133.541 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 400.0$$
$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(69.0 + 0.0)$$
$$= 552.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[133.541, 400.0, 552.0] + 69.0$$
$$= 202.541 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 69.0 + ((0.0 \cdot 1.0) / 398.631)$$
$$= 69.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 231.6 + 77.0) / (\sqrt{(4606.0 + 69.0) \cdot 69.0}))$$
$$= 0.951$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 69.0 \cdot 398.631 \cdot \max((0.951/5)^{0.85}, 1)$$
$$= 27505.570 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 13013.0 + 2020.158$$
$$= 15033.157 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 20.0^2$$
$$= 200.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 27505.57 + 0.944(15033.157 + 0.0) + 200.0 + 0.0 + 0.0$$
$$= 41892.871 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 77.0 / \ln[1 + 77.0/231.6]$$
$$= 268.261 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 69.0 / \ln[1 + 69.0/2303.0]$$
$$= 2337.334 \text{ mm}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.593 \cdot 268.261 \cdot 202.541 \\ &= 249511.5 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.593 \cdot 2337.334 (398.631 + 77.0) \\ &= 5105189.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.593 \cdot 2337.334 \cdot 231.6 \\ &= 2485878.2 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_N + f_Y + f_N) / P \\ &= (5105189 + 2485878 + 249511.5) / 4.593 \\ &= 1707370.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 234.6 / (2 \cdot 1707370 / 41892.871 - 2337.334 / 69.0) \\ &= 4.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 156.4 [69.0 / 2337.334] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(4.925, 4.617) \\ &= 4.617 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (249511.516 + 5105189 + 2485878) / 41892.871 \\ &= 187.174 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.593 \cdot 2337.334 / 69.0 \\ &= 155.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 187.174 - 155.571, 155.571) \\ &= 218.8 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	218.78	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (231.6 + 77.0) / 231.6 \\ &= 1.332 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (R_n + t_n) \\ &= \pi / 2 \cdot (231.6 + 77.0) \end{aligned}$$



= 484.748 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 14.142, 0.000, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 14.142, must be >= min(0.7 * 77.000, 6.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
 = min(2485878•1.33, 1.5•147.6(15033.157+0.0), $\pi/4 \cdot 4.6 \cdot 231.6^2 \cdot 1.33^2$)
 = 343480.562 N

Weld Stress Value [τ]:
 = fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
 = 343480.562/(484.748(0.49•14.142 + 0.6•20.0 + 0.49•0.0))
 = 37.435 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -40 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 12 °C
 Temperature reduction per Graph 3.13 : 53 °C
 The computed stress ratio is : 0.385

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (6.537 • 1.0)/(20.0 - 3.0 - 0.0)
 = 0.385

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -40 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 20.7970 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 492.7971 mm

Input Echo, WRC297 Item 1, Description: D :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	4600.000	mm	
Shell Thickness	72.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	457.200	mm	
Nozzle Thickness	80.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	4.59	MPa
Radial Load	P	40500.02	N
Circumferential Shear	Vc	29250.00	N
Longitudinal Shear	Vl	29250.00	N
Circumferential Moment	Mc	0.405E+08	N-mm
Longitudinal Moment	Ml	0.405E+08	N-mm
Torsional Moment	Mt	0.466E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.715) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 617.2 + 2 \cdot 1.65 \cdot \sqrt{(2337.5(72.0 - 3.0))}$
 $= 1942.500 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4675.000 mm
Nozzle Outside Diameter	(d) =	617.200 mm
Vessel Thickness used	(T) =	69.000 mm
Nozzle Thickness used	(t) =	77.000 mm
T / t	=	0.896
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.087$$

Nr/P	=	0.118
Mr/P	=	0.186
M0/P	=	0.061
N0/P	=	0.120
MrD/Mc	=	0.283
NrDL/Mc	=	0.135
M0d/Mc	=	0.104
N0DL/Mc	=	0.038
MrD/Ml	=	0.188
NrDL/Ml	=	0.104
M0D/Ml	=	0.068
N0DL/Ml	=	0.122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-3	3	-3	3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-5	5	5	-5
Normal Pressure Stress	151	155	151	155
Outplane Stress Summary	139	161	154	153

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	75	75	75	75
Inplane Stress Summary	48	98	81	69

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	139	161	154	153

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-3	3	-3	3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	75	75	75	75
Outplane Stress Summary	62	85	80	69

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-23	23	23	-23
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	151	155	151	155
Inplane Stress Summary	115	185	165	142

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	115	185	165	142

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	11	17	14	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-9	18	13	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	21	19	15	20

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	12	18	13	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-15	23	18	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	27	23	18	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	151	155	151	155	151	155	151	155
Circ. Pl (SUS)	-2	-2	0	0	-1	-1	1	1
Circ. Q (SUS)	-8	8	2	-2	-32	32	14	-14
Long. Pm (SUS)	75	75	75	75	75	75	75	75
Long. Pl (SUS)	-1	-1	1	1	-1	-1	-1	-1
Long. Q (SUS)	-24	24	6	-6	-11	11	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	151.0	155.0	151.0	155.0	151.0	155.0	151.0	155.0
Pm+Pl (SUS)	149.0	153.0	151.0	155.0	150.0	154.0	152.0	156.0
Pm+Pl+Q (Total)	141.0	161.0	153.0	153.0	118.0	186.0	166.0	142.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	155.00	156.40	Passed
Pm+Pl (SUS)	156.00	234.60	Passed
Pm+Pl+Q (TOTAL)	186.01	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	19	14	19	14	19	14	19
Circ. Pl (SUS)	-2	-2	0	0	-1	-1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-14	14	4	-4	-19	19	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0
Pm+Pl (SUS)	12.0	17.0	14.0	19.0	13.0	18.0	13.0	18.0
Pm+Pl+Q (Total)	21.1	19.4	14.6	19.1	27.1	24.2	18.2	18.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	147.60	Passed
Pm+Pl (SUS)	19.00	221.40	Passed
Pm+Pl+Q (TOTAL)	27.07	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-1

From: 210

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		42889.97	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

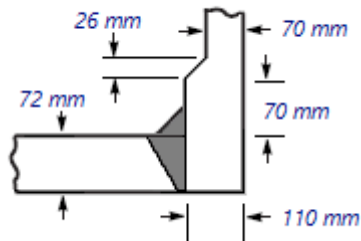
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 9.7278 + 3.0 + 0.0 = 12.7278 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$$

$$= 29.069 - 0.0 = 29.0687 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 602.91 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 813.43 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 3.0$$

$$= 107.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 70.0 - 3.0$$

$$= 67.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.26$$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 307.8)$$

$$= 398.631 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 69.0, 0.0] + \sqrt{307.8 \cdot 107.0}$$



= 181.479 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 400.0
= 400.000 mm

Thickness Limit Candidate [LH3]:

= $8(t + t_e)$
= $8(69.0 + 0.0)$
= 552.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[181.479, 400.0, 552.0] + 69.0$
= 250.479 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
= $69.0 + ((0.0 \cdot 1.0) / 398.631)$
= 69.000 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 307.8 + 107.0) / (\sqrt{(4606.0 + 69.0) \cdot 69.0}))$
= 1.272

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $69.0 \cdot 398.631 \cdot \max((1.272/5)^{0.85}, 1)$
= 27505.570 mm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= A2A + A2C
= 14873.0 + 7504.835
= 22377.834 mm²

Area Contributed by the Outside Fillet Weld [A41]:

= $0.5 \cdot Leg^2$
= $0.5 \cdot 10.0^2$
= 50.000 mm²

The total area contributed by A1 through A43 [AT]:

= $A1 + frn(A2 + A3) + A41 + A42 + A43$
= $27505.57 + 0.944(22377.834 + 0.0) + 50.0 + 0.0 + 0.0$
= 48674.297 mm²

Nozzle Radius for Force Calculation [Rxn]:

= $tn / \ln[1 + tn/Rn]$
= $107.0 / \ln[1 + 107.0/307.8]$
= 358.644 mm

Shell Radius for Force Calculation [Rxs]:

= $teff / \ln[1 + teff/Reff]$
= $69.0 / \ln[1 + 69.0/2303.0]$
= 2337.334 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
= $1.5 \cdot 156.4 \cdot 1.0$
= 234.6 MPa

Force acting on the Nozzle [fN]:

= $P \cdot Rxn \cdot LH$
= $4.593 \cdot 358.644 \cdot 250.479$
= 412529.3 N

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.593 \cdot 2337.334(398.631 + 107.0) \\ &= 5427194.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.593 \cdot 2337.334 \cdot 307.8 \\ &= 3303770.8 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (5427194 + 3303771 + 412529.3)/4.593 \\ &= 1991093.8 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 1991094/48674.297 - 2337.334/69.0) \\ &= 4.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[69.0/2337.334] \\ &= 4.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.894, 4.617) \\ &= 4.617 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (412529.312 + 5427194 + 3303771)/48674.297 \\ &= 187.867 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.593 \cdot 2337.334/69.0 \\ &= 155.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 187.867 - 155.571, 155.571) \\ &= 220.2 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	220.16	MPa
Maximum Allowable Working Pressure	Pmax	4.62	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (307.8 + 107.0)/307.8 \\ &= 1.348 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (307.8 + 107.0) \\ &= 651.566 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 107.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(3303771 \cdot 1.35, 1.5 \cdot 147.6(22377.834 + 0.0), \pi/4 \cdot 4.6 \cdot 307.8^2 \cdot 1.35^2) \end{aligned}$$



= 620565.750 N

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
= $620566 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= $37.113 < \text{or} = \text{to } 156.400$ Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:

= $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
= $(8.688 \cdot 1.0) / (70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 37.7132 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 509.7133 mm

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Input, Nozzle Desc: B

From: 220

Pressure for Reinforcement Calculations	P	4.5926	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Hemispherical Head	D	4632.00	mm
Head Finished (Minimum) Thickness	t	40.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

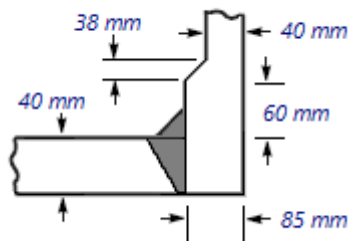
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	38.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		85.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.575	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(4.593/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 9.7278 + 3.0 + 0.0 = 12.7278 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 37.0 + 615.6) / 615.6) - 0.0$$

$$= 16.755 - 0.0 = 16.7547 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 37.300 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 40.005 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.4,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 363.4 MPa	Passed
Occasional	: 18.0,	Allowable	: 196.3 MPa	Passed
Shear	: 5.3,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 85.0 - 3.0$$

$$= 82.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 40.0 - 3.0$$

$$= 37.000 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4632.0/2 + 3.26$$



$$= 2319.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(2319.0 \cdot 37.0)}, 2 \cdot 307.8) \\ &= 292.921 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((37.0 + 0.0)/82.0)^{0.35}, 1) \\ &= \min(0.757, 1) \\ &= 0.757 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(37.0 + 0.0 + 0.757 \cdot \sqrt{(307.8 \cdot 82.0)}, 400.0 + 37.0) \\ &= 157.248 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / LR) \\ &= 37.0 + ((0.0 \cdot 1.0) / 292.921) \\ &= 37.000 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 37.0 \cdot 292.921 \\ &= 10838.095 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A_{2A} + A_{2C} \\ &= 7954.0 + 3079.863 \\ &= 11033.864 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\ &= 10838.095 + 0.944(11033.864 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 21301.127 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 82.0 / \ln[1 + 82.0/307.8] \\ &= 347.188 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/\text{Reff}] \\ &= 37.0 / \ln[1 + 37.0/2319.0] \\ &= 2337.443 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.593 \cdot 347.188 \cdot 157.248 \\ &= 250708.5 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 4.593 \cdot 2337.443 \cdot (292.921 + 82.0) / 2$$
$$= 2012203.1 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$
$$= (4.593 \cdot 2337.443 \cdot 307.8) / 2$$
$$= 1651962.1 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (2012203 + 1651962 + 250708.5) / 4.593$$
$$= 852505.8 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$
$$= 234.6 / (2 \cdot 852506 / 21301.127 - 2337.443 / (2 \cdot 37.0))$$
$$= 4.8 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$
$$= 2 \cdot 156.4 [37.0 / 2337.443]$$
$$= 5.0 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(4.841, 4.951)$$
$$= 4.841 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (250708.516 + 2012203 + 1651962) / 21301.127$$
$$= 183.803 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$
$$= 4.593 \cdot 2337.443 / (2 \cdot 37.0)$$
$$= 145.1 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 183.803 - 145.066, 145.066)$$
$$= 222.5 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	222.54	MPa
Maximum Allowable Working Pressure	Pmax	4.84	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (307.8 + 82.0) / 307.8$$
$$= 1.266 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$
$$= \pi / 2 \cdot (307.8 + 82.0)$$
$$= 612.296 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions:

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 82.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(1651962 \cdot 1.27, 1.5 \cdot 147.6(11033.864 + 0.0), \pi / 4 \cdot 4.6 \cdot 307.8^2 \cdot 1.27^2)$$
$$= 548016.938 \text{ N}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T (0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 548017 / (612.296 (0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0)) \\ &= 34.876 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.235

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (8.688 \cdot 1.0) / (40.0 - 3.0 - 0.0) \\ &= 0.235 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 33.0387 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 473.0388 mm

Input Echo, WRC107/537 Item	1,	Description: B	:
Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	4632.000	mm
Vessel Thickness	Tv	40.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	609.600	mm
Nozzle Thickness	Tn	85.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa
Design Internal Pressure	Dp	4.593	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

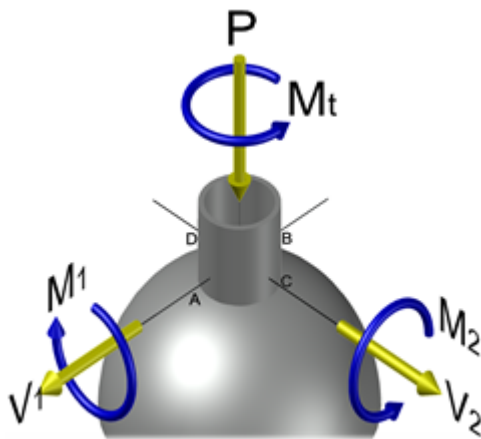
Radial Load	(SUS)	P	53979.1	N
Longitudinal Shear	(SUS)	(V1) V1	38985.0	N
Circumferential Shear	(SUS)	(Vc) V2	38985.0	N
Circumferential Moment	(SUS)	(Mc) M1	72003792.0	N-mm

Longitudinal Moment (SUS) (M1) M2 72003792.0 N-mm
 Torsional Moment (SUS) Mt 82804296.0 N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 779.6 + 2 \cdot 1.65 \cdot \sqrt{(2337.5 (40.0 - 3.0))}$
 $= 1750.089 \text{ mm}$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	53979.1	N
Circumferential Shear	(VC) V2	38985.0	N
Longitudinal Shear	(VL) V1	38985.0	N
Circumferential Moment	(MC) M1	72003792.0	N-mm
Longitudinal Moment	(ML) M2	72003792.0	N-mm
Torsional Moment	MT	82804296.0	N-mm

Dimensionless Param: U = 1.33 TAU = 5.00 (4.25) RHO = 0.45

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.03026	(A,B,C,D)
M(x) / P	SP 1	0.02937	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.03167	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.04337	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.03167	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.04337	(A,B,C,D)
N(y) * T / P	SP 1	0.04570	(A,B,C,D)
M(y) / P	SP 1	0.00831	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.04350	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.01379	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.04350	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.01379	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Rad. Bend. P		-6.9	6.9	-6.9	6.9	-6.9	6.9	-6.9	6.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-5.7	-5.7	5.7	5.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-46.5	46.5	46.5	-46.5
Rad. Memb. ML		-5.7	-5.7	5.7	5.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-46.5	46.5	46.5	-46.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-60.3	46.6	44.0	-35.1	-60.3	46.6	44.0	-35.1
Tang. Memb. P		-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8
Tang. Bend. P		-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.8	-7.8	7.8	7.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-14.8	14.8	14.8	-14.8
Tang. Memb. ML		-7.8	-7.8	7.8	7.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-14.8	14.8	14.8	-14.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-26.3	7.2	18.8	-6.9	-26.3	7.2	18.8	-6.9
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tot. Shear		3.2	3.2	1.5	1.5	1.5	1.5	3.2	3.2
Str. Int.		60.6	46.9	44.1	35.2	60.4	46.7	44.4	35.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		145.1	145.1	145.1	145.1	145.1	145.1	145.1	145.1
Rad. Pl (SUS)		-6.9	-6.9	4.5	4.5	-6.9	-6.9	4.5	4.5
Rad. Q (SUS)		-53.5	53.5	39.6	-39.6	-53.5	53.5	39.6	-39.6
Long. Pm (SUS)		145.1	145.1	145.1	145.1	145.1	145.1	145.1	145.1
Long. Pl (SUS)		-9.6	-9.6	6.0	6.0	-9.6	-9.6	6.0	6.0
Long. Q (SUS)		-16.8	16.8	12.8	-12.8	-16.8	16.8	12.8	-12.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Pm (SUS)		145.1	145.1	145.1	145.1	145.1	145.1	145.1	145.1
Pm+Pl (SUS)		138.5	138.5	151.4	151.4	138.5	138.5	151.4	151.4
Pm+Pl+Q (Total)		119.0	191.9	189.2	138.3	118.8	191.7	189.5	138.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145.07	156.40	Passed
Pm+Pl (SUS)	151.43	234.60	Passed
Pm+Pl+Q (TOTAL)	191.94	469.20	Passed



The $Pm+PI+Q$ allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The $Pm+PI+Q$ allowable was based on a temperature range cycling from ambient to design temperature.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more Nozzles failed Code Requirements!**

Please review all reports carefully!

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